

NXP_SR250T_UCI_Specification

SR250T

Rev. 2.0.23 — 19 February 2025

User manual
CONFIDENTIAL

Document Information

Information	Content
Keywords	SR250T,UCI
Abstract	SR250T Proprietary UCI Specification.



Change Log

Table 1. Change History (Unified UCI Specification)

Revision	Date	Description	Author
2.0.23	2025.02.19	- Editorial Changes - Updation of OCPD angle information - Removed Binary logging related debug options - Adding New App config Parameter to Enable and Disable FoM Processing - ReWording of CORE_DEVICE_INIT_CMD related info - Addition and removal of reason codes - Adding new reason code to report LO gattting calibration error - RADAR_PRESENCE_DET_CFG is marked as active parameter and can be allowed during session active state. - removed WTX_COUNT_CONFIG from the external UCI document - Editorial changes for RSSI_CALIB_CONSTANT_PER_ANTENNA - Addition of Multi target AoA reporting - Correction of values in CIR Header Tap[5] and Tap[7]	
2.0.23		- Editorial Changes - Updating Slot index description in PSDU_LOG_NTF to handle slot index roll over. - Adding TX Antenna Information in RFRAME_LOG_NTF - Corrected the GID of CCC_DEBUG_INFO_NTF - Updated the description of ERROR_AOA_NOT_SUPPORTED_IN_SINGLE_RX - Added a note in the description of RADAR_PRESENCE_DET_CFG - Updated the descriptions of few CCC_RANGE_DATA_NTF parameters. - Consolidated all Aliro UCI changes	
2.0.22		Editorial Changes	
2.0.21	2024.09.18	- Editorial Changes - Addition of new assert cause to the list of UCI asserts. - Enable Radar mode 0 - Improve description of HW security registers - Addition of support of new preambles in UCI specification.	

Table 1. Change History (Unified UCI Specification)...continued

Revision	Date	Description	Author
		Removal of UART references.	
2.0.20	2024.09.13	- Editorial Changes - Editorial Updated for the ANTENNAS_CONFIGURATION_RX Table - Update the default value for CCC_DEBUG_INFO_CONFIG - Add a new status code STATUS_REJECTED_BUFFER_IN_USE - Note has been added to the description of ERROR_INVALID_STS_CONFIG - Updation of note for CLK_CONFIG_CTRL - Value of TX_PULSE_SHAPE_CONFIG updated.	
2.0.19	2024.08.5	Editorial Changes	
2.0.18	2024.07.18	Editorial Changes	
2.0.17	2024.07.15	Editorial Changes	
2.0.16	2024.07.08	Editorial Changes	
2.0.15	2024.07.02	- Editorial Changes - Removed the WIFI_COEX_FEATURE legacy parameter. - New tag id assigned for WIFI_COEX_FEATURE_ALL_CH. - Updated the GID/OID and name of WiFi_CoEx notifications. - Addition of note in SET_SECURE_CALIBRATION_CMD [UP-3338] Rename WIFI_COEX_FEATURE_ALL_CH as WIFI_COEX_FEATURE	
2.0.14	2024.07.10	- Editorial Changes - Added missing info about RX_TIMESTAMP	
2.0.13	2024.05.25	- Editorial Changes - Update default value of XTAL_32K_CAP_CTRL	
2.0.12	2024.05.25	- Editorial Changes - Addition of WIFI_COEX_FEATURE_CH5 and WIFI_COEX_FEATURE_ALL_CH - Addition of Device status code STATUS_FATAL_ERROR - Correction in Vendor Specific data length in UL-TDoA - Addition of CCC Pulse Shapes - Addition of note for calibration parameters [UP-2916] OCPD Detection value and Distance Support in RADAR_RX_NTF	

Table 1. Change History (Unified UCI Specification)...continued

Revision	Date	Description	Author
2.0.11	2024.04.01	• Editorial Changes • Addition of time sync • Addition of DPD wakeup src • Set and Get GPIO pin state is applicable only to customer lifecycle • Addition of note regarding invalid SFD IDs • Addition of Max Supported Channels for AoA Calibration at a time column. • Addition of Module Maker. • Enabling CCC/CSA configs. • Addition of UL-TDOA vendor extension for H2 • Radar Mode 4 update from RFU to Close distance.	
2.0.10	2024.02.6	• Addition of proprietary test configuration carrier frequency offset • Update the minimum value supported by DPD_ENTRY_TIMEOUT from 10ms to 20ms • Allow access for SESSION_INFO_NTF_FILTER_NUM during active ranging • Updated the description of Device configuration parameters. • Updated the description of KDF NTF in UCI specification with CMAC of keys. • Editorial changes	
2.0.9	2024.02.06	• Editorial Changes • Update in fractional CIR Offset • Addition of UWB_SLEEP_PREVENTION_NTF • [UP-2329] Added a note for Single Pin Bit in Radar Mode Switching • Default values of TX_PA_GAIN, TX_PPA_GAIN, TX_DIG_GAIN, DAC_GAIN, TX_MIX_GAIN is set to 0 • Rejecting Dual STS config with an error code. • Enabling Radar Section • Enabling all major IOT features. • Addition of note in DEFAULT_SESSION_VENDOR_APP_CONFIG • Removal of channel 10 support.	
2.0.8	2024.2.01	• Editorial Changes	
2.0.7	2023.12.08	• Unify both RSSI_CALIB param for high power and low power into one parameter as RSSI_CALIB_CONSTANT_PER_ANTENNA • Modified DPD_ENTRY_TIMEOUT value range • Editorial Changes	

Table 1. Change History (Unified UCI Specification)...continued

Revision	Date	Description	Author
		• RADAR_DRIFT_COMPENSATION is enabled by default • Added new parameter SESSION_INFO_NTF_FILTER_NUM • Addition of rframe length in rframe debug notification • Added new status code STATUS_DEVICE_LOW_VBAT • Additional note added for few calibration parameters	
2.0.6	2023.10.09	• Editorial Changes • Allowing default antenna settings as part of CORE_SET_CONFIG_CMD • Updated description of AoA / PDoA measurements per RX entry • Updated description of RADAR_RX_NTF Helios 1 CIR_SAMPLE • Added Missing description for RADAR_RX_NTF Helios 1 CIR_SAMPLE • Added TEST_CHAINED_TX command • Updated SESSION_INFO_NTF section	
2.0.5	2023.09.25	• Added missing parameters to One Way (TDOA) Ranging Measurement table • Description of rx mode for rx-antenna-info is corrected • Editorial Changes	
2.0.4	2023.09.05	Editorial Changes • Updated URSK_DELETE_NTF • Changes w.r.t GID/OID table	
2.0.3	2023.08.07	Android U Drop 7 RC1	

1 Introduction

This document specifies all extensions supported by SR250. Those extensions are NXP proprietary commands supported on top of standard UCI specification protocol.

1.1 Objectives

UCI is defined to meet the following requirements

- Be independent of a specific transport layer (independent of the physical connection and any associated link protocol).
- Provides most configurable way to program the UWB device for the needs of application and platform.
- Be extensive, to allow future extensions.

1.2 Scope

The following figure outlines a typical architecture of UCI interface.

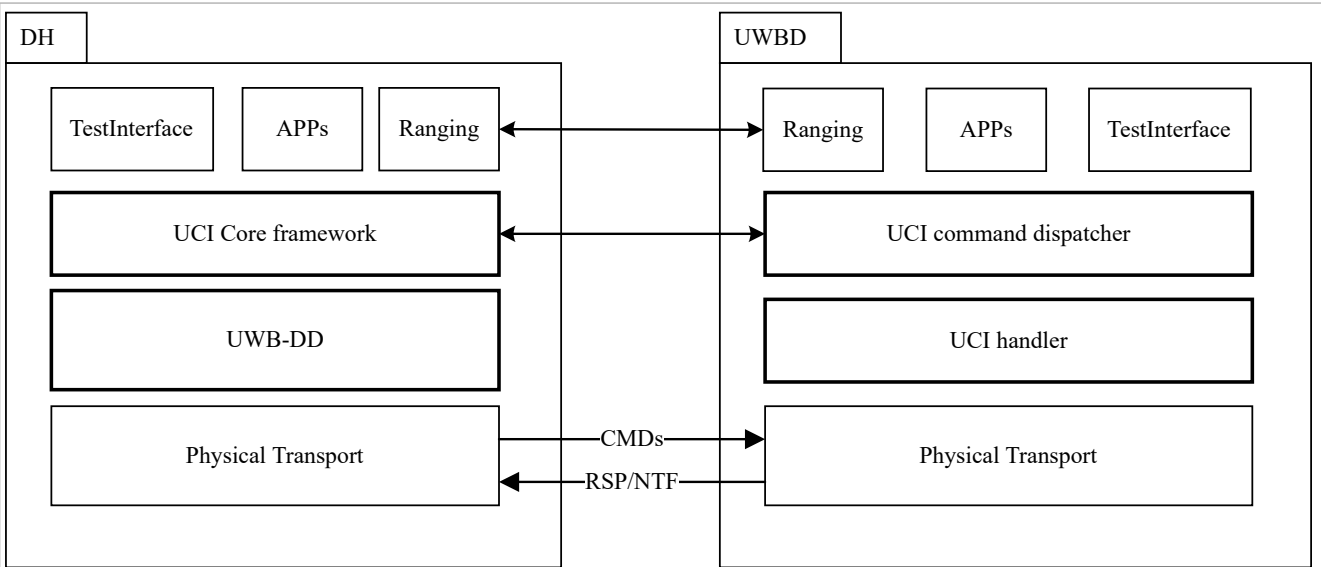


Figure 1. UCI architecture

1.3 Audience

This document is intended for internal use to allow user understanding extensions supported on top of generic UCI specification.

1.4 Abbreviations

Table 2. Abbreviations

Abbreviations	Description
DH	Device Host
GID	Group Identifier
LSB	Least Significant Byte

Table 2. Abbreviations...continued

Abbreviations	Description
lsb	least significant bit
MSB	Most Significant Byte
msb	most significant bit
MT	Message Type
OID	Opcode Identifier
PDU	Protocol Data Unit
PTY	Proprietary
UWBS	Ultra-Wide Band Subsystem
WTX	Waiting Time Extension
URSK	UWB Ranging Session Key

2 SR250 Architecture Overview

SR250 is an UWB transceiver device. SR250 system is depicted in below figure:

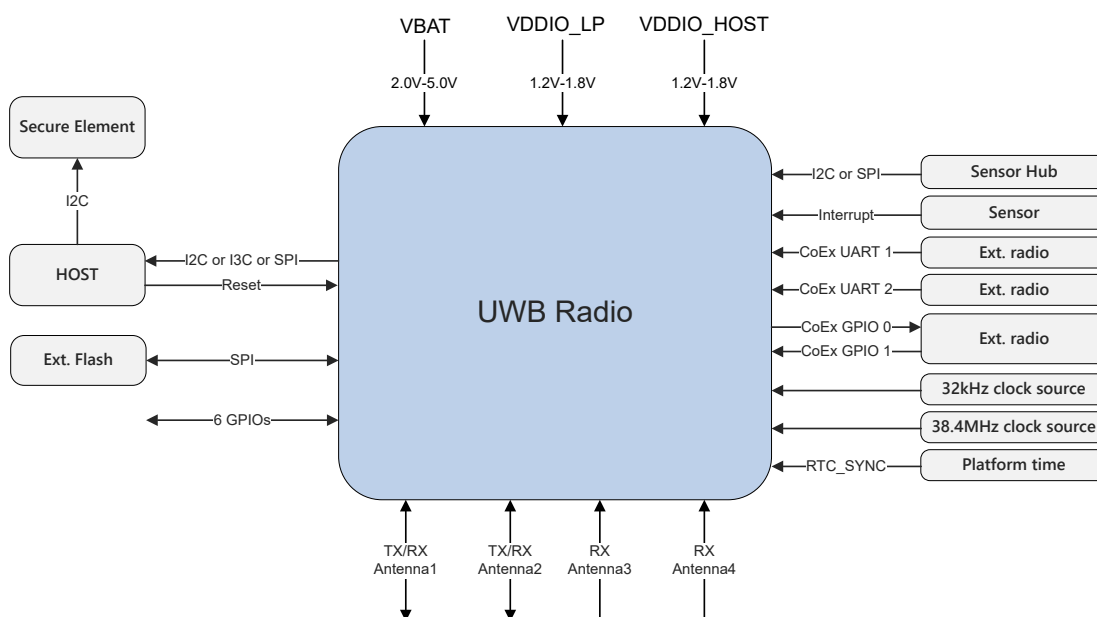


Figure 2. SR250 Architecture

SR250 is connected to the DH through a [physical interface](#). SR250 firmware supports [UCI-FORUM] specifications and are proprietary to NXP.

The firmware also provides support for additional extensions that are not covered in the [UCI-FORUM] specification.

Note: UWB system to Host Interface shall be any of the following

- SPI

3 UCI Overview

The aim of this section is to give an overview of the key points of the [UCI-FORUM] specification.

3.1 Overview

The following figure outlines a reference architecture of UWB enabled device. This document is meant for UCI as logical transport layer.

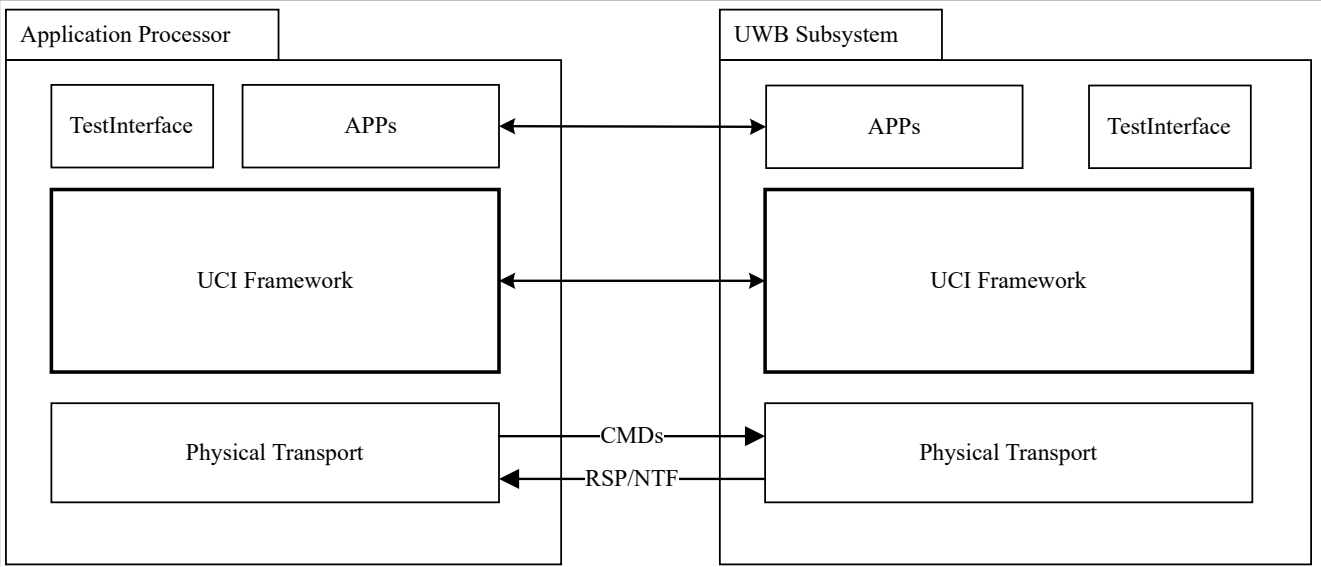


Figure 3. UCI Architecture

3.2 UCI Concepts

3.2.1 Control Messages

Refer [UCI-FORUM] for more details.

3.3 UCI Packet Format

3.3.1 Common packet Header

Refer [UCI-FORUM] for more details.

3.3.2 Format of Control Packets

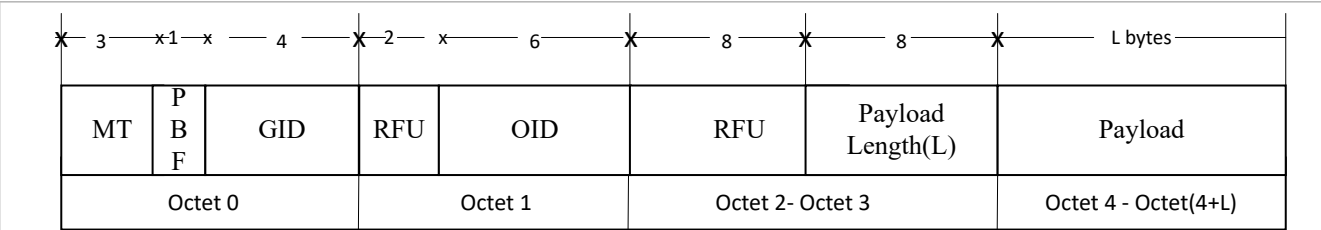


Figure 4. UCI Control packet format

Each Control Packet shall have a 4-bytes Packet Header and may have additional payload for carrying a Control Message payload or a segment of a Control Message payload or a segment of a Control Message payload.

NOTE: In the case of an ‘empty’ Control Message, only the Packet Header is sent.

Message Type (MT)

Refer to [UCI-FORUM] Specification for details of the MT field

Packet Boundary Flag (PBF)

Refer to [UCI-FORUM] Specification for details of the PBF field.

Group Identifier (GID)

The UCI supports Commands, Responses are categorized according their individual groups. The Group Identifier (GID) indicates the categorization of the message and shall be a 4-bit field containing one of the values listed in [UCI] specification

Opcode Identifier (OID)

The Opcode Identifier (OID) indicates the identification of the Control Message and shall be a 6-bit field that is a unique identification of a set of Command, Response Messages within the group. OID values are defined along with the definition of the respective Control Messages described in [UCI-FORUM] specification.

Payload Length (L)

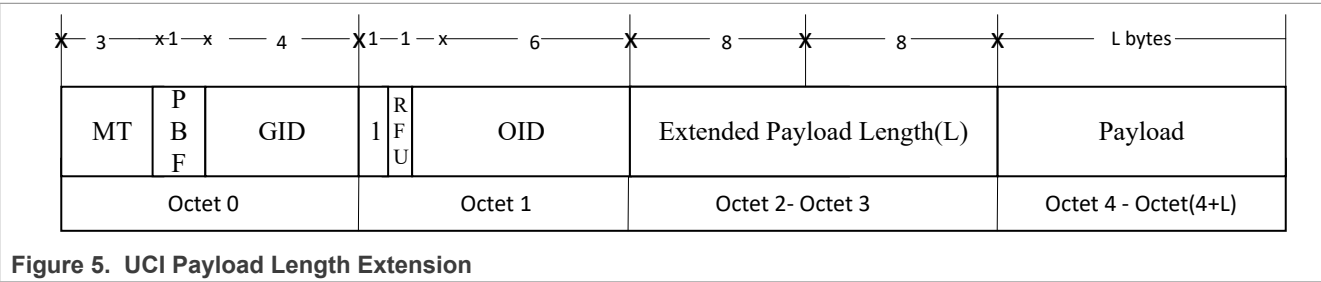
The Payload Length shall indicate the number of octets present in the payload. The Payload Length field shall be an 8-bit field containing a value from 0 to 255 or as specified by max control packet payload size.

3.3.3 UCI Payload Extension

UWBS and DH shall exchange the large UCI packets by using extended payload length (Octet[2] & Octet[3]) as shown in below UCI Header.

Extended payload length is indicated by using RFU bit[7] in Octet[1],If this bit is set to 1, DH and UWBS shall interpret Octet[2] & Octet[3] as extended payload length.

UWBS shall indicate the large packet payload. The maximum control packet from DH is limited by max UCI packet size supported by UWBS. UWBS shall indicate the maximum supported UCI packet size in capability information.



4 Compliance with [UCI-FORUM] and SR250 extension

The SR250 is a secure ranging chip, which offers additional features including UWB proprietary configuration, debug test capabilities and other compliance modes.

To configure those feature, platform specific commands shall be used with UCI proprietary extensions [SR250-UCI] and extensions to [UCI-FORUM] Control messages

[Proprietary GID and GID Definition](#) contains all the additional commands/notifications in [SR250-UCI].

4.1 [SR250-UCI] extensions to [UCI-FORUM] parameters space

[UCI-FORUM] defines a parameter space with a size of 255 parameters in each GID and OID group

[UCI-FORUM] defines that 29 IDs are allocated for proprietary use

Table 3. Parameter Space

Parameters space sub-sections	IDs Space
Assigned & reserved for [UCI-FORUM]	0x00 : 0xE2
Reserved for Proprietary Use	0xE3 : 0xFF

To extend this space, the solution chosen is to define a space of IDs coded on 16 bits, instead of 8 bits. These extended Tags will always start by the value 0xE3, which is the first value available in the Proprietary range. This allows adding 255 new parameters.

Table 4. Extended TLV for proprietary parameters

Payload	Field	Length	Description
m+3 Octets	Tag = 0xE3XX	2 Octets	Extended tag identifier.
	Len	1 Octet	The length of Val (m).
	Val	m Octets	Value of the configuration parameter.

5 Data Download Commands

Host will send the data in form of **extended control packets** to UCI.

In addition to support data download, FW will support additional two commands:

1. Query command : To report details of all data (or specific data) stored in FW.
2. Delete command : Host MW will delete the data from FW when it is no longer required for any use-case.

5.1 DATA_DOWNLOAD_START

Table 5. DATA_DOWNLOAD_START_CMD

Payload Field	Length (Octet)	Value/Description
DATA_ID	1	This is unique number give to each data to be downloaded from Host. Range: 4-255 Value 1-3 are used for default models stored in flash memory. Default Value: 1
DATA_TYPE	1	Type of data: Model, binary.

Table 5. DATA_DOWNLOAD_START_CMD...continued

Payload Field	Length (Octet)	Value/Description
		0x1: Model Data 0x2: Library Data 0x3: Generic Data Default Value: 0x1
DATA_SIZE	4	Size of data to be downloaded Min Value: 256 bytes. Max Value: 32KB.
DATA_CHUNK_SIZE	2	Size of each data chunk. It could be 256bytes - 2k range.

Table 6. DATA_DOWNLOAD_START_RSP

Payload Field	Length (Octet)	Value/Description
STATUS	1	0x31 : DATA_PAYLOAD_SIZE_NOT_SUPPORTED 0x32 : DATA_ID_NOT_SUPPORTED 0x33 : DATA_TYPE_NOT_SUPPORTED 0x34 : DATA_CHUNK_NOT_SUPPORTED 0x35 : DATA_DOWNLOAD_SUCCESS 0x37 : DATA_DOWNLOAD_MEM_ALLOC_FAIL

5.2 DATA_DOWNLOAD

Table 7. DATA_DOWNLOAD_CMD

Payload Field	Length (Octet)	Value/Description
Data	Variable	Actual Data

Table 8. DATA_DOWNLOAD_RSP

Payload Field	Length (Octet)	Value/Description
STATUS	1	0x35: DATA DOWNLOAD SUCCESS 0X36: DATA DOWNLOAD FAILURE

Table 9. DATA_DOWNLOAD_NTF

Payload Field	Length (Octet)	Value/Description
STATUS	1	0x35: DATA DOWNLOAD SUCCESS 0X36: DATA DOWNLOAD FAILURE

5.3 DATA_DOWNLOAD_STOP

Table 10. DATA_DOWNLOAD_STOP_CMD

Payload Field	Length (Octet)	Value/Description
DATA_ID	1	This is an unique number given to each data to be downloaded from Host.

Table 11. DATA_DOWNLOAD_STOP_RSP

Payload Field	Length (Octet)	Value/Description
STATUS	1	0x32 : DATA_ID_NOT_SUPPORTED 0x35 : DATA_DOWNLOAD_SUCCESS 0x36 : DATA_DOWNLOAD_FAILURE 0x39 : DATA_DOWNLOAD_NOT_PRESENT

5.4 DATA_DOWNLOAD_COMPLETE

Table 12. DATA_DOWNLOAD_COMPLETE_NTF

Payload Field	Length (Octet)	Value/Description
STATUS	1	0x35: DATA_DOWNLOAD_SUCCESS 0x36: DATA_DOWNLOAD_FAILURE

5.5 DATA_DOWNLOAD_DELETE

Table 13. DATA_DOWNLOAD_DELETE_CMD

Payload Field	Length (Octet)	Value/Description
DATA_ID	1	This is an unique number given to each data to be downloaded from Host.

Table 14. DATA_DOWNLOAD_DELETE_CMD

Payload Field	Length (Octet)	Value/Description
STATUS	1	0x32: DATA_ID_NOT_SUPPORTED 0x35: DATA_DOWNLOAD_SUCCESS 0x38: DATA_DOWNLOAD_IS_IN_USE 0x39: DATA_DOWNLOAD_NOT_PRESENT

5.6 DATA_DOWNLOAD_QUERY

Table 15. DATA_DOWNLOAD_QUERY_CMD

Payload Field	Length (Octet)	Value/Description
DATA_ID	1	0: Send complete list of all data ids. 4-255: Send info of specific data id. Default Value: 0

Table 16. DATA_DOWNLOAD_QUERY_RSP

Payload Field	Length (Octet)	Value/Description		
STATUS	1	0x32 : DATA_ID_NOT_SUPPORTED 0x35 : DATA_DOWNLOAD_SUCCESS 0x39 : DATA_DOWNLOAD_NOT_PRESENT Note: Only on status as SUCCESS, FW will report the structure list for the Data id/ids		
Number of data Id list	1 Octet	Number of models present at FW (N).		
Structure of each data id.	N * 8 octets	Data Id	1 Octet	This is a unique number given to each

Table 16. DATA_DOWNLOAD_QUERY_RSP...continued

Payload Field	Length (Octet)	Value/Description		
				data to be downloaded from Host.
		Data Type	1 Octet	Type of data
		Data Size	4 Octet	Size of data (in bytes)
		Data Use Count	1 Octet	Indicated data being used by multiple sessions.
		Data Download Status	1 Octet	Status of current data to be downloaded. 0x0: Not present 0x1: Ongoing 0x2: Completed

5.7 Data Download Error Codes

Table 17. Error Codes

ERROR CODES	Values	Description
DATA_PAYLOAD_SIZE_NOTE_SUPPORTED	0x31	Data Payload not supported
DATA_ID_NOT_SUPPORTED	0x32	Data Id not supported
DATA_TYPE_NOT_SUPPORTED	0x33	Data type not supported
DATA_CHUNK_NOT_SUPPORTED	0x34	Data chunk not supported
DATA_DOWNLOAD_SUCCESS	0x35	Data download success response
DATA_DOWNLOAD_FAILURE	0x36	Data download failure response
DATA_DOWNLOAD_MEM_ALLOC_FAIL	0x37	Host is trying to download data at FW, but free memory not available
DATA_DOWNLOAD_IS_IN_USE	0x38	Host is trying to delete data that is being used by FW
DATA_DOWNLOAD_NOT_PRESENT	0x39	Host starts a session with a data which is not present in FW

6 Initialization and Configuration

6.1 Platform variant selection

CORE_DEVICE_INIT_CMD may be used to initialize the UWBS for specific platform variant.

The Major and Minor Version parameters are shown .:

- [Board Variant Parameters](#)

Table 18. CORE_DEVICE_INIT_CMD

Payload Field(s)	Length	Value/Description
Major Version	1 Octet	Platform identifier
Minor Version	1 Octet	Variant identifier

Table 19. CORE_DEVICE_INIT_RSP

Payload Field(s)	Length	Value/ Description
Status	1 Octet	Status values. Refer Status Codes

Table 20. Board Variant Parameters

Major Version	Minor Version	Selected Platform
0x00	0x00	Default value

Table 21. Proprietary device state on top of [UCI-FORUM] devices state

Status	Value/ Description
DEVICE_STATUS_NTF	
0x01	STATUS_READY UWBD is ready for performing UWB session
0x02	STATUS_ACTIVE UWBS is busy running uwb session
0x03 : 0xFC	RFU
0xFD	STATUS_FATAL_ERROR Firmware write is corrupted and needs re-flashing
0xFF	STATUS_ERROR Error occurred within UWBS. Refer [UCI-FORUM] Specification for more details

6.1.1 Handling unified Antennas Configuration

WARNING: When using Unified antennas configuration feature, UWBS may return STATUS_READY, but the UWBS is not yet ready to do ranging, because none of the antannes are defined yet.

As of release of this document, the order of commands have to be:

1. Host -> UWBS : CORE_DEVICE_INIT_CMD
2. Host <- UWBS : CORE_DEVICE_INIT_RSP
3. Host <- UWBS : CORE_DEVICE_STATUS_NTF. (Although status is STATUS_READY, device is not yet ready to do Ranging)
4. Host -> UWBS : [ANTENNA_RX_IDX_DEFINE](#)
5. Host -> UWBS : [ANTENNA_TX_IDX_DEFINE](#)

Table 22. Board Variant Parameters

Major Version	Minor Version	Selected Platform
0x00	0x00	Default value

Table 23. Channel IDs

Channel IDs
5
9

6.2 Device specific information

SR250 sends device specific information as "**Manufacturer Specific Information**" in CORE_GET_DEVICE_INFO_RSP. Refer [Table 108](#) for device information parameters description.

6.3 Device configuration parameters

Device Configuration Parameters are the device specific configuration parameters related to the UWBS and these parameters shall be set by the Host before creating first session in the UWBS device.

The CORE_SET_CONFIG_CMD/CORE_GET_CONFIG_CMD commands are used to set and get the device configuration parameters.

CORE_SET_CONFIG_CMD will be rejected if any of the session is already initialized.

Refer for [Table 110](#) device configuration parameters description.

6.3.1 Clock Settings

CLK_CONFIG_CTRL parameter shall be used as part of CORE_SET_CONFIG_CMD to define the clock options provided by the platform to the UWBS.

This UCI entry is stored in retained in FLASH data area and it's not overwritten when UWBS FW is updated.

This parameter must be written one time in OEM factory. For being considered by UWBS, device must be reboot one time after the parameter update (i.e. RSTn pin toggle is required).

There are following two clocks:

38.4MHz - Used during RF operations

32.768KHz - Base clock, used for time tracking during DPD phases.

Each of the above clock can be either derived from external platform clock or from onboard crystal.

6.3.1.1 RF clock (38.4 MHz clock)

6.3.1.1.1

Platform Clock option:

CLK_REQ2 pin is used to inform Host platform that UWBS requires the external RF clock.

UWBS will assert CLK_REQ2 pin when external clock input is needed and set level to "low" when not needed.

On board crystal option:

UWBS (SR250) shall use the on-board crystal option to derive the RF clock.

CLK_REQ2 shall be not connected.

6.3.1.2 Slow clock (32.768 KHz clock)

6.3.1.2.1

Platform Clock option:

CLK_REQ1 pin is used to let the Host know that the UWBS (SR250) needs the external 32.768 kHz clock.

UWBS will assert CLK_REQ1 pin when external 32.768 kHz clock input is needed and level "low" when not needed.

On board crystal option:

UWBS shall use the on board crystal option to derive the 32.768 kHz clock.

CLK_REQ1 shall be not connected.

6.3.2 CLOCK_PRESENT_WAITING_TIME Configuration

6.3.2.1

This configuration shall be used to wait a dedicated time for the RF clock to become stable.

This duration is platform dependent, therefore Host shall define this parameter with respect to the maximum time its platform may require to provide a stable clock to the UWBS.

Same parameter is used in case of on-board crystal or platform clock. In case of platform clock, it defines the duration between the assertion of CLK_REQ2 pin and the moment UWBS will consider the clock is stable.

6.3.3 SLOW_CLK_PRESENT_WAITING_TIME Configuration

This configuration shall be used to wait a dedicated time for 32kHz clock to become stable.

This time is platform dependent, therefore Host shall define this parameter with respect to the maximum time its platform may require to provide a stable clock to the UWBS.

Same parameter is used in case of crystal or platform clock. In that case of platform clock, this parameter defines the duration between the assertion of CLK_REQ1 pin and the moment UWBS will consider the clock is stable.

6.3.4 Controlee RX window offset Configuration

When UWBS acting as responder/controlee, this config shall be used to set the offset for RX antenna to turn on to receive the first message of a ranging round, i.e. RCM or RIM.

Background: If counterpart device (controller/initiator) has too huge clock drift between two ranging cycles, controlee might miss the RCM/RIM leading to a failed ranging round. This can be avoided by enabling Rx much earlier before expected RCM/RIM. However, the earlier Rx is enabled (=bigger Rx on window), the higher is the power consumption so a trade-off must be defined between power consumption and interoperability.

The INITIAL_RX_ON_OFFSET_ABS and INITIAL_RX_ON_OFFSET_REL parameters allows customer to fine-tune offset value. The description for the parameters is defined at [Device Configuration Parameters](#).

6.4 DPD wake up source configuration and DPD Handling

DH shall configure the wake-up source with DPD_WAKEUP_SRC config by using CORE_SET_CONFIG_CMD. This UCI command is defined in [UCI-FORUM] specification.

Default value: 0x00

The wake-up sources are defined as below

Table 24. DPD wakeup sources

Bit mask								Descriptions
7	6	5	4	3	2	1	0	
							X	RFU
						x		GPIO 3
X	X	X	X	X	X			RFU

Note:

1. GPIO 3 can also be used for the purpose of antenna selection. When GPIO 3 is configured as DPD_WAKEUP_SRC, it cannot be configured for antenna selection.

6.4.1 DPD Handling

This section describes the low power mode handling of UWBS.

UWBS can enter into low power mode if device configuration parameter LOW_POWER_MODE config is enabled.

UWBS shall enter into DPD immediately after DEVICE_STATUS_NTF with STAUS_READY is sent from UWBS and DH has received. UWBS shall also enter into low power mode during STAUS_READY (ranging round).

If the host receives CORE_GENERIC_ERROR_NTF with Status Retry then it shall retry to send the command as the UWBS was in DPD when the previous attempt of sending the UCI command occurred.

The DPD_ENTRY_TIMEOUT is started if the wake-up reason is host interface and it shall restart the timer if it receives subsequent UCI commands and once the timer is expired UWBS shall enter into DPD

6.5 Vendor Application configuration parameters

UWBS offers Vendor application configurations used to configure the UWB session. The [Table 112](#) contains the proprietary application configuration which are specific to SR250 device

DH shall configure Vendor APP parameters by using SET_VENDOR_APP_CONFIG_CMD.

6.6 Capability information

AP shall use CORE_GET_CAPS_INFO_CMD command to get the capability of the UWBS. The capability information is vendor specific and AP shall use capability information to communicate with UWBS. Refer [Table 109](#) for capability parameters.

6.7 Query on temperature

AP shall use QUERY_TEMPERATURE_CMD to query the instantaneous temperature of the UWBS. Temperature value is reported via QUERY_TEMPERATURE_RSP by UWBS.

Table 25. QUERY_TEMPERATURE_CMD

Payload Field(s)	Length	Value/ Description
Command	0 Octet	Query thermal sensor reading

Table 26. QUERY_TEMPERATURE_RSP

Payload Field(s)	Length	Value/ Description
Status	1 Octet	Status values. Refer Status Codes . Below field is valid if status is STATUS_OK
Temperature value	1 Octet	Signed temperature value in degree Celsius

6.8 Notification of high temperature to host

UWBS shall send GENERIC_ERROR_NTF with status STATUS_DEVICE_TEMP_REACHED_THERMAL_RUNAWAY when UWBS detects over temperature. FW will send DEVICE_STATUS_NTF with state DEVICE_STATE_READY once internal temperature is back to its operating range. It is recommended to do a POR and re-configure the session.

6.9 Notification of low VBat to host

UWBS shall send GENERIC_ERROR_NTF with status STATUS_DEVICE_LOW_VBAT when UWBS detects VBat dropped below its operational range. FW will send DEVICE_STATUS_NTF with state DEVICE_STATE_READY once VBat is back to its operating range. It is recommended to do a POR and re-configure the session.

6.10 True Random GEN

The following UCI command(s) are used to generate the Random Number for TRNG_DATA

Table 27. TRNG_DATA_CMD

Payload Field(s)	Length	Value / Description
Length	1 Octet	Number of bytes of True Random Number to be generated.

Table 28. TRNG_DATA_RSP

Payload Field(s)	Length	Value / Description
Status	1 Octet	Status code as per FIRA UCI Specification

Table 28. TRNG_DATA_RSP...continued

Payload Field(s)	Length	Value / Description
Length	m Octet	True Random number generated.

6.11 Module Maker Info

Following UCI commands are used to access the OTP memory for writing and reading the module maker information.

6.11.1 Write Module Maker Info

Table 29. WRITE_MODULE_MAKER_INFO_CMD

Payload Field(s)	Length	Value / Description
Module Maker Info	2 Octet	Module maker information to be written <i>.in FLASH. Can be written only once.</i> Note: Supported in Customer Lifecycle.

Table 30. WRITE_MODULE_MAKER_INFO_RSP

Payload Field(s)	Length	Value / Description
Status	1 Octet	UCI Status

6.11.2 Read Module Maker Info

Table 31. READ_MODULE_MAKER_INFO_CMD

Payload Field(s)	Length	Value / Description
NA	0 Octet	NA

Table 32. READ_MODULE_MAKER_INFO_RSP

Payload Field(s)	Length	Value / Description
Status	1 Octet	UCI Status
Module Maker Info	2 Octet	Module maker information read <i>from the FLASH.</i> Note: Supported in all Lifecycle.

6.12 Profile Parser

6.12.1 Profile Parser Requirement Details

1. Phone receives the accessory configuration data, phone configures and sends the shareable configuration data to configure the accessory.
2. This accessory information termed as 'Profile' or 'Blob' is a session configuration data which is received over BLE.

6.12.2 Profile Parser UCI Details

Following are the profile parser UCI commands.

Table 33. PROP_SET_PROFILE_CMD

Payload Field(s)	Length (Bytes)	Value / Description
Profile ID	1	MW Data (0x0B)
Device Type	1	MW Data DEVICE_TYPE (FiRa)
Source Mac Address	2	MW Data DEVICE_MAC_ADDRESS (FiRa)
Device Role	1	MW Data DEVICE_ROLE (FiRa)
Version (Major, Minor)	4	MW Data <u>Profile Data versioning format</u> Byte[1:0] : Major version - Major number in version is set to 0x0001 Byte[3:2] : Minor version - Minor number in version is set to 0x0000 / 0x0001 (based on IOS version)
Profile Data Length	1	MW Data 0x17 if Minor version is set to 0x0000 0x19 if Minor version is set to 0x0001
Profile Data	23 or 25	Profile or Blob Data

Table 34. PROP_SET_PROFILE_RSP

Payload Field(s)	Length (Bytes)	Value / Description
Status	1	UCI Status Session Handle field shall be present if UCI Status is STATUS_OK
Session Handle	4	Session Handle is a 32bit unique number generated by the UWBS and it shall be used to uniquely identify the UWBS Session. The Host and the UWBS shall use a Session Handle as a reference for a UWBS session in any session specific command/response/notification.

6.12.3 Profile Parser Debug Enable

To debug the profile parser, the following SET_VENDOR_APP_CONFIG/CORE_SET_CONFIG parameters can be enabled.

Table 35. SET_NXP_APP_CONFIG Parameters

Parameter Name
CIR_LOG_NTF
PSDU_LOG_NTF

Table 35. SET_NXP_APP_CONFIG Parameters...continued

Parameter Name
CIR_CAPTURE_WINDOW
ANTENNAS_CONFIGURATION_RX
TX_ADAPTIVE_PAYLOAD_POWER

Table 36. CORE_SET_CONFIG Parameters

Parameter Name
NXP_EXTENDED_NTF_CONFIG

Steps to enable the debug for profile praser :

1. Send PROP_SET_PROFILE_CMD
2. Send SESSION_SET_VENDOR_APP_CONFIG/CORE_SET_CONFIG (with the required parameters)

6.12.4 Profile Parser Implementation

1. INITIAL_RX_ON_OFFSET_ABS : This will be set to 0 when Blob parser command is received.
2. INITIAL_RX_ON_OFFSET_REL : CLOCK_DRIFT will be mapped to this UCI parameter

It is offset in PPM relative to ranging interval, So if INITIAL_RX_ON_OFFSET_REL is set to 100ppm and ranging interval is 100ms, then relative offset will be 10us. $(\text{INITIAL_RX_ON_OFFSET_REL} * \text{Ranging Interval} / 100000)$

Example :

- RI (Ranging Interval) = 100ms and CLOCK DRIFT = 100 ppm will lead to RX ON window = 10 us
- RI = 5 seconds and CLOCK DRIFT = 1 ppm will lead to RX ON window = 5us
- RI = 5 seconds and CLOCK DRIFT = 300 ppm will lead to RX ON window = 1500 us or 1.5 m

7 UWB session

7.1 UWB Session Initiation

Initiation of a UWB Session

The Host shall establish a session by sending the SESSION_INIT_CMD command to create the new UWB session with parameter session ID and type of the session. The UWBS shall respond with SESSION_INIT_RSP with Status set to STATUS_OK and the Session Handle field if the command is successfully accepted by the UWBS, otherwise the UWBS shall send the SESSION_INIT_RSP with proper status code which indicates a failure. The UWBS shall send the SESSION_INIT_RSP with STATUS_OK and a Session Handle after the session is created followed by sending SESSION_STATUS_NTF with SESSION_STATE_INIT Session State. The UWBS and Host shall use the Session Handle as a reference for a UWB session and use the Session Handle in any session specific UCI Messages during communication. The Host shall receive the SESSION_STATUS_NTF notification on every UWB session state change. The Host shall wait for SESSION_STATUS_NTF notification with Session State of SESSION_STATE_INIT before applying application configurations for a UWB session.

The UWBS shall respond with STATUS_ERROR_SESSION_NOT_EXIST error status when the Host tries to Configure/Start/Stop the UWB session without initializing the UWB session.

Table 37. SESSION_INIT_CMD

Payload Field(s)	Length	Value/Description
Session ID	4 Octets	Session ID is 4 Octets random number generated by application. Note: Session ID = 0x00000000: Reserved for Device Test Mode session
Session Type	1 Octet	Type of session 0x00 = Ranging session (no in-band data) 0x01 = Ranging and in-band data session 0x02 = Dedicated data transfer session 0x03 = Ranging-only Secondary Session 0x04 = In-band data Secondary Session 0x05 = Ranging-with-data Secondary Session 0x9F = HUS Primary Session 0xA0 = Reserved for CCC 0xA2 = Aliro Ranging Session 0x06-0x9E = RFU 0xA1& 0xA3-0xCF = Reserved for Vendor Specific use case 0xD0 = Device Test Mode 0xD1-0xDF = RFU 0xE0-0xFF = Vendor Specific use

Table 38. SESSION_INIT_RSP

Payload Field(s)	Length	Session ID
Status	1 Octet	Status code as per Status Codes Session Handle field shall be present if Status is STATUS_OK
Session Handle	4 Octets	Session Handle is a 32bit unique number generated by the UWBS and

Table 38. SESSION_INIT_RSP...continued

Payload Field(s)	Length	Session ID
		it shall be used to uniquely identify the UWB Session. The UWBS shall set the Session Handle most significant bit (msb) of the Most Significant Byte (MSB) to 0b1. The Host and the UWBS shall use a Session Handle as a reference for a UWB session in any session specific command/response/notification

7.2 Get the list of UWB sessions

Below proprietary command shall be used to get all the sessions in UWBS. UWBS shall send GET_ALL_UWB_SESSIONS_RSP with all the sessions (Init/Active/Idle) in UWBS.

Table 39. GET_ALL_UWB_SESSIONS_CMD

Payload Field(s)	Length	Value/ Description
Command	0 Octets	Get the list of UWB sessions

Table 40. GET_ALL_UWB_SESSIONS_RSP

Payload Field(s)	Length	Value/ Description		
Status	1 Octet	Status values. Refer Status Codes		
Session Count	1 Octet		Number of sessions(n)	
Session(1..n)	n * 6 Octets	Session Handle	Byte[3:0]	Session handle
		Session Type	Byte[4]	Session type
		Session State	Byte[5]	Session state, Refer generic spec for different session states

7.3 Scheduler status notification

UWBS shall SCHEDULER_STATUS_NTF notification when SCHED_STATUS_NTF is enabled. UWBS shall send SCHED_STATUS_NTF notification when scheduler meets either SESSION_SCHEDULER_ATTEMPTS or SESSION_SYNC_ATTEMPTS configuration criteria.

Table 41. Control message for schedule notification

Payload Field(s)	Length	Value/ Description		
SCHEDULER_STATUS_NTF				
Num of sessions	1 Octet		Num of sessions (m)	
Session Data	M x 14 octets	Session Handle	4 Octets	Session Handle
		Scheduler status	1 Octet	Scheduler status. Refer Table 42 for status codes

Table 41. Control message for schedule notification...continued

Payload Field(s)	Length	Value/ Description		
		Num of successful scheduling	4 Octets	Number of successfully scheduled session
		Num of unsuccessful scheduling	4 Octets	Number of failed scheduled session
		Priority	1 Octets	Current priority of the session

Table 42. Scheduler status codes

Codes	Name	Description
0x00	SCHD_STATUS_SESSION_SUCCESS	Session scheduled successfully
0x01	SCHD_STATUS_SESSION_CANNOT_SCHEDULE	There is not enough time slots available to schedule the session
0x02	SCHD_STATUS_SESSION_SYNC_FAILURE	Controlee session couldn't receive any message from Controller for a long time

7.4 Extension to reason code in SESSION_STATUS_NTF

Table 43. Extension to reason code in SESSION_STATUS_NTF

Value	Description
SESSION_STATUS_NTF Reason codes. Refer section Error Code description for description on error code)	
0x80	ERROR_INVALID_ANTENNA_CFG
0x81	ERROR_INVALID_RX_MODE
0x82	ERROR_FAIL_DYNAMIC_STS_NOT_ALLOWED
0x83	ERROR_FEATURE_NOT_SUPPORTED_FOR_MODEL
0x84	ERROR_RX_MODE_TOA_MODE_MISMATCH
0x85	ERROR_INSUFFICIENT_MEMORY_FOR_INBAND_DATA
0x86	ERROR_INVALID_DATA_TRANSFER_MODE
0x87	ERROR_INVALID_MAC_CFG
0x88	ERROR_ANTENNA_DEFINES_NOT_CONFIGURED
0x89	ERROR_INVALID_MAX_TDOA_SESSION_COUNT_REACHED
0x8A	ERROR_LOOPBACK_MODE_TX_POWER_TOO_HIGH
0x8B	ERROR_WRONG_SESSION_TYPE_FOR_INBAND_DATA
0x8C	ERROR_AOA_NOT_SUPPORTED_IN_SINGLE_RX
0x8D	ERROR_DUPLICATE_DST_MAC_ADDRESS_DETECTED

Table 43. Extension to reason code in SESSION_STATUS_NTF...continued

Value	Description
0x8E	ERROR_INVALID_ADAPTIVE_HOPPING_THRESHOLD
0x8F	ERROR_UNSUPPORTED_RANGING_LIMIT
0x90	ERROR_INVALID_HOPPING_MODE
0x91	ERROR_RNG_INVALID_DEVICE_ROLE
0x92	ERROR_KEY_ROTATION_NOT_SUPPORTED
0x93	ERROR_TEST_KDF_NOT_SUPPORTED
0x96	ERROR_RNG_INVALID_DEVICE_TYPE
0x98	ERROR_INVALID_CHANNEL_ID
0xA1	ERROR_URSK_TTL_MAX_VALUE_REACHED
0xA2	ERROR_CCC_TERMINATION_ON_MAX_STS_INDEX
0xB0	ERROR_RADAR_CIR_MAX_TAP_IDX_EXCEEDED
0xB1	ERROR_RADAR_ANTENNA_CONFIG_RX_NOT_OK
0xB2	ERROR_RADAR_PRESENCE_DETECTION_RANGE_EXCEEDED
0xB3	ERROR_RADAR_RX_GAIN_INDEX_NOT_OK
0xB4	ERROR_RADAR_DRIFTCOMP_ANTENNA_CONFIG_NOT_OK
0xB6	ERROR_RADAR_LOGATING_CALIBRATION_INVALID_SETUP
0xB7	ERROR_SESSION_STOPPED_DUE_TO_FCC_LIMIT_REACHED
0xC0	ERROR_CSA_INVALID_CFG
0xC1	ERROR_INVALID_RESPONDER_SLOT_INDEX_CONFIGURED

NOTE:

- [0xB0 - 0xBF]: Radar specific use
- [0xC0 – 0xCF]: CCC/CSA/Alro Specific Use

7.4.1 Error Code description

UWBS shall report below error codes when required application configurations are not met.

7.4.1.1 List of Error Code(s)

ERROR_RX_MODE_INVALID_RX_ANT_PAIR_SELECTION

- With RX_MODE = 0(Dual), RX_ANTENNA_PAIR_SELECTION should not be 0 (single Rx)

ERROR_INVALID_RX_MODE

- With DUAL_AOA_PREAMBLE_STS = 1, RX_MODE != 0 (Dual Rx) is invalid

ERROR_FAIL_DYNAMIC_STS_NOT_ALLOWED

- For Dynamic STS, if UWBS and ESE is not bound and locked
- If Dynamic STS config is set in factory FW

ERROR_FEATURE_NOT_SUPPORTED_FOR_MODEL

- If requested feature is not supported in given model ID

ERROR_RX_MODE_TOA_MODE_MISMATCH

- RX_MODE = 1 (Rx0) should have the TOA_MODE as 0
- RX_MODE = 2 (RX1) should have the TOA_MODE as 1
- RX_MODE = 0 (Dual Rx) should have the TOA_MODE as 2

ERROR_INSUFFICIENT_MEMORY_FOR_INBAND_DATA

- This reason code shall be sent when requested buffer blocks via below config are not available in UWBS:
 - SESSION_INBAND_DATA_TX_BLOCKS
 - SESSION_INBAND_DATA_RX_BLOCKS

ERROR_INVALID_DATA_TRANSFER_MODE

- This reason code shall be sent if HOST tries to use Connection Oriented data transfer mode

ERROR_INVALID_MAC_CFG

- MAC_CFG !=3 for RANGING_METHOD = SS_TWR or DS_TWR is not supported

ERROR_INVALID_MAX_TDOA_SESSION_COUNT_REACHED

- When SCHEDULED_MODE = 1, For RANGING_METHOD = ONE_WAY_RANGING, only one TDOA session allowed at a time

ERROR_LOOPBACK_MODE_TX_POWER_TOO_HIGH

- This reason code shall be sent if HOST tries to start a Loopback mode with a TX gain index below 80

ERROR_WRONG_SESSION_TYPE_FOR_INBAND_DATA

- This reason code shall be sent when requested buffer blocks via below config and session type is not 'Ranging and in-band data session' or 'Data transfer session':
 - SESSION_INBAND_DATA_TX_BLOCKS
 - SESSION_INBAND_DATA_RX_BLOCKS

ERROR_AOA_NOT_SUPPORTED_IN_SINGLE_RX

- – AoA should be enabled only in RX Antenna Configuration Modes : AOA / AOA_RFM / AOA_CSA

ERROR_DUPLICATE_DST_MAC_ADDRESS_DETECTED

- Reported when DST_MAC_ADDRESS List should not contain duplicate entries

ERROR_INVALID_ADAPTIVE_HOPPING_THRESHOLD

- When SCHEDULED_MODE = 1, With DEVICE_TYPE = CONTROLLER and HOPPING_MODE = FIRA_ADAPTIVE_HOPPING / CCC_ADAPTIVE_MODULO_HOPPING / CCC_ADAPTIVE_AES_HOPPING, then ADAPTIVE_HOPPING_THRESHOLD should be < NUMBER_OF_CONTROLEES

ERROR_UNSUPPORTED_RANGING_LIMIT

- In ONE_WAY_RANGING, TDOA mode has only SP1 packet and Distance Limit is not applicable (i.e SESSION_INFO_NTF_CONFIG should not be 2)

ERROR_INVALID_HOPPING_MODE

- ROUND_HOPPING should be disabled during contention based ranging
- Restricting HOPPING_MODES with respect to session-type:
 - For Ranging Sessions:
 - 0x00 : Hopping Disable
 - 0x01 : FiRa Hopping Enable
 - 0xA0 : NXP Adaptive Hopping Enable
 - For CCC Sessions:
 - 0 : No Hopping
 - 2 : Adaptive Hopping using MODULO
 - 3 : Continuous Hopping using MODULO
 - 4 : Adaptive Hopping using AES
 - 5 : Continuous Hopping using AES

ERROR_RNG_INVALID_DEVICE_ROLE

- DEVICE_ROLE with 0x02 (Master Anchor – TdoA Synchronizer) when RANING_ROUND_USAGE is not equal to 0x00 (One Way Ranging)
- DEVICE_ROLE with 0x05 (Advertiser) / 0x06 (Observer) when RANING_ROUND_USAGE is not equal to 0x06 (OWR for AoA Measurement)
- DEVICE_ROLE is not 0x05 (Advertiser) or 0x06 (Observer) and RANING_ROUND_USAGE is equal to 0x06 (OWR for AoA Measurement)

ERROR_KEY_ROTATION_NOT_SUPPORTED

- The key rotation feature is enabled only during Dynamic STS ranging (STS_CONFIG = 0x01).

ERROR_TEST_KDF_NOT_SUPPORTED

- For STS_CONFIG = 2 (PHY ACCESS), bTest KDF feature Enable should not be enabled.

ERROR_RNG_INVALID_DEVICE_TYPE

- UWBS shall report this error code when the required application configurations are-
- SESSION_TYPE=CCC Session and DEVICE_TYPE is 0x00 or 0x01
- SESSION_TYPE=RNG Session and DEVICE_TYPE is 0xA0 or 0xA1

ERROR_INVALID_NUM_OF_STS_SEGMENTS

- NUM_STS_SEGMENTS>=2 will be rejected with error code SSN_RSNCODE_ERROR_INVALID_NUM_OF_STS_SEGMENTS = 0x32

ERROR_CSA_INVALID_CFG

- CSA active number of ranging rounds can be set for CCC session only.
- CSA offset to be in the range of 1 to Nround-1.

ERROR_INVALID_STS_CONFIG

- STS_CONFIG should be STATIC when contention based ranging is enabled
- Returned by FW when Dynamic STS is selected but WRDS is not set

ERROR_RADAR_LOGATING_CALIBRATION_INVALID_SETUP

This reason code shall be sent if the set of parameters specified for LO gating calibration is not consistent.

This may be because:

- Radar mode is not appropriate, can only be 5 or 6 for LO gating calibration
- More than one Rx path is enabled during calibration
- RADAR_SINGLE_FRAME_NTF is not set to 1

7.4.2 UWB session stop by in-band signaling

UWBS shall notify SESSION_STATUS_NTF with below reason code for in-band stop of the UWB session. The AP Shall de-initialize the session after received session status notification with idle state due to in-band stop signal over UWB.

Reason Code:

- 0x05

7.5 Dynamic Application Configuration Parameters

Below is the list of Application configuration parameters, which can be dynamically modified, meaning while the session is active.

Table 44. Dynamic APP configuration parameter List

Parameter Name
ANTENNAS_CONFIGURATION_RX
ANTENNAS_CONFIGURATION_TX
DATA_LOGGER_NTF

Table 44. Dynamic APP configuration parameter List...continued

Parameter Name
CIR_LOG_NTF
PSDU_LOG_NTF
SESSION_INFO_NTF_FILTER_NUM

7.6 Extension to SESSION_INFO_NTF

The vendor extension for SESSION_INFO_NTF can be added at the end of the generic SESSION_INFO_NTF as shown below. UWBS shall append the vendor extension only if NXP_EXTENDED_NTF_CONFIG core device configuration parameter is set to 0x01. Refer [Table 110](#) for details on NXP_EXTENDED_NTF_CONFIG core configuration parameter.

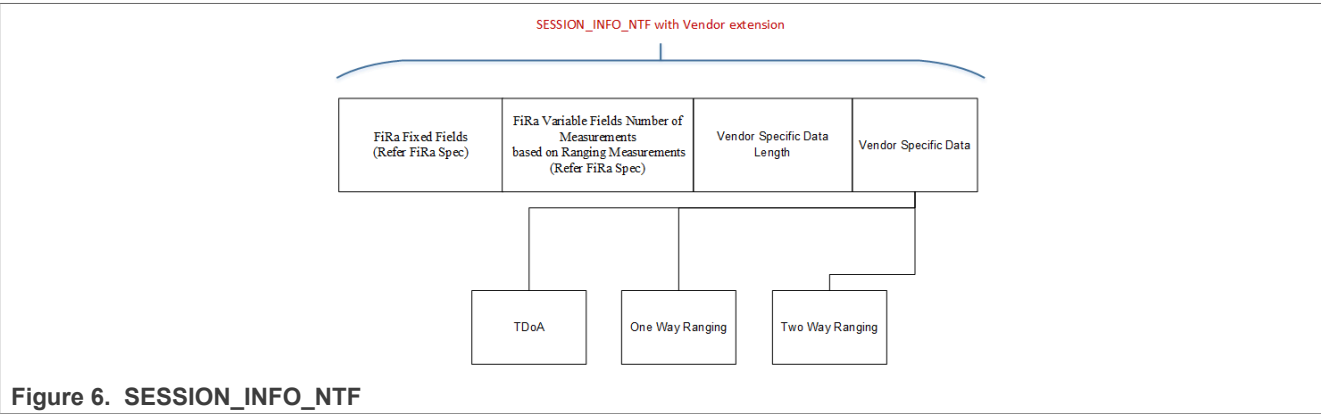


Figure 6. SESSION_INFO_NTF

Table 45. SESSION_INFO_NTF

Fields Description	UL-TDoA	Two Way Ranging / One Way Ranging
FiRa Fixed Fields	Refer FiRa Spec	Refer FiRa Spec
FiRa Variable Fields - Number of Measurements based on Ranging Measurements	Refer to Table OWR UL-TDoA Measurement Result from FiRa spec, if Ranging Measurement Type = 0x00	- Refer to Table Two Way Ranging Measurement Result from FiRa spec, if Ranging Measurement Type = 0x01 - Refer to Table DL-TDoA Ranging Measurement Result from FiRa spec, if Ranging Measurement Type = 0x02 Note: No vendor extension for DL-TDoA - Refer to Table OWR for AOA Measurement Result from FiRa spec, if Ranging Measurement Type = 0x03
Vendor Specific Data Length	2 Octets - Vendor Data Length for vendor specific ranging data, Vendor Specific Data field is not present if Vendor Data Length = 0. - This Vendor Data Length field is present if NXP_EXTENDED_NTF_CONFIG is set to 0x01(Vendor extended notifications)	2 Octets - Vendor Data Length for vendor specific ranging data, Vendor Specific Data field is not present if Vendor Data Length = 0. - This Vendor Data Length field is present if NXP_EXTENDED_NTF_CONFIG is set to 0x01(Vendor extended notifications)

Table 45. SESSION_INFO_NTF...continued

Fields Description	UL-TDoA	Two Way Ranging / One Way Ranging
Vendor Specific Data	N Octets Vendor Specific extension for SESSION_INFO_NTF (UL-TDoA Specific). Refer Table 46 for details	N Octets Vendor Specific extension for SESSION_INFO_NTF for TWR Measurements and OWR Advertisement. Refer Vendor Specific extensions in SESSION_INFO_NTF for details

Table 46. Vendor Specific extensions in SESSION_INFO_NTF for UL-TDoA

Parameter Name	Length	Value/ Description
Ranging Measurements vendor extension	M*ranging meas. Size	Refer Vendor specific extension for UL-TDoA Measurement at Table 47 if Ranging Measurement Type = 0x00.

Table 47. Vendor specific extension for One Way Ranging (UL-TDoA) Measurement

Parameter Name	Length	Value/ Description
Vendor Extension Length	1 Octet	Vendor extension length for TDOA Measurements below.
RSSI RX1	2 Octets	Received Signal Strength Indicator for RX1 in dB. It is expressed in Q8.8 format, Non zero values are always interpreted as negative value.
RSSI RX2	2 Octets	Received Signal Strength Indicator for RX2 in dB. It is expressed in Q8.8 format, Non zero values are always interpreted as negative value.
Number of PDoA Measurements	1 Octet	Number of PDoA measurements to follow (N).
PDoA Measurements	N x 4 Octets	Where N is the number of PDoA measurements to follow and Each measurement includes the PDoA estimate (2 octets) and the corresponding PDoA index (2 Octets) at which the estimate has been taken, Refer Vendor Specific PDoA Measurements for more details
Message Control Extension	2 Octets	Bitfield for specific extensions, bit0=1: RX3 extension, others RFU.
RX3 Extension info length	1 Octets	Number of bytes for RX3-related extension.
RSSI RX3	2 Octets	Received signal strength RX3

Table 48. Vendor Specific PDoA Measurements (UL-TDoA)

Parameter Name	Length	Value/ Description
PDoA [1...N]	2 Octets	Estimation of phase difference in degrees from antenna pair [1...N]. It is a signed degree in Q9.7 format.

Table 48. Vendor Specific PDoA Measurements (UL-TDoA)...continued

Parameter Name	Length	Value/ Description
		This field is zero if AOA_RESULT_REQ = 0. Value ranges from -180 to 180 degrees.
PDoA Index [1...N]	2 Octets	CIR index estimate at which PDoA [1...N] has been detected. This is vendor specific. This field is zero if AOA_RESULT_REQ = 0.

Table 49. Vendor Specific extensions in SESSION_INFO_NTF Continued for UL-TDoA

Parameter Name	Length	Value/ Description
Antenna Pair Info	4 Octets	Antenna Rx Configuration information used in current ranging round. - Octet[0] : Configuration Mode as per ANTENNAS_CONFIGURATION_RX supported configuration mode are 0, 1 0: Configuration Mode 0: ToF only mode. 1: Configuration Mode 1: 3D/2D AoA, implicit ToA mode usecase. - Octet[1] : - Config mode(s) 0: - Antenna ID 1 as per ANTENNA_RX_IDX_DEFINE - Config mode(s) 1: - Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE - Octet[2] : - Config mode(s) 0: - Antenna ID 2 as per ANTENNA_RX_IDX_DEFINE - Config mode(s) 1: - Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE - Octet[3] : - Config mode(s) 0: - Antenna ID 3 as per ANTENNA_RX_IDX_DEFINE - Config mode(s) 1: - Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE
WiFi-CoEx Status Code	1 Octet	Status code for WLAN during ranging RR 0x00 : WLAN_UWB_IND_LOW 0x01 : WLAN_UWB_IND_HIGH 0x02 : WLAN_UWB_IND_ERR 0xFF - 0x03 : RFU
Authenticity Info Presence	1 Octet	Indicator for presence of Authentication Tag. 0x00 : Authenticity info is not present 0x01 : Authenticity info is present This field is 0x00 if AUTHENTICITY_TAG = 0x00 (disable). Authenticity Tag is applicable when STS_CONFIG is set to 0x01 or 0x02 otherwise this field is always 0x00

Table 49. Vendor Specific extensions in SESSION_INFO_NTF Continued for UL-TDoA...continued

Parameter Name	Length	Value/ Description
Authentication Tag	16 Octets	Generated Authentication Tag for RANGE_DATA_NTF and this field is present when Authenticity Info Presence is set to 0x01

Table 50. Vendor Specific extensions in SESSION_INFO_NTF for TWR Measurements and OWR Advertisement

Parameter Name	Length	Value/ Description
Ranging Measurements vendor extension	M*ranging meas. Size	Refer Vendor specific extension for Two-way measurement at Vendor specific extension for Two Way Ranging Measurement Table if Ranging Measurement Type = 0x01. Refer Vendor specific extension for One-way Ranging Advertisement at Vendor specific extension for One Way Ranging (AoA) Advertisement Table if Ranging Measurement Type = 0x03.

Table 51. Vendor specific extension for Two Way Ranging Measurement

Parameter Name	Length	Value/ Description
NXP Specific Data type	1 Octet	This parameter indicates the type of vendor specific data 0x00 : Specific Data V1 for TWR .. 0xA0 : FoV Specific Data 0xA2 to 0xFF : reserved for UWB chipset vendor specific usage
NXP Specific Data (FIXED PART)	1 Octet	WiFi CoEx Status WiFi co-existence status Wlan State as defined in WLAN_UWB_IND_STATUS_NTF 0x00 = WLAN_UWB_IND_LOW 0x01 = WLAN_UWB_IND_HIGH_AT_RR_START 0x02 = WLAN_UWB_IND_HIGH_DURING_RR Others = RFU Note: If pin state from Wlan to UWB is low will send WLAN_UWB_IND_LOW(0x00).
...	2 + M Octets	Rx Antenna Info for AoA Measurements Antenna used for RX Octet[0] : RX mode - ToA mode = 0 (following ids are RX antenna IDs) - AoA mode = 1 (following ids are RX pair IDs)

Table 51. Vendor specific extension for Two Way Ranging Measurement...continued

Parameter Name	Length	Value/ Description	
			-Radar mode = 2 -TOA RFM Mode = 3 - AoA RFM mode = 4 (following ids are RX pair IDs) - Octet[1] : Number of following RX antennas (M >=1) - Octet[2] : RX antenna ID (or RX pair ID) - Octet[3] : RX antenna ID (or RX pair ID) ... Following values (AoA, PDoA, PDoA Index Etc..) shall be matched according to the order of RX antenna Info (values are filled with 0 if RX mode is set to ToA mode).
...	2 + N Octets	Rx Antenna Info For Debug Notifications	Antenna used for RX operations: Octet[0] : RX mode - ToA mode = 0 (following ids are RX antenna IDs) - Octet[1] : Number of following RX antennas (N >=1) - Octet[2] : RX antenna ID - Octet[3] : RX antenna ID ... Following values (Snr First, SnrMain, Etc..) shall be matched according to the order of RX antenna Ids
Repetition part (This fields are repeated for each responder)	M * 7 Octets(If Vendor Specific Data Type is 0xA0)	AoA / PDoA/FoV measurements per RX entry	Octet[1-0] : Angle of arrival (2 Octets signed value in Q9.7 format) Octet[3-2] : Phase difference of arrival (2 Octets signed value in Q9.7 format) Octet[5-4] : Phase difference of arrival index in the whole CIR (2 Octets unsigned integer value)

Table 51. Vendor specific extension for Two Way Ranging Measurement...continued

Parameter Name	Length	Value/ Description	
			Octet[6] : Field of View Flag (Value 0 indicates that Peer Device is not present in FoV and Value 1 indicates that Peer Device is present in FoV). Note: If DUAL RX is configured, AoA FoV Elevation Flag shall be ignored by host. (This is present only if Vendor Specific Data Type is 0xA0) Note: This field will be reported when ANTENNA_CONFIG_RX is AoA mode or RFM mode for AoA, in all other cases of ANTENNA_CONFIG_RX this field will not be reported
...	N * 6 Octets	SNRFirst / SNRMain / FirstIndex / Main Index measurements per RX entry	Octet[0] : Signal-to-Noise (SNR) of the First Path in dB (1 Octet unsigned integer value) compared to noise floor in the same RX entry Octet[1] : Signal-to-Noise (SNR) of the Main Path in dB (1 Octet unsigned integer value) compared to noise floor in the same RX entry Octet[3-2] : First path index in the whole CIR Octet[5-4] : Main path index in the whole CIR
...	2 Octets	Distance_2	Distance between the device and target measured with the antenna enabled on RFM (2 Octets unsigned integer value) Note1: This field is reported only in AOA_RFM and TOA_RFM modes Note2: In Initiator device role, UWBS reports the value 0x0000 as this measurement is not applicable to this role

Table 52. Vendor specific extension for One Way Ranging (AoA) Advertisement

Parameter Name	Length	Value/ Description	
NXP SPECIFIC DATA TYPE	1 Octet	This parameter indicates the type of vendor specific data 0x00 : Specific Data V1 for OWR .. 0xA0 to 0xFF : reserved for UWB chipset vendor specific usage	
VENDOR_ SPECIFIC_DATA	2 + N Octets	Rx Antenna Info for AoA Measurements	Antenna used for RX Octet[0] : RX mode • ToA mode = 0 (following ids are RX antenna IDs) • AoA mode = 1 (following ids are RX pair IDs) • Radar mode = 2 • TOA RFM Mode = 3 • AoA RFM mode = 4 (following ids are RX pair IDs) Octet[1] : Number of following RX antennas (N >=1) Octet[2] : RX antenna ID (or RX pair ID) Octet[3] : RX antenna ID (or RX pair ID) Following values (AoA, PDoA, PDoA Index Etc..) shall be matched according to the order of RX antenna Info (values are filed with 0 if RX mode is set to ToA mode).
...	N * 6 Octets	AoA / PDoA measurements per RX entry	Octet[1-0] : Angle of arrival (2 Octets signed value in Q9.7 format) Octet[3-2] : Phase difference of arrival (2 Octets signed value in Q9.7 format) Octet[5-4] : Phase difference of arrival index in the whole CIR (2 Octets unsigned integer value) Note: This field will be reported when ANTENNA_CONFIG_RX is AoA mode or RFM mode for AoA, in all other cases of ANTENNA_CONFIG_RX this field will not be reported
...	2 Octets	RSSI	RSSI in dB (signed value expressed in Q8.8 format)

Table 53. One Way Ranging (UL-TDOA) Measurement(For Legacy Devices)

Payload Field(s)	Length	Value/ Description
One Way (UL-TDOA) Ranging Measurement		
MAC Address	2/8 Octets	MAC address of the initiator TAG device. 2 bytes short address (if MAC addressing mode indicator = 0x00) 8 bytes extended address (if MAC addressing mode indicator = 0x01)
Status	1 Octet	For various status values refer to Status Codes

Table 53. One Way Ranging (UL-TDOA) Measurement(For Legacy Devices)...continued

Payload Field(s)	Length	Value/ Description
		If status field is other than STATUS_OK, then all other values shall be ignored.
Message Control	1 Octet	b0-1: Presence and length of UL-TDoA Device ID. 0b00 = UL-TDoA Device ID is not present 0b01 = 16-bit UL-TDoA Device ID 0b10 = 32-bit UL-TDoA Device ID 0b11 = 64-bit UL-TDoA Device ID b2-3: Presence and length of TX timestamp of the received Blink or Synchronization UTM. 0b00 = TX timestamp is not present 0b01 = 40-bit TX timestamp 0b10 = 64-bit TX timestamp 0b11 = RFU b4-5: Length of the measured RX timestamp by the UWBS that receives the UTM. 0b00 = 40-bit RX timestamp 0b01 = 64-bit RX timestamp 0b10-0b11 = RFU b6-7: RFU
Frame Type	1 Octet	0x00 : BLINK message to indicate that data notification received from Tag 0x01 : SYNC message to indicate that data notification received from Master Anchor
NLoS	1 Octet	Indicates if the ranging measurement was in Line of sight or non-line of sight. 0x00 : LoS 0x01 : NLoS 0x02 - 0xEF : RFU 0xFF : Unable to determine
AoA Azimuth	2 Octets	AoA Azimuth in degrees and it is a signed value in Q9.7 format. This field is zero if AOA_RESULT_REQ = 0.
AoA Azimuth FOM	1 Octet	The AoA Azimuth FOM field conveys the reliability of the estimated AOA in the Azimuth. FOM goes from 0 to 100. Higher values indicate better quality AoA estimates. Lower values suggest decreased reliability in the estimates.
AoA Elevation	2 Octets	AoA Elevation in degrees and it is a signed value in Q9.7 format. This field is zero if AOA_RESULT_REQ = 0.
AoA Elevation FOM	1 Octet	The AoA Elevation FOM field conveys the reliability of the estimated AOA Elevation. FOM goes from 0 to 100. Higher values indicate better quality AoA estimates. Lower values suggest decreased reliability in the estimates.
Timestamp	8 Octets	Timestamp of UWB RFRAME in 15.65ps ticks

Table 53. One Way Ranging (UL-TDOA) Measurement(For Legacy Devices)...continued

Payload Field(s)	Length	Value/ Description
Blink Frame Number	4 Octets	Number received in UWB blink frame from Tag or SYNC frame from Master Anchor
RFU	12/6 Octets	Reserved for future use 12 bytes RFU if MAC Address is 2 Octets 6 bytes RFU if MAC Address is 8 Octets
Device Specific Information Size	1 Octet	Size of Device Specific Information(N) to follow
Device Specific Information	N Octets	Device specific data. This field is present if Device Specific Information Size >= 1
Blink Payload Data Size	1 Octet	Size of Blink Payload Data(M) to follow.
Blink Payload Data	M Octets	Blink payload data as part of Blink Frame over UWB. This field is present if Blink Payload Data Size >= 1

7.7 Selection of Antennas for a given Session

The process to use arbitrary antenna for a given session is as follows.

Step 1: Define Individual Antenna (Core Set Config). See [ANTENNA_TX_IDX_DEFINE](#) and [ANTENNA_RX_IDX_DEFINE](#)

Step 2: Define RX Antennas Group (Core Set Config). See [ANTENNAS_RX_PAIR_DEFINE](#)

Step 3: Store Calibration information for RX Groups

This is required per antennas pair (H and V), per channel (5 and 9)

Step 4: For a given session, select the Antennas to be used.

This is optional. TX Antenna ID 1 is default for Tx, RX Pair ID 1 is for RX-H, RX Pair ID 2 is for RX-V.

See [ANTENNAS_CONFIGURATION_TX](#) to define TX Antenna(s).

See [ANTENNAS_CONFIGURATION_RX](#). Select RX Antenna, RX-H Pair and RX-V Pair to be used. The RX-H and RX-V pair IDs are as defined by [ANTENNAS_RX_PAIR_DEFINE](#).

Table 54. ANTENNA_SELECTION_GPIO_BITMASK

GPIO Bit Masks for Antennas Selection

Pin	Mask	b[7]	b[6]	b[5]	b[4]	b[3]	b[2]	b[1]	b[0]
EF1	1								X
EF2	2							X	
GPIO0	4						X		
GPIO1	8					X			
GPIO2	16				X				
GPIO3	32			X					
GPIO4	64		X						

Table 54. ANTENNA_SELECTION_GPIO_BITMASK ...continued

GPIO Bit Masks for Antennas Selection

Pin	Mask	b[7]	b[6]	b[5]	b[4]	b[3]	b[2]	b[1]	b[0]
RFU	-	X							

Note: Bit(s) from b[15] to b[8] are reserved for future use.

Table 55. Max number of entries per Product Variant

Variant	# TX Antennas	# RX Antennas (GPIO)	# Antenna Pairs / Single (Functional)	# Max Calibraton entries for AoA per Antennas Pair per Channel	# Max Supported Channels for AoA Calibration at a time
SR250	4	12	8	4	3

- All the IDs that are used and defined must to be sequential.
 - Partial update/set configurations are not permitted. UWBS Does not implement parsing/traversal logic and hence
- The FW internally can use this Indexes to "optimize" access to the look up table without traversing to any array. Or find any empty or filled slots.

If the HOST calls one the APIs with the Same ID, the FW may choose to over-write the values and the behavior may be non deterministic for advanced use cases. Hence the Host has to ensure all the slots are required filled before exercising any use case.

The FW fills up slots based on the indexes given as the ID parameters of these set configs.

In some cases IDs 1 and 2 have special purpose. See the description in [ANTENNAS_RX_PAIR_DEFINE](#).

Maximum supported Channel for calibration at a time is variant specific (See [Table 55](#)).

7.8 Antenna Switching During Active Ranging

UWBS shall use SESSION_SET_VENDOR_APP_CONFIG(["ANTENNAS_CONFIGURATION_RX"](#)) command to perform the dynamic switch based on session specific antenna pair to be used based on [ANTENNAS_CONFIGURATION_RX](#) application configuration. [ANTENNAS_CONFIGURATION_RX](#) application configuration is configured based on [ANTENNAS_RX_PAIR_DEFINE](#) and [ANTENNA_RX_IDX_DEFINE](#) core set configuration parameters.

Application shall use this command to switch between configuration modes such as Dual Rx Pair, Dual Rx and Single Rx.

AoA Result Req during Active Ranging

If the antenna configuration specified via [ANTENNAS_CONFIGURATION_RX](#), [ANTENNAS_RX_PAIR_DEFINE](#) and [ANTENNA_RX_IDX_DEFINE](#) is SINGLE RX and application sets AOA Res Req to enabled, than FW will ignore this setting and implicitly treat AoA Res Req to be disabled

7.9 UWB Session Vendor App Configurations

The Host may retrieve current proprietary application configurations for a given UWB session at any session states.

The Host may establish new UWB sessions and configure UWB sessions during the DEVICE_STATE_READY/DEVICE_STATE_ACTIVE Device State of the UWBS.

7.9.1 Setting Vendor App Configurations

The Host shall use the SET_VENDOR_APP_CONFIG_CMD command to set APP Configuration Parameters for the requested UWB session. The UWBS shall respond with the SET_VENDOR_APP_CONFIG_RSP with Status indicating that APP Configuration Parameters are applied or not.

Table 56. SET_VENDOR_APP_CONFIG_CMD

Payload Field(s)	Length	Value/Description		
Session Handle	4 Octets	Unique session identifier of a session for which the Vendor APP Configuration Parameters are to be set		
Number of Vendor App Configurations	1 Octet	The number (n) of APP Configuration Parameters (TLV) to follow. Each APP Configuration Parameter follows TLV structure as below.		
Vendor APP Configuration Parameter [1...n]	(2+m)*n Octets	Tag	1 Octet	Vendor APP configuration Parameter ID
		Length	1 Octet	Length of Value(m)
		Value	m Octets	Value of the Vendor APP Configuration parameter

Table 57. SET_VENDOR_APP_CONFIG_RSP

Payload Field(s)	Length	Value/Description		
Status	1 Octet	Status code		
Number of Vendor APP Configurations	1 Octet	If the Status is STATUS_OK, the value shall be 0x00 and no Vendor APP Configuration Parameters are listed. If the Status is something other than STATUS_OK, the value shall indicate the number (n) of failed Vendor APP Configuration Parameters to follow.		
Failed Vendor APP Configuration Parameter [1...n]	2*n Octets	Vendor APP Config ID	1 Octet	The identifier of the failed Application Configuration Parameter.
		Status	1 Octet	STATUS_INVALID_PARAM / STATUS_INVALID_RANGE/ STATUS_ERROR_SESSION_ACTIVE

7.9.2 Retrieve Vendor App Configurations

The Host shall use the GET_VENDOR_APP_CONFIG_CMD command to read APP Configuration Parameters for the requested UWB session. The UWBS shall respond with the GET_NXP_APP_CONFIG_RSP with Status indicating that APP Configuration Parameters are applied or not.

Table 58. GET_VENDOR_APP_CONFIG_CMD

Payload Field(s)	Length	Value/Description	
Session Handle	4 Octets	Unique session identifier of a session for which the Vendor APP Configuration Parameters are to be set	
Number of Vendor App Configurations	1 Octet	The number (n) of APP Configuration Parameters (TLV) to follow. Each vendor APP Configuration Parameter follows TLV structure as below.	
Vendor App Configuration Parameter [1...n]	n Octets	Vendor APP config IDs	This field is required if Number of APP Configurations field is > 0.

Table 59. GET_VENDOR_APP_CONFIG_RSP

Payload Field(s)	Length	Value/Description		
Status	1 Octet	Status code		
Number of Vendor App Configurations	1 Octet	If the Status is STATUS_OK, the value shall be 0x00 and no Vendor APP Configuration Parameters are listed. If the Status is something other than STATUS_OK, the value shall indicate the number (n) of failed vendor APP Configuration Parameters to follow.		
Vendor App Configuration Parameter [0...n]	(2+m)*n Octets	Type	1 Octet	Vendor APP Configuration Parameter Tag ID.
		Length	1 Octet	The length (m) of Value.
		Value	m Octets	The value of the Vendor APP Configuration Parameter.

7.10 Radar Session

AP shall create the data session using SESSION_INIT_CMD with Session Type as "Radar".

7.10.1 Radar Session Type

Table 60. SESSION_INIT_CMD : Session Type extension

Payload Field(s)	Length	Value/Description
SESSION_INIT_CMD		
Session ID	4 Octets	Session ID is 4 Octets unique random number generated by application Note: Session ID = 0x00000000: Reserved for Device Test Mode session
Session Type	1 Octet	Type of session 0xF0: RADAR

7.10.2 Configuration of a Radar Session

Parameters marked with (*) can be set during an active session.

AP shall set mandatory APP Configuration Parameters defined in [Table 61](#) for a Radar Session and shall wait for SESSION_STATUS_NTF notification with Session State of SESSION_STATE_IDLE to make sure that session is ready for Radar.

Table 61. Mandatory APP Configurations for a Radar Session

Parameter Name
RADAR_MODE
ANTENNAS_CONFIGURATION_RX (<i>must not be updated during radar session</i>)

Below application configurations are relevant for a Radar Session. Intended behavior may not be achieved if below APP Configuration Parameters are not properly set. For a Radar Session, both [ANTENNAS_CONFIGURATION_RX](#) and ANTENNAS_CONFIGURATION_TX are applied at the same time.

Table 62. APP Configurations for a Radar Session

Parameter Name
CHANNEL_NUMBER
SLOT_DURATION
RANGING_INTERVAL
PREAMBLE_CODE_INDEX
SLOTS_PER_RR
TX_ADAPTIVE_PAYLOAD_POWER
PRF_MODE
SESSION_PRIORITY
HOPPING_MODE
MAX_NUMBER_OF_MEASUREMENTS
TX_POWER_DELTA_FCC
ANTENNAS_CONFIGURATION_TX (<i>must not be updated during radar session</i>)

Below new application configurations are added for a Radar Session.

Table 63. Radar Session Extensions

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Description
RADAR_MODE	1	0xA0	MODES: • 0x01: Medium distance, e.g. used for vital sign detection • 0x02: Close distance, e.g. used for hand gesture recognition • 0x03: Far distance, e.g. used for presence detection • 0x04: Close distance, e.g. used for single pin antenna support. • 0x05: Medium distance, e.g. used for static object support • 0x06: Far distance, e.g. used for static object support • 0x020: Test mode for antenna isolation (Default value: 0x01) NOTE: • MODE: 0x20 – The Radar session stops after sending 1 Radar Rx Notification with antenna isolation report. – Only Single RX Antenna is supported, without Antenna switching.
RADAR_RX_GAIN (*)	4	0xA4	• bits[31:24] – 0: Enable AGC – 1: Force gain index values – 2: Force AGC to be re-executed • bits[23:16]: Gain index for RXA (0-31) • bits[15:8] : Gain index for RXB (0-31) • bits[7:0] : Gain index for RXC (0-31) Default: 0x00000000 (AGC Enabled)
RADAR_SINGLE_FRAME_NTF (*)	1	0xA5	Defines the mode used to transmit CIR samples. 0x00: use a buffer to transmit the CIR samples if the buffer is full or session switch will be performed. 0x01: transmit every single CIR sample (Default: 0)
RADAR_CIR_NUM_SAMPLES	1	0xA7	Number of TAPs to be captured (Default = 128, MIN = 16, MAX = 128)
RADAR_CIR_START_OFFSET (*)	1 + 2 * N (N = 3)	0xA8	Defines the start index of the CIR in case of Radar modes using HPRF: Octet[0]: Length in Words (Default: 3) • Word[1] start index for RXC for antenna pair 0 • Word[2] start index for RXB for antenna pair 0 • Word[3] start index for RXA for antenna pair 0 (Default: 0) All other values except 3 for length are invalid. Start index of 0 triggers the automatic calculation of the start index in case of Radar Mode 1,2,3; Any other value sets the start index (Radar Mode 1,2,3 range is [1:727], Radar Mode 4,5,6 Range is [0-127]);
RADAR_RFRI (*)	7	0xA9	RFRI (Radar Frame Repetition Interval):

Table 63. Radar Session Extensions...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Description																		
			Octet[3:0] : Ranging Interval Duration between the start of two, consecutive Radar rounds in ms Octet[5:4] : Slot Duration in RSTU Duration between the start times of two consecutive Radar frames in RSTU. Octet[6] : Slots per RR Number of Radar frames within the Radar round NOTE: This parameter will have precedence over and RANGING_DURATION, SLOT_DURATION, SLOTS_PER_RR. If this parameter is set via SET_APP_CONFIG_CMD in INIT/IDLE/ACTIVE states, then UWBS shall override the RANGING_DURATION, SLOT_DURATION, SLOTS_PER_RR parameters.																		
RADAR_PRESENCE_DET_CFG*	10	0xAA	Octet[0]: Presence Detection Mode Bit 0: Presence detection enabled Bit 1: Distance and AoA reporting enabled Bit 2: report all targets (Default: 0: Disabled) Bit 4: Detection trigger value reporting Note: If Bit 4 is set, also Bit 1 has to be set. Octet[1]: Periodic data reporting Bit 0: sending raw CIRs to the host Bit 2-1: 00: no periodic presence reporting 01: presence reporting every 50ms 10: presence reporting every 400ms 11: presence reporting every 1600ms (Default: 0:Disabled) Octet[2] - Presence Detection Sensitivity in Q4.4 Format (4 bits for Fractional, 4 bits for integer) (Default: 3.75, Min: 1, Max 7) Octet[3]: Set to enable GPIO to notify presence Bit 3-0: GPIO line to be used, 0 disabled Table 64. <table><tr><td>2</td><td>GPIO2</td></tr><tr><td>4</td><td>GPIO4</td></tr><tr><td>5</td><td>GPIO5</td></tr><tr><td>6</td><td>GPIO6</td></tr><tr><td>7</td><td>GPIO7</td></tr><tr><td>8</td><td>GPIO8</td></tr><tr><td>9</td><td>GPIO9</td></tr><tr><td>10</td><td>GPIO10</td></tr><tr><td>11</td><td>GPIO11</td></tr></table>	2	GPIO2	4	GPIO4	5	GPIO5	6	GPIO6	7	GPIO7	8	GPIO8	9	GPIO9	10	GPIO10	11	GPIO11
2	GPIO2																				
4	GPIO4																				
5	GPIO5																				
6	GPIO6																				
7	GPIO7																				
8	GPIO8																				
9	GPIO9																				
10	GPIO10																				
11	GPIO11																				

Table 63. Radar Session Extensions...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Description								
			Table 64. ...continued <table><tr><td>12</td><td>GPIO12</td></tr><tr><td>13</td><td>GPIO13</td></tr><tr><td>14</td><td>GPIO14</td></tr><tr><td>15</td><td>GPIO15</td></tr></table> Note: Host should not configure the same GPIO for Presence Detection as used for Antenna Defines.(GPIO2 and GPIO4 are used for Antenna Defines as well) Note : - GPIO pins should not be used, if already in use for the below - SPI - GPIOs 6,7,8 and 10 - SE_I2C - GPIOs 13 and 14 - External Clock (Slow) - GPIO12 - External Clock (RF) - GPIO9 (Enabled for Mobile Platforms) Bit 6-4: RFU Bit 7: 1 to disable RADAR_RX_NTF(Presence Detection) (Default:0) Note: Both GPIO and PD notification should not be disabled at one time. FW will return invalid range error. Octet[4-5]: Presence Detection Distance Min in cm (Default: 50, Min: 50, Max - 800) Octet[6-7]: Presence Detection Distance Max in cm (Default: 800, Min: 50, Max - 800) Octet[8-9]: Presence Detection Hold Delay (Default: 1600, Min: 400, Max:5000) delay until presence is reported as lost in ms Octet[10]: minimum angle, (Default: -90, range -90, +90, int8_t format), default means that no minimum angle is used Octet[11]: maximum angle, (Default: +90, range -90, +90, int8_t format), needs to be greater than minimum angle, default means that no maximum angle is used if Octets [10-11] are set to not default, a second receiver must be configured, else session start has to report a fail (*) only following parameters are allowed to be changed during an active session: Octet[0]: Bit 1, Octet[1]: Bit 2-1, Octet[2], Octet[4-11] Changing Octet[0]: Bit 0 will be rejected by FW. Changing other bits which are not allowed to change might result in behavioral change of OCPD algo. This UCI parameter shall check the below following if Presence Detection Mode is Enabled RADAR_MODE with value set as 1	12	GPIO12	13	GPIO13	14	GPIO14	15	GPIO15
12	GPIO12										
13	GPIO13										
14	GPIO14										
15	GPIO15										

Table 63. Radar Session Extensions...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Description																
			RADAR_CIR_NUM_SAMPLES with value must be greater than (MaxDistance in cm/15cm) Note: • OCPD only supports 50ms RFR! • Drift Compensation should always be enabled.																
RADAR_PERFORMANCE	1	0xAD	0b000001: DC Freeze Enabled 0b000010: Bandwidth increase (Default: 0x03)																
RADAR_DRIFT_COMPENSATION	2	0xB2	bit[15:0] : Alpha in Q1.15 format (Default: 0x01) Note: Value 1 denotes that Drift compensation is enabled																
RADAR_CFG	1+N*17	0xB3	1 byte - Number of Radar Mode Configs(N) (0-2) RADAR_SESSION_CFG[0]: Octet[0] - RADAR_MODE Octet[1] - PREAMBLE_CODE_INDEX Octet[2] - TX_ANT_ID Octet[3-4]: Start Index for RXC Octet[5-6]: Start Index for RXB Octet[7-8]: Start Index for RXA Octet[9]: ANT_CFG for RXC Octet[10]: ANT_CFG for RXB Octet[11]: ANT_CFG for RXA Octet[12]: Gain index for RXC Octet[13]: Gain index for RXB Octet[14]: Gain Index for RXA Octet[15]: • 0: Enable AGC • 1: Force gain index values • 2: Force AGC to be re-executed Default: 0x0 Octet[16]: RFU, has to be set to 0																
RADAR_LPRF_CALIB	2	0xB4	Octet[0]: Residue threshold, using the following LUT: Table 65. Residue threshold, using the following LUT <table><tr><th>Octect Value</th><th>Threshold</th></tr><tr><td>0</td><td>.04%</td></tr><tr><td>1</td><td>.07%</td></tr><tr><td>2</td><td>.10%</td></tr><tr><td>3</td><td>.14%</td></tr><tr><td>4</td><td>.19%</td></tr><tr><td>5</td><td>.29%</td></tr><tr><td>6</td><td>.39%</td></tr></table>	Octect Value	Threshold	0	.04%	1	.07%	2	.10%	3	.14%	4	.19%	5	.29%	6	.39%
Octect Value	Threshold																		
0	.04%																		
1	.07%																		
2	.10%																		
3	.14%																		
4	.19%																		
5	.29%																		
6	.39%																		

Table 63. Radar Session Extensions...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Description																		
			Table 65. Residue threshold, using the following LUT...continued <table><tr><td>7</td><td>.58%</td></tr><tr><td>8</td><td>.78%</td></tr><tr><td>9</td><td>1.17%</td></tr><tr><td>10</td><td>1.56% (default)</td></tr><tr><td>11</td><td>2.34%</td></tr><tr><td>12</td><td>3.12%</td></tr><tr><td>13</td><td>4.68%</td></tr><tr><td>14</td><td>6.25%</td></tr><tr><td>15</td><td>9.37%</td></tr></table> Octet[1]: Residue window length in number of samples. Valid range: 1-32 Default value: 20 Note: <i>This parameter will trigger a LPRF configuration on chip, based on the set parameters. Radar mode 5 or 6 has to be set for this mode. When the calibration is done, the resulting values will be reported and the Radar session will be stopped.</i> This parameter is optional, by default it should not be present.	7	.58%	8	.78%	9	1.17%	10	1.56% (default)	11	2.34%	12	3.12%	13	4.68%	14	6.25%	15	9.37%
7	.58%																				
8	.78%																				
9	1.17%																				
10	1.56% (default)																				
11	2.34%																				
12	3.12%																				
13	4.68%																				
14	6.25%																				
15	9.37%																				

7.10.3 Radar Notification

Table 66. RADAR_RX_NTF

Payload Field(s)	Length	Value/ Description
RADAR_RX_NTF		
Session Handle	4 Octets	Session Handle for which Data is received
Status code	1 Octet	Status code as per Table 128 (generic specification) RADAR data type/field contain data if status is STATUS_OK
RADAR data type	1 Octet	Type of RADAR data: 0x00: CIR samples 0x01: Presence Detection 0x20: Antena Isolation Report 0x21: LPRF Calibration Report
RADAR data field	N Octets	CIR samples received are transmitted as CIR payload Header: - Byte[1:0] : Number of CIR samples following (LSB first); - Byte[2] : Length of 1 CIR sample in number of TAPs - Byte[3] : RFU Payload

Table 66. RADAR_RX_NTF...continued

Payload Field(s)	Length	Value/ Description
		If Radar Data Type is 0: (each TAP are coded on uint32bit_LSB) • TAP[0]: Counter of CIRs per receiver (starting from 2 if AGC has been performed, else starting from 1); • TAP[1]: Timestamp when CIR has been received in snapshot timer ticks; • TAP[2]: – bits[31:28]: RFU – bits[27:26]: TAP resolution in bits – 0b00: 16 bit complex (default) – 0b01: 24 bit complex – 0b10: 32 bit complex – bits[25:24]: TAP resolution in cm – 0b00: 15 cm (default) – 0b01: 7.5 cm – bits[23:16]: Tx pulse shape ID – bits[15:8]: Tx gain index – bits[7:0]: Rx gain index • TAP[3]: Timestamp when RADAR frame has been send in snapshot timer ticks; • TAP[4]: – bits[31:24]: Tx Antenna index – bits[23:16]: Rx Antenna index – bits[15:14]: RFU – bits[13]: DC Freeze (Disabled:0, Enabled: 1) – bits[12]: Single Pin (Disabled:0, Enabled: 1) – bits[11-8]: Radar Mode – bits[7:0] Rx path used (RXC: 1, RXB: 2, RXA: 3) • TAP[5] (For Radar Mode 1,2,3): – bits[7:0]: pulse difference distance between pulses (8ns) – bits[15:8]: bit shift during symbol correlation – bits[21:16]: log2(delta_L) – bits[31:22]: NSync / 8 • TAP[5] (For Radar Mode 4,5,6): – RFU • TAP[6] (For Radar Mode 1,2,3):: – bits[15:0]: peak value of real part in the CIR – bits[31:16]: number of pulse used • TAP[6] (For Radar Mode 4,5,6): – RFU • TAP[7]: – bits[15:0]: – 0: Start offset of the CIR's first TAP had been set by the host; – 1-32767: Q0.15 value of the fractional CIR Offset which is computed via Zero Distance Calibration;

Table 66. RADAR_RX_NTF...continued

Payload Field(s)	Length	Value/ Description
		- bits[31:16]: Start offset of the CIR's first TAP within the complete CIR (TAP index 1 to 727 for radar mode 1/2/3; TAP index 1 to 127 for radar mode 4/5/6) - TAP[N:8]: CIR samples
		If Radar Data Type is 0x01: (each TAP are coded on uint32bit_LSB) - TAP[0]: - Bit 0: presence detected - Bit 7-1: RFU - Bit 15-8: Presence detection Mode - Bit 23-16: Number of detections N (max. 5) - Bit 31-24: RFU - NOTE: Minimum number of targets reported will be 1 [- TAP[1+2*n]: - Bit 15—0 : Distance of detection in cm, else set to 0xFFFF of Nth target - Bit 23—16 : angle of detection as signed degree (int8_t format) if a second receiver is configured and Presence detection mode bit 1 is set and angle is detected, else set to 0x7F - Bit 31-24 : RFU - TAP[2+2*n]: - If Detection trigger value (Bit 4 of Presence Detection Mode) is enabled, send - SNR. SNR will be reported as a 32 bit floating value calculated using magnitude and noise $SNR = \text{magnitude/noise}$ $n = \text{target Index } 0:(N-1)$] repeat by N (number of detections)
	4 Octets	For RADAR data type 0x20: 1 Octet Tx antenna 1 Octet Rx Antenna 2 Octets Antenna Isolation in dB Note: The Radar session stops after sending Radar Rx NTF with antenna isolation report. The Radar NTF will have only this field when Radar mode is set to 0x20 (Antenna isolation)
RADAR data field	3 + N*3 Octets	For RADAR data type 0x21: Octet[0]: Tx antenna IDs as defined by ANTENNA_TX_IDX_DEFINE Octet[1]: channel as defined by CHANNEL_NUMBER Octet[2]: Number of receivers N. Octet[3]: Rx antenna ID as per ANTENNA_RX_DEFINE Octet[4]: delay calibration for Rx antenna defined in octet[3].

Table 66. RADAR_RX_NTF...continued

Payload Field(s)	Length	Value/ Description
		Octet[5]: width calibration calibration for Rx antenna defined in octet[3]. Octet[3+3*n]: nth Rx antenna ID as per ANTENNA_RX_DEFINE Octet[4+3*n]: delay calibration for Rx antenna defined in octet[3+3*n]. Octet[5+3*n]: width calibration calibration for Rx antenna defined in octet[3+3*n]. All delays and widths are expressed in nanoseconds, in unsigned Q6.2 format. (Range : 0..63.75ns, resolution of .25ns) Note: <i>The Radar session stops after sending Radar Rx NTF with LPRF calibration report.</i> <i>The Radar NTF will have only this field when Radar mode is set to 0x21 (LPRF calibration). In that case, on CIRs are reported.</i>

7.10.4 Radar Session specific default values and changed description for application configuration parameters

Some descriptions have been changed to reflect the RADAR behavior. Possible value range and default values for a RADAR session might be also different compared to a ranging or data transfer session.

Table 67. Radar session configuration parameters

Parameter Name (*): Can be applied during session active state as well	Length (Octets)	Tag (IDs)	Description
SLOT_DURATION	2	0x08	Unsigned integer that specifies duration of a RADAR frame in the unit of RSTU. This is also called Radar Frame Repetition Interval (RFRI). As the maximum value is 54ms, any higher RFRI values need to be defined in a combination of SLOT_DURATION and RANGING_INTERVAL Default = 12000 (10ms)
RANGING_INTERVAL	4	0x09	Ranging interval is time in the unit of 1200 RSTU (=1ms) between beginning of one RADAR frame interval to the beginning of the next. Minimum Ranging interval should be at least the duration of a SLOT_DURATION. RANGING_INTERVAL should be a multiple of SLOT_DURATION* SLOTS_PER_RR. Default: 40 (40ms)
PREAMBLE_CODE_INDEX	1	0x14	Ci Code index: [25:32] : HPRF, used for RADAR_MODE 1, 2 and 3 [95:102] : LPRF used for RADAR_MODE 5 and 6 Default: 26

Table 67. Radar session configuration parameters...continued

Parameter Name (*): Can be applied during session active state as well	Length (Octets)	Tag (IDs)	Description
SLOTS_PER_RR	1	0x1B	Number of RADAR frames within one ranging round. This config is used to specify the RADAR Duration in multiple of SLOT_DURATION (Default:4)
PRF_MODE	1	0x1F	This parameter is used to configure the mean PRF. 0x00 : 62.4 MHz PRF (BPRF mode) 0x01 : 124.8 MHz PRF (HPRF mode) 0x02 : 249.6 MHz PRF (HPRF mode with data rate 27.2 and 31.2 Mbps) 0x03 : RADAR (default) [0x04 : 0xFF] : RFU <i>Note: Refer section 8.3.1 on usage this parameter</i>
SESSION_PRIORITY (*)	1	0x25	Priority value for a Session. [1:100]: Range allowed [255]: Maximum value for a static priority (Default: 50)
MAX_NUMBER_OF_MEASUREMENTS	2	0x32	Configuration parameter to set maximum number of RADAR blocks to be executed in a session. The session is stopped and moved to SESSION_STATE_IDLE Session State when configured ranging blocks are elapsed. [0x00]: Unlimited (Default) [0x01 - 0xFFFF]: Number of Ranging Blocks

7.11 CCC Range Data NTF

While the UWB session is ongoing, the UWBS shall send ranging result with the CCC_RANGE_DATA_NTF notification to the Host when the ranging round is complete.

Table 68. CCC_RANGE_DATA_NTF

Payload	Length	Value/Description
Session Handle	4 Octet	Session ID of the currently active session
Ranging status	1 Octet	For the CCC session, the lower nibble represents the ranging status of the controller and the higher nibble represents the ranging status of the Controllee. For other sessions, this contains only the status of the Controllee. 0x0: Success 0x1: Transaction overflow 0x2: Transaction expired

Table 68. CCC_RANGE_DATA_NTF...continued

Payload	Length	Value/Description
		0x3: Incorrect frame 0xD: Responder is in listen only mode (Not available for the controller) 0xF: Ranging Control Message lost (Not available for the controller)
STS index	4 Octet	STS index received in final data message
RR Index	2 Octet	The ranging round in which the next ranging cycle will be executed
Block Index	2 Octets	Index of the Ranging block for which the following measurement result applies
Distance	2 Octet	Distance between Initiator and Anchor in cm
Uncertainty Anchor (FoM)	1 Octet	Ranging timestamp uncertainty of Controllee
Uncertainty Initiator (FoM)	1 Octet	Ranging timestamp uncertainty of Controller
CCM TAG	8 Octets	CCM* TAG calculated over all payload fields. CCM* TAG can be set to all 0xFF if not used.
AOA Azimuth	2 Octets	AoA Azimuth in degrees measured at the device that is sending this SESSION_INFO_NTF notification. It is a signed value in Q9.7 format. Allowed values range from -180° to +180°.
AOA Azimuth FOM	1 Octet	The AoA Elevation FOM field conveys the reliability of the estimated AOA Elevation. FOM goes from 0 to 100. Higher values indicate better quality AoA estimates. Lower values suggest decreased reliability in the estimates. This field shall be set to zero and ignored if AOA_RESULT_REQ = 0
AOA Elevation	2 Octets	AoA Elevation in degrees measured at the device that is sending this SESSION_INFO_NTF notification. It is a signed value in Q9.7 format. Allowed values range from -90 to +90.
AOA Elevation FOM	1 Octet	The AoA Elevation FOM field conveys the reliability of the estimated AOA Elevation. FOM goes from 0 to 100. Higher values indicate better quality AoA estimates. Lower values suggest decreased reliability in the estimates. This field shall be set to zero and ignored if AOA_RESULT_REQ = 0

Table 68. CCC_RANGE_DATA_NTF...continued

Payload	Length	Value/Description
Antenna Pair Info	4 Octets	Antenna Rx Configuration information used in current ranging round. • Octet[0] : Configuration Mode as per ANTENNAS_CONFIGURATION_RX supported configuration modes are 0, 1, 5 and 6 – 0 : Configuration Mode 0: ToF Only Mode – 1 : Configuration Mode 1: Configuration Mode for 3D/2D AoA, test/loopback session and implicit ToA mode usecase – 5 : Configuration Mode 5: Configuration Mode for CSA ToA mode use case – 6 : Configuration Mode 6: Configuration Mode for CSA AoA mode use case • Octet[1] : – Config mode(s) 0: Antenna ID 1 as per ANTENNA_RX_IDX_DEFINE – Config mode(s) 1: Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE – Config mode(s) 5: Antenna ID as per ANTENNA_RX_IDX_DEFINE mapped to current active RR – Config mode(s) 6: Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE mapped to current active RR • Octet[2] : – Config mode(s) 0 : Antenna ID 2 as per ANTENNA_RX_IDX_DEFINE – Config mode(s) 1: Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE – Config mode(s) 5: Antenna ID as per ANTENNA_RX_IDX_DEFINE mapped to current active RR – Config mode(s) 6: Antenna Pair ID as per ANTENNAS_RX_PAIR_DEFINE mapped to current active RR • Octet[3] : – RFU
Number of PDoA Measurements	1 Octet	Number of PDoA measurements to follow (M).
PDoA Measurements	M x 4 Octets	Where M is the number of PDoA measurements to follow and Each measurement includes the PDoA estimate (2 octets) and the

Table 68. CCC_RANGE_DATA_NTF...continued

Payload	Length	Value/Description
		corresponding PDoA index (2 Octets) at which the estimate has been taken • PDoA – 2 Octets – Estimation of phase difference in degrees from antenna pair . – It is a signed degree in Q9.7 format. – This field is zero if AOA_RESULT_REQ = 0. – Value ranges from -180 to 180 degrees. • PDoA Index – 2 Octets – CIR index estimate at which PDoA has been detected. – This is vendor specific. – This field is zero if AOA_RESULT_REQ = 0. This field shall not be present when Number of PDoA Measurements field is 0
Number of RSSI measurements	1 Octet	'Number of RSSI measurements' to follow = N. (Each measurement = RSSI_RX1, RSSI_RX2). Where N = 2 (Measured at Poll and Final).
RSSI measurements	N x 4 Octets	Where N is the number of RSSI measurements. Each measurement is reported in the sequence according to antenna pair in Antenna Pair Info field. Each measurement includes the RSSI from RX1 and RSSI from RX2. RSSI_RX1 [1...N] -2 Octets Received Signal Strength Indicator for RX1 in dB. It is signed value expressed in Q8.8 format RSSI_RX2 [1...N] -2 Octets Received Signal Strength Indicator for RX2 in dB. It is signed value expressed in Q8.8 format For example: With SWAP_ANTENNA_PAIR_3D_AOA enabled RSSI-RX1 RSSI-RX2 for Pair 1 for POLL RSSI-RX1 RSSI-RX2 for Pair 2 for

Table 68. CCC_RANGE_DATA_NTF...continued

Payload	Length	Value/Description
		FINAL With SWAP_ANTENNA_PAIR_3D_AOA disabled For RX Pair- RSSI-RX1 RSSI-RX2 for Pair 1 for POLL RSSI-RX1 RSSI-RX2 for Pair 1 for FINAL For single RXRSSI- RX1 for RX1, RSSI-RX2 will be 0, for POLL and FINAL. RSSI-RX2 for RX2, RSSI-RX1 will be 0, for POLL and FINAL
Number of SNR measurements	1 Octect	Number of SNR measurements to follow = N x K K indicates the number of RX port pins. K = 2 (RX1 and RX2)
SNR measurements	N x K x 5 Octets	This field contains the SNR measurements. Each SNR measurement element contains 5 octets as below: [• Octet [0]: Slot index and Antenna mapping details – bits[5:0] - RFRAME slot index – bits[7:6] : – 0b00: RXC – 0b01: RXB – 0b10: RFU – 0b11: RXA • Octet [1]: SNR_MAIN_PATH Signal-to-Noise Ratio (SNR) of the main path and it is reported as unsigned value in dB. • Octet [2]: SNR_FIRST_PATH Signal-to-Noise Ratio (SNR) of the first path and it is reported as unsigned value in dB. • Octets [4-3]: SNR_TOTAL Signal-to-Noise Ratio (SNR) of configured RX in Q8.8 dB.] Repeat up to N x K x 5 For example: With SWAP_ANTENNA_PAIR_3D_AOA enabled: SNR-RX1 SNR-RX2 for Pair 1 for POLL SNR-RX1 SNR-RX2 for Pair 2 for FINAL

Table 68. CCC_RANGE_DATA_NTF ...continued

Payload	Length	Value/Description
		With SWAP_ANTENNA_PAIR_3D_AOA disabled: For RX Pair- SNR-RX1 SNR-RX2 for Pair 1 for POLL SNR-RX1 SNR-RX2 for Pair 1 for FINAL For single RX SNR-RX1 for RX1, SNR-RX2 will be 0, for POLL and FINAL. SNR-RX2 for RX2, SNR-RX1 will be 0, for POLL and FINAL. Note: Thisfields shall contain the SNR values measured for both“ Poll and Final FRAMES”

Note: CSA Controlee NTF will be as per existing proprietary CCC_RANGE_DATA_NTF.

8 UCI Test Specification Extensions

The UWBS shall not send any proprietary information in any response/notification if NXP_EXTENDED_NTF_CONFIG is set to 0x00.

8.1 TEST_PER_RX_NTF

Additional fields which are highlighted below are added TEST_PER_RX_NTF.

Table 69. TEST_PER_RX_NTF Extensions

Payload Field(s)	Length (Octets)	Value/ Description
Status	1	Notify host after receiving NUM_PACKETS. Refer Generic Status Codes
ATTEMPTS	4	No. of RX attempts
ACQ_DETECT	4	No. of times signal was detected
ACQ_REJECT	4	No. of times signal was rejected
RX_FAIL	4	No. of times RX did not go beyond ACQ stage
SYNC_CIR_READY	4	No. of times sync CIR ready event was received
SFD_FAIL	4	No. of time RX was stuck at either ACQ detect or sync CIR ready
SFD_FOUND	4	No. of times SFD was found
PHR_DEC_ERROR	4	No. of times PHR decode failed
PHR_BIT_ERROR	4	No. of times PHR bits in error
PSDU_DEC_ERROR	4	No. of times payload decode failed
PSDU_BIT_ERROR	4	No. of times payload bits in error
STS_FOUND	4	No. of times STS detection was successful
EOF	4	No. of times end of frame event was triggered
Vendor Specific Data Length	2	Vendor Data Length, Vendor Data field is not present if Vendor Data Length = 0. Otherwise = (N+1) This Vendor Data Length field is present if NXP_EXTENDED_NTF_CONFIG is set to 0x01 (Vendor extended notifications) and Vendor presence is based on Vendor Data Length
NXP Specific Data type	1	This parameter indicates the type of vendor specific data: 0x00 : Specific Data V1 for TEST_PER_RX_NTF
NXP Specific Data	N	Specific Data Refer : TEST_PER_RX_NTF Vendor Specific Data for more details

Table 70. TEST_PER_RX_NTF Vendor Specific Data

Parameter Name	Length	Value/ Description
RX antenna information	1 + 1 + N	RX mode and number of antenna used Octet[0] - 0x00 (TOA RX mode) Octet[1] - Number of RX antenna to follow - Octet[2] RX1 antenna ID - Octet[3] RX2 antenna ID ... - Octet[N] RXN antenna ID
RSSI_RX1	2 Octets	Average value of the computed RSSI value on RX1 path. It is signed value expressed in Q8.8 format
RSSI_RX2	2 Octets	Average value of the computed RSSI value on RX2 path. It is signed value expressed in Q8.8 format
RSSI_RX[N]	2 Octets	Average value of the computed RSSI value on RXN path. It is signed value expressed in Q8.8 format
SNR_RX1	2 Octets	Averaged value of the SNR computed on RX1 path. It is a signed value in Q 8.8 format
SNR_RX2	2 Octets	Averaged value of the SNR computed on RX2 path. It is a signed value in Q 8.8 format.
SNR_RX[N]	2 Octets	Averaged value of the SNR computed on RXN path. It is a signed value in Q 8.8 format.
RX_CFO_EST	2 Octets	Average value of the computed carrier frequency offset estimate from the reception in PPM. It is a signed value in Q 5.11 format

8.2 TEST_RX_NTF

Table 71. TEST_RX_NTF Extensions

Payload Field(s)	Length	Value/ Description
TEST_RX_NTF		
Status	1 Octet	Refer Generic Status Codes
RX_DONE_TS_INT	4 Octet	Integer part of timestamp 1/124.8Mhz ticks.
RX_DONE_TS_FRAC	2 Octet	Fractional part of timestamp in 1/128 * 499.2Mhz ticks.
AoA Azimuth	2 Octet	AoA Azimuth in degrees and it is a signed value in Q9.7 format. This field is zero if AOA_RESULT_REQ = 0.
AoA Elevation	2 Octet	AoA Elevation in degrees and it is a signed value in Q9.7 format. This field is zero if AOA_RESULT_REQ = 0.

Table 71. TEST_RX_NTF Extensions...continued

Payload Field(s)	Length	Value/ Description
ToA Gap	1 Octet	ToA of main path minus ToA of first path in nanoseconds.
PHR	2 Octet	Received PHR (bits 0-12 as per IEEE spec)
PSDU Data Length	2 Octet	Length of PSDU Data(N) to follow
PSDU_Data	N Octet	Received PSDU Data[0:N] bytes 0 ≤ N ≤ 127 for BPRF 0 ≤ N ≤ 4095 for HPRF
Vendor Specific Data Length	2 Octets	Vendor Data Length, Vendor Data field is not present if Vendor Data Length = 0. Otherwise = (N+1) This Vendor Data Length field is present if NXP_EXTENDED_NTF_CONFIG is set to 0x01 (Vendor extended notifications) and Vendor presence is based on Vendor Data Length
NXP Specific Data type	1 Octet	This parameter indicates the type of vendor specific data: 0x00 : Specific Data V1 for TEST_RX_NTF
NXP Specific Data	N Octet	Specific Data Refer : TEST_RX_NTF Vendor Specific Data for more details

Table 72. TEST_RX_NTF Vendor Specific Data

Parameter Name	Length	Value/ Description
RX antenna information	1 + 1 + N	RX mode and number of antenna used Octet[0] - 0x00 (TOA RX mode) Octet[1] - Number of RX antenna to follow - Octet[2] RX1 antenna ID - Octet[3] RX2 antenna ID ... - Octet[N] RXN antenna ID (or RX antenna pair N)
RSSI_RX1	2 Octets	Average value of the computed RSSI value on RX1 path. It is signed value expressed in Q8.8 format
RSSI_RX2	2 Octets	Average value of the computed RSSI value on RX2 path. It is signed value expressed in Q8.8 format
RSSI_RX[N]	2 Octets	Average value of the computed RSSI value on RXN path. It is signed value expressed in Q8.8 format
SNR_RX1	2 Octets	Averaged value of the SNR computed on RX1 path. It is a signed value in Q 8.8 format
SNR_RX2	2 Octets	Averaged value of the SNR computed on RX2 path. It is a signed value in Q 8.8 format.
SNR_RX[N]	2 Octets	Averaged value of the SNR computed on RXN path. It is a signed value in Q 8.8 format.

8.3 TEST_LOOPBACK_NTF

Table 73. TEST_LOOPBACK_NTF Extensions

Payload Field(s)	Length	Value/ Description
TEST_LOOPBACK_NTF		
Status	1 Octet	Refer Generic Status Codes
TX_TS_INT	4 Octet	Integer part of timestamp in 1/124.8 μ s resolution
TX_TS_FRAC	2 Octet	Fractional part of timestamp in 1/124.8/512 μ s resolution
RX_TS_INT	4 Octet	Integer part of timestamp in 1/124.8 μ s resolution
RX_TS_FRAC	2 Octet	Fractional part of timestamp in 1/124.8/512 μ s resolution
AoA Azimuth	2 Octet	AoA Azimuth in degrees and it is a signed value in Q9.7 format. This field is zero if AOA_RESULT_REQ = 0.
AoA Elevation	2 Octet	AoA Elevation in degrees and it is a signed value in Q9.7 format. This field is zero if AOA_RESULT_REQ = 0.
PHR	2 Octet	Received PHR
PSDU Data Length	2 Octet	Length of PSDU DATA (N) to follow
PSDU_DATA	N Octet	Received PSDU Data bytes
Vendor Specific Data Length	2 Octets	Vendor Data Length, Vendor Data field is not present if Vendor Data Length = 0. Otherwise = (N+1) This Vendor Data Length field is present if NXP_EXTENDED_NTF_CONFIG is set to 0x01 (Vendor extended notifications) and Vendor presence is based on Vendor Data Length
NXP Specific Data type	1 Octet	This parameter indicates the type of vendor specific data: 0x00 : Specific Data V1 for TEST_LOOPBACK_NTF
NXP Specific Data	N Octet	Vendor Data Refer TEST_LOOPBACK_NTF Vendor Specific Data for more details

Table 74. TEST_LOOPBACK_NTF Vendor Specific Data

Parameter Name	Length	Value/ Description
RX antenna information	1 + 1 + N	RX mode and number of antenna used Octet[0] - 0x00 (TOA RX mode) Octet[1] - Number of RX antenna to follow - Octet[2] RX1 antenna ID - Octet[3] RX2 antenna ID ...

Table 74. TEST_LOOPBACK_NTF Vendor Specific Data...continued

Parameter Name	Length	Value/ Description
		- Octet[N] RXN antenna ID
RSSI_RX1	2 Octets	Average value of the computed RSSI value on RX1 path. It is signed value expressed in Q8.8 format
RSSI_RX2	2 Octets	Average value of the computed RSSI value on RX2 path. It is signed value expressed in Q8.8 format
RSSI_RX[N]	2 Octets	Average value of the computed RSSI value on RXN path. It is signed value expressed in Q8.8 format
SNR_RX1	2 Octets	Averaged value of the SNR computed on RX1 path. It is a signed value in Q 8.8 format
SNR_RX2	2 Octets	Averaged value of the SNR computed on RX2 path. It is a signed value in Q 8.8 format.
SNR_RX[N]	2 Octets	Averaged value of the SNR computed on RXN path. It is a signed value in Q 8.8 format.
RX_CFO_EST	2 Octets	Average value of the computed carrier frequency offset estimate from the reception in PPM. It is a signed value in Q 5.11 format

8.4 Periodic TX Test

8.4.1 Periodic TX details

Minimum T_GAP supported is 350us. (HELIOS1/HELIOS2)

8.4.2 Periodic TX Test extension mode

If Periodic TX test is executed with the UCI test configuration NUM_PACKETS equal to 0, SR250 will transmit only the continuous wave for the configured channel. For stopping this mode, Host shall send a Stop test mode command.

9 Debug Configurations

9.1 CIR log configuration

Below proprietary config parameter is used to enable the CIR log dump feature. DH shall use SET_VENDOR_APP_CONFIG_CMD command to enable the CIR log dump feature with CIR_LOG_NTF config.

Once CIR_LOG_NTF is set to non-zero, the UWBS starts sending below CIR log notifications to DH.

Table 75. CIR log Notification Packets (CIR_LOG_NTF)

Payload Field(s)	Length	Value/ Description
Session Handle	4 Octets	Session Handle for which data CIR NTF belongs
Slot Index	1 Octet	Slot index for which received RFrame
RX Antenna ID	1 Octet	RX Antenna ID for which received RFrame
Number of CIR sample	2 Octets	Number of CIR sample to follow
CIR0 log sample	(n * 4) Octets	CIR samples payload

9.2 Data logger configuration

DATA_LOGGER_NTF config parameter is used to enable and configure the data logger feature. DH shall use SESSION_SET_VENDOR_APP_CONFIG_CMD command to enable the data logger feature with DATA_LOGGER_NTF config.

Once DATA_LOGGER_NTF is enabled (set to 0x01), the UWBS starts sending data logger DBG_DATA_LOGGER_NTF notification to DH

Table 76. DBG_DATA_LOGGER_NTF

Payload Field(s)	Length	Value/ Description
Session Handle*	4 Octets	Session Handle for which data logger info belongs
Data logger info	n Octet(s)	Data logger payload

*Session Handle is not part of the datalogger notification as of now since there is no plan to support multiple data logger notifications in multisession scenarios. This may change in the future.

9.3 PSDU log configuration

PSDU_LOG_NTF config parameter is used to enable/disable PSDU payload logging feature. DH shall use SET_VENDOR_APP_CONFIG_CMD command to enable this feature via PSDU_LOG_NTF configuration paramter.

Once PSDU_LOG_NTF is set to 0x01, the UWBS starts sending PSDU_LOG_NTF notification to DH. UWBS shall not send DBG_PSDU_LOG_NTF notification when DBG_PPDU_LOG_NTF application config is set to 0x00.

Table 77. PSDU_LOG_NTF

Payload Field(s)	Length	Value/ Description
Session Handle	4 Octets	Session Handle for which PSDU payload belongs

Table 77. PSDU_LOG_NTF...continued

Payload Field(s)	Length	Value/ Description		
PSDU log Info	N * (3 + M) Octets	Slot Index	1 Octet	Slot Index of the PSDU in the Ranging Round. After slot index 255 this field will role over to 0. DH should interpret it as 256+Current Value of Slot Index after roll over. e.g. if last received slot index was 255 and current slot index is 0 then it should be considered as 256th slot.
		PSDU Size	2 Octet	PSDU size in bytes
		PSDU Bytes	M Octets	PSDU payload (MFR+Payload) <i>Note: MFR is optional</i>

9.4 RFRAME log configuration

RFRAME_LOG_NTF config parameter is used to enable/disable RFRAME details feature. DH shall use SET_VENDOR_APP_CONFIG_CMD command to configure this feature via RFRAME_LOG_NTF parameter.

Once RFRAME_LOG_NTF is set to 0x01, the UWBS starts sending RFRAME details as DBG_RFRAME_LOG_NTF notification to DH.

Table 78. RFRAME Data Notification

Payload Field(s)	Length	Value/ Description
DBG_RFRAME_LOG_NTF		
Session Handle	4 Octets	Session Handle for which RFRAME details belongs
Sequence Number	4 Octets	Sequence Number as reported in Ranging Data Notification. Only valid if Ranging Data Notifications are enabled, otherwise this field is set to 0.
TX Antenna Information	1 Octet	Indicates the TX antenna used for transmission, has to be ignored for RX slots
Number of RFRAME measurements "N"	1 Octet	N: Number of RFRAME measurements to follow
Size of RFRAME measurement	1 Octet	Length of each RFRAME measurement

Table 78. RFRAME Data Notification...continued

Payload Field(s)	Length	Value/ Description
Message Control for fixed Fields	1 Octet	RFU for future additions
RFRAME log Info	N * RFRAME measurement	RFRAME measurement details
Message Control for Variable RFRAME Log Info fields	1 Octet	RFU for future additions

Table 79. RFRAME Notification Details

Payload Field(s)	Length	Value/ Description
RFRAME measurement details		
Mapping	1 Octet	bits[5:0] - RFRAME slot index bits[7:6] : For H2 • 0b00 :RXC • 0b01: RXB • 0b10: RFU • 0b11: RXA
Dec Status	1 Octet	Decode status of the RFRAME. Refer Table 80 for different decode status codes
NLoS	1 Octet	Indicates if the ranging measurement was in Line of sight or Non-Line of Sight. 0x00: LoS 0x01: NLoS 0xFF: Invalid
First Path Index	2 Octets	Estimated first path index in nanoseconds expressed in unsigned Q10.6 format
Main Path Index	2 Octets	Estimated main path index in nanoseconds expressed in unsigned Q10.6 format
SNR Main Path	1 Octet	Signal-to-Noise Ratio (SNR) of the main path in dB
SNR First Path	1 Octet	Signal-to-Noise Ratio (SNR) of the first path in dB
SNR Total	2 Octets	Signal-to-Noise Ratio (SNR) of configured RX in Q8.8 dB, it is used for internal purpose only.
RSSI	2 Octets	Received Signal Strength Indicator in dB. It is signed value expressed in Q8.8 signed format
CIR Main Power	4 Octets	CIR main power : It represents $cir_real^2 + cir_imag^2$ of the main path. This is unsigned 32bit value.
CIR first path power	4 Octets	CIR first path power : It represents $cir_real^2 + cir_imag^2$ of the first path. This is unsigned 32bit value.

Table 79. RFRAME Notification Details...continued

Payload Field(s)	Length	Value/ Description
Noise variance	2 Octets	Noise variance in the CIR
CFO	2 Octets	CFO estimate in PPM. It is signed value coded in Q5.11 format.
AoA Phase	2 Octets	Phase in degrees expressed in Q9.7 format, used for calculating phase difference
CIR samples	16*2*2 Octets	CIR samples from (first path index-8) to (first path index+7) Octet[4*i+1:4*i+0] = Real value Octet[4*i+3:4*i+2] = Imaginary value
RX_TIMESTAMP	6 Octets	Octet [0:3] - Integer part of timestamp when RMARKER has been received Octet [4:5] - Fractional part of timestamp when RMARKER has been received. Fractional part is coded in Q7.9 format

Table 80. Rx decode status codes in RFRAME

Codes	Description
0x00	STATUS_RX_ACQ_FAILURE: Signal acquisition failed. No Signal Received.
0x01	STATUS_RX_SECDEC_FAILURE: For PHR Decoding 'Single Error Correction Double Error Detection'(SECDED) is used. If PHR Decode is unsuccessful, this error code is returned.
0x02	STATUS_RX_RSDEC_FAILURE : For PSDU Decoding 'Reed Solomon'(RS) is used. If PSDU Decode is unsuccessful, this error code is returned.
0x03	STATUS_RX_DEC_FAILURE : Generic error for Packet Decode Failure. Message Code to indicate Rx Decode Chain Failure.
0x04	STATUS_RX_DEC_SUCCESS : If CRC is correct and Packet Decode is successful, this status code is returned. Else, PH_UCI_STATUS_RANGING_RX_MAC_DEC_FAILED is returned.
0x05	STATUS_RX_DEC_NO_DATA : No Payload available on the received frame. It would be (SYNC+STS) frame also called as (SP3 Frame).
0x06	STATUS_PHY_RX_ERR : Generic error for Rx Data Buffer Overflow.
0x07	STATUS_RX_STS_FAILURE: Error code for STS Mismatch -> SFD Detected, STS Not Matching
0x08	STATUS_RX_TOA_DETECT_FAILURE:

Table 80. Rx decode status codes in RFRAME....continued

Codes	Description
	ToA Detect Failure happens when CIR Signal Amplitude is Low and the signal is not detectable by ToA.
0x09	STATUS_RX_PHR_DEC_FAILURE: Generic error for PHR Decode Failure. Occurs if one of the below condition is true: a) If PHR is available and if PHR Length is Zero b) If PHR is available and if PHR Length exceeds Maximum Frame Length.
0x0A	STATUS_RX_SYNC_SFD_FAILURE: Expecting SYNC or SFD. Failing to receive them will result in this error.
0x0B	STATUS_PHR_DATA_RATE_ERROR: If the Data Rate of Raw PHR and Decoded PHR are not equal, this error code is returned.
0x0C	STATUS_RX_PHR_RANGING_ERROR: PHR Ranging Setting Error

9.5 KDF debug notification

UWBS shall communicate output of KDF functionality to HOST through the UWB_SESSION_KDF_NTF notification.

The generation of UWB_SESSION_KDF_NTF notification shall be enable via TEST_KDF_FEATURE configuration.

KDF debug feature shall enable/disabled by using TEST_KDF_FEATURE proprietary configuration. As KDF feature is specific to session and shall be enabled using SESSION_SET_VENDOR_APP_CONFIG_CMD command with TEST_KDF_FEATURE config.

UWBS shall keep generating UWB_SESSION_KDF_NTF notification and send it those notifications DH when TEST_KDF_FEATURE is set to 0x01 and when KEY_ROTATION_RATE condition satisfies

9.5.1 FingerPrint(CMAC) of the keys in KDF Notification

FingerPrint = CMAC(Key, Context_Data)

This section describes the computation of Context Data, based on the configuration described in next table, all the paramters in the table are concatenated to form the input to CMAC function.

Table 81. Context Data

Parameter	Length	Value
Counter	4 Bytes	0x00000001
Label	8 Bytes	0x544553545f4b4446 (TEST_KDF in ASCII)
EmptyContext	16 Bytes	0x00000000000000000000000000000000
Length	4 Bytes	0x00000080

Table 82. UWB_SESSION_KDF_NTF

Payload Field(s)	Length		Value/ Description	
Number of Parameters	1 Octet		The number of Parameter fields to follow (n)	
Parameter [1..n]	m+2 Octets	ID	1 Octet	The identifier of the configuration parameter. Refer Table 83
		Length	1 Octet	The length of Val (m).
		Value	m Octets	The value of the parameter Refer Table 83

Table 83. Parameters in UWB_SESSION_KDF_NTF

Parameter Name	Length (Octets)	Tag (IDs)	Description
KDF_BLOCK_INDEX	2	0x00	Block Index on which key rotation was applied
KDF_STS_INDEX	4	0x01	STS index value used for KDF
KDF_FIRA_CONFIG_DIGEST	17	0x02	Config digest value (Applicable in FiRa Ranging session)
KDF_FIRA_DERIVED_AUTH_IV	16	0x03	CMAC of KDF Context Data using Derived Authentication IV (Applicable in FiRa Ranging session)
KDF_FIRA_DERIVED_AUTH_KEY	16	0x04	CMAC of Derived Authentication Key in FiRa UWB session (Applicable in FiRa Ranging session)
KDF_FIRA_DERIVED_PAYLOAD_KEY	16	0x05	CMAC of KDF Context Data using Derived Payload Key (Applicable in FiRa Ranging session)
KDF_FIRA_DATA_PROTECTION_KEY	16	0x06	CMAC of FiRa data protection key (Applicable in FiRa Ranging session)
KDF_FIRA_PRIVACY_KEY	16	0x08	CMAC of KDF Context Data using FIRA data privacy key (Applicable in FiRa Ranging session)
KDF_FIRA_NOTIFICATION_KEY	16	0x09	CMAC of KDF Context Data using FIRA data notification key (Applicable in FiRa Ranging session)
KDF_CCC_RANGING_CONFIGURATION	17	0xA0	Negotiated Ranging Configuration data (Applicable in CCC Ranging session)
KDF_CCC_MUPSK1	16	0xA1	CMAC of mUPSK1 key. mUPSK1 is used in order to encrypt the Type 2 Pre-POLL. (Applicable in CCC Ranging session)
KDF_CCC_DURSK	16	0xA2	CMAC of dURSK key.

Table 83. Parameters in UWB_SESSION_KDF_NTF...continued

Parameter Name	Length (Octets)	Tag (IDs)	Description
			Derived UWB Ranging Secret Key (Applicable in CCC Ranging session)
KDF_CCC_DUDSK	16	0xA3	CMAC of dUDSK key. Derived UWB Data Secret Key (Applicable in CCC Ranging session)
KDF_CCC_SALTED_HASH	16	0xA4	CMAC of Salted Hash. Salted Hash of configuration (Applicable in CCC Ranging session)
KDF_CCC_MURSK	16	0xA5	CMAC of mURSK key. Master UWB Ranging Secret Key (Applicable in CCC Ranging session)
KDF_CCC_MUPSK2	16	0xA6	CMAC of mUPSK2 key. mUPSK2 is used to derive the UWB addresses. (Applicable in CCC Ranging session)

9.6 Error log information

SR250 notifies device panic state as STATUS_ERROR in DEVICE_STATUS_NTF. DH shall use

DBG_GET_ERROR_LOG_CMD command to fetch the error log from SR250 in STATUS_ERROR state. SR250 responds by DBG_GET_ERROR_LOG_RSP with error log data.

Table 84. Control messages to fetch error log (DBG_GET_ERROR_LOG_CMD)

Payload Field(s)	Length	Value/ Description
Command	0 Octet	Get error log information

The crash log buffer will have the following information:

Table 85. DBG_GET_ERROR_LOG_RSP

Payload Field(s)	Length	Value/ Description
Exception Type	4 Octets	As defined in the Exception types
R0	4 Octets	Cortex-M33 CPU core R0 register
R1	4 Octets	Cortex-M33 CPU core R1 register
R2	4 Octets	Cortex-M33 CPU core R2 register
R3	4 Octets	Cortex-M33 CPU core R3 register
R12	4 Octets	Cortex-M33 CPU core R12 register
LR	4 Octets	Cortex-M33 CPU core Link Register
PC	4 Octets	Cortex-M33 CPU core Program Counter register
PSR	4 Octets	Cortex-M33 CPU Program Status Register

Table 85. DBG_GET_ERROR_LOG_RSP...continued

Payload Field(s)	Length	Value/ Description
STACK_WORDS	128 Octets	Crash stack words
S_CFSR	4 Octets	Cortex-M33 Secure Configure Fault Status Register
NS_CFSR	4 Octets	Cortex-M33 Non-secure Configure Fault Status Register
HFSR	4 Octets	Cortex-M33 Hard Fault Status Register
SFAR	4 Octets	Cortex-M33 Secure Fault Address Register
SFSR	4 Octets	Cortex-M33 Secure Fault Status Register
BFAR	4 Octets	Cortex-M33 Bus Fault Address R Register
MMFAR	4 Octets	Cortex-M33 Memory Management Fault Address Register
FW_STAT	4 Octets	Firewall Status Register
FW_BUS_ERR_INFO	28 Octets	Firewall bus error information Registers
FW_SEC_ERR_INFO	8 Octets	Firewall security error information Registers

Table 86. Exception types

Type	Description
0x00	No Fault
0x01	Hard Fault
0x02	Bus Fault
0x04	Secure Fault
0x08	Usage Fault
0x10	Watchdog
0x20	CoolFlux Fault
0x40	Assert Fault log. Refer Assert fault log information

Note: When No Fault occurs DBG_GET_ERROR_LOG_RSP will be only 4 octets long

Table 87. Assert fault log information

Payload Field(s)	Length	Value/ Description
Exception Type	4 Octets	0x00000040
Assertion Code	2 Octets	Assertion Error code, refer Table 88
Line Number	2 Octets	Line number from the file below

Table 88. Assertion Error Code

Assertion Code	Value	Description
VCO_LOCK_FAIL	34	PLL not locked.
RF_CLK_ABSENT	36	RF Clock is not available after waiting time.

Table 89. Device Specific Assertion Error Codes

Assertion Code	Value	Description
EVT_FLAG_DELETE_FAIL	77	Failed to delete OS event flag
DIGITAL_CLOCK_STUCK_DETECTED	78	Dig. Clock is stuck after enabling SysPLL
NSC_API_FAIL	79	Checks failed in NSC API
HASH_INTEGRITY_FAILED	80	Failed to check hash integrity
SYS_TIMER_FAILURE	81	Failed to read the systimer tick
UCI_HIF_INIT_FAILURE	82	Failed to init the HAL HIF UCI layer

9.7 WiFi Co-Existence Feature

9.7.1 Debug notification for WIFI CoEx status

UWB_WLAN_IND_STATUS_NTF notification is used to notify the UWB_WLAN_IND GPIO toggle status. This notification shall be sent when *GPIO* based CoEx feature with debug verbose is enabled via *WIFI_COEX_FEATURE* config

9.7.1.1 GPIO based CoEx feature debug notification

Below is the UWB_WLAN_IND_STATUS_NTF notification format is received when *GPIO* or *OneWire* based CoEx feature with debug verbose is enabled via *WIFI_COEX_FEATURE* config

Table 90. UWB_WLAN_IND_STATUS_NTF

Payload Field(s)	Length	Value/ Description
Status	1 Octet	UWB_WLAN_IND GPIO Status 0x00 : High to Low 0x01 : Low to High
Time index	4 Octets	Time index where the GPIO change occurred
Session Handle	4 Octets	Session Handle with which UWB_WLAN_IND_STATUS_NTF notification belongs.

9.7.2 Debug notification for WLAN_UWB_IND status during ranging round

WLAN_UWB_IND_STATUS_NTF notification is used to notify the WLAN_UWB_IND GPIO status during ranging round. This notification shall be sent when *WIFI_COEX_FEATURE* config is set to 0x02 (Enable with Debug Verbose)

Table 91. WLAN_UWB_IND_STATUS_NTF

Payload Field(s)	Length	Value/ Description
Status	1 Octet	0x01: WLAN_UWB_IND_HIGH_AT_RR_START WLAN_UWB_IND GPIO is high at the start of the ranging round and ranging round shall be attempted 0x02: WLAN_UWB_IND_HIGH_DURING_RR WLAN_UWB_IND GPIO changed to high during Ranging round <i>Note : Don't need to report status 0x00 for this notification</i>

Table 91. WLAN_UWB_IND_STATUS_NTF...continued

Payload Field(s)	Length	Value/ Description
Time index	4 Octets	Time index where the GPIO change occurred

Note:

When Wifi Coex feature is enabled, Wlan State will be checked twice

One before ranging and one during ranging.

- *Pre Check (before Ranging) - If pin state from Wlan to UWB is high, Wlan state set to WLAN_UWB_IND_HIGH_AT_RR_START(0x01),*
- *Post Check (during Ranging) - If Wlan state is not set during Pre Check, Wlan State is set as WLAN_UWB_IND_HIGH_DURING_RR(0x02) in Post check Else, will retain the state set in Pre Check.*

9.7.3 Maximum Grant duration exceeded warning notification

The UWBS shall send warning UWB_WLAN_COEX_MAX_GRANT_DURATION_EXCEEDED_WRN_NTF notification when maximum active grand duration is exceeded.

This notification will be sent for OneWire CoEx when required ranging runtime > MAX_WIFI_BLOCKAGE_DURATION.

Table 92. UWB_WLAN_COEX_MAX_GRANT_DURATION_EXCEEDED_WRN_NTF

Payload Field(s)	Length	Value/ Description
Status	1 Octet	0x00: MAX Active Grant Duration Exceeded Warning Notification

9.8 Hardware Security Error detected

Table 93. TRIGGER_HW_SECURITY_CHECK_ERROR_NTF

Payload Field(s)	Length	Value/ Description
Security error	4 Octets	Please refer to below description

9.8.1 Description

Bits [31:0] – Security error code

9.9 UWBS_SLEEP_ENTRY_PREVENTION_NTF

Table 94. UWBS_SLEEP_ENTRY_PREVENTION_NTF

Payload Field	Length	Value / Description
Task ID	1 Octect	The ID of the task which prevents sleep entering. The corresponding bit is set for a task which has prevented entering the sleep state. b0: Task 1 b1: Task 2 b2: Task 3

Table 94. UWBS_SLEEP_ENTRY_PREVENTION_NTF

Payload Field	Length	Value / Description
		b3: Task 4 b4: Task 5 b5: Task 6 b6: Task 7 b7: Task 8

This notification data is to monitor whether UWBS doesn't enter a low power mode what task let to an abnormal state.

For this kind of management, we described more details as shown below.

1. This can be configured using the UWBS_SLEEP_ENTRY_PREVENTION_NTF_CONFIG. Please refer to [App config](#)
2. When to send UWBS_SLEEP_ENTRY_PREVENTION_NTF
 - There are only two cases where UWBS should check the low power mode status.
case 1: From Power on to DEVICE_STATE_READY
case 2: From DEVICE_STATUS_ACTIVE to DEVICE_STATE_READY
 - For UWBS state please refer to FiRa Generic UCI Specification for more info.

10 UWBS Device Calibration

10.1 Performing Device Calibration

10.1.1 Calibration of CHIP

Below commands are used to perform the calibration for requested parameter. UWBS shall perform calibration process by itself and return the calibrated value via DO_CHIP_CALIBRATION_NTF notification after calibration process is completed.

The UWBS shall not accept another DO_CHIP_CALIBRATION_CMD if DO_CHIP_CALIBRATION_NTF is pending. UWBS shall reject command by DO_CHIP_CALIBRATION_RSP with STATUS_CALIBRATION_IN_PROGRESS indicating that calibration process is under progress.

Table 95. DO_CHIP_CALIBRATION_CMD

Payload Field(s)	Length	Value/Description
DO_CHIP_CALIBRATION_CMD		
Channel ID	1 Octet	Channel IDs

Table 96. DO_CHIP_CALIBRATION_RSP

Payload Field(s)	Length	Value/Description
DO_CHIP_CALIBRATION_RSP		
Status	1 Octet	Status values. Refer Status Codes . Below fields are valid if status is STATUS_OK

Table 97. DO_CHIP_CALIBRATION_NTF

Payload Field(s)	Length	Value/Description
DO_CHIP_CALIBRATION_NTF		
Status	1 Octet	Status values. Refer Status Codes . Below fields are valid if status is STATUS_OK
Value *** Do_CHIP_Calibration_Cmd triggers a write operation on SR250's Flash memory ***	2 Octets	Return the calibration outcome per channel. bit[8:0] : Calibration code bit[13:9] : RFU bit[15:14] : Calibration status as below 0b10: Calibration is valid - Other: Calibration is invalid

10.2 Setting Calibration Parameters

Below UCI commands are used to set the calibration values for requested parameter and channel. Generally the calibrated values shall be set during UWBS power on sequence.

Table 98. SET_DEVICE_CALIBRATION_CMD

Payload Field(s)	Length	Value/Description
Channel ID	1 Octet	Channel ID for which calibration to be applied. Channel IDs

Table 98. SET_DEVICE_CALIBRATION_CMD...continued

Payload Field(s)	Length	Value/Description
Tag	1 Octet	Calibration parameter for which calibration to be configured. Refer Calibration Parameters for Tag IDs and Values.
Length	Length depends on the parameter	Refer Calibration Parameters
Value	N Octets	Calibration value Refer Calibration Parameters

Table 99. SET_DEVICE_CALIBRATION_RSP

Payload Field(s)	Length	Value/Description
Status	1 Octet	Status values. Refer Status Codes

10.3 Setting Secure Calibration Parameters

Some sensitive calibration parameters can't be modified when UWBS is in protected mode with the SET_CALIBRATION_CMD command.

Below UCI commands are used to set secure calibration parameters in all UWBS lifecycle modes.

Table 100. SET_SECURE_CALIBRATION_CMD

Payload Field(s)	Length	Value/Description
Channel ID	1 Octet	Channel ID for which calibration to be configured Channel IDs
Tag	1 Octet	Calibration parameter for which calibration values to be configured. Refer to Calibration Parameters for their format description For background compatibility, those register are accessible through different tag IDs in secure: 0x02: RF_CLK_ACCURACY_CAL 0x0E: TX_ANT_DELAY_CALIB 0x0F: RX_ANT_DELAY_CALIB 0x5C: PLATFORM_ID 0x5D: CONFIG_VERSION
Value	n Octets	Calibration value
ConfigVersion	4 Octets	default is 0
Cryptotag	16 Octets	Cryptographic tag protecting the parameter value

Note:

- *Crypto tag is validated before validating Config Version*
- *When there is no Platform Id in customer area, Config version will not be checked*

Table 101. SET_SECURE_CALIBRATION_RSP

Payload Field(s)	Length	Value/Description
Status	1 Octet	Status values. Refer Status codes

10.4 Retrieving Calibration Parameters

Below UCI commands are used to get the configured calibration values for requested parameter and channel. The UWBS shall respond via GET_DEVICE_CALIBRATION_RSP with configured calibration values.

Table 102. GET_DEVICE_CALIBRATION_CMD

Payload Field(s)	Length	Value/Description
Channel ID	1 Octet	Channel IDs
Tag	1 Octet	Calibration parameter for which calibration to retrieved. Refer Calibration Parameters for Tag IDs and Values.
RX Antenna Pair ID	1 Octet	Only applicable when calibration parameter ID is AOA_ANTENNAS_PDOA_CALIB. This Octet shall be set to Rx Antenna Pair ID as defined by ANTENNAS_RX_PAIR_DEFINE for which the PDoA Calibration table needs to be retrieved. For the other Calibration Parameter IDs this field is not required to be used.

Table 103. GET_DEVICE_CALIBRATION_RSP

Payload Field(s)	Length	Value/Description
GET_CALIBRATION_RSP		
Status	1 Octet	Status values. Refer Status Codes . Below fields are valid if status is STATUS_OK
Calibration state	1 Octet	Refer Calibration Parameter States for possible Calibration Parameter State
Tag	1 Octet	Tag of the calibration parameter
Length	Refer Calibration Parameters	Length of the value following
Value	n Octets	Calibration value for requested channel and parameter. <i>Value field is shared if Calibration state is different from INVALID_STATE</i> Note: FW will return the calibration data for maximum Rx antenna pairs (ANTENNAS_RX_PAIR_DEFINE) or Rx antennas (ANTENNA_RX_IDX_DEFINE) or Tx Antennas (ANTENNA_TX_IDX_DEFINE) as per Table 55 , Host has to ignore all the entries beyond the entries configured via CORE_SET_CONFIG

Table 104. Calibration Parameter States

Value	Calibration state	Description
0x00	DEFAULT	Calibration parameter carries default value by UWBS
0x01	CUSTOM_NOT_INTEGRITY_PROTECTED	Indicates this calibration parameter is updated to value other than its default value and is not integrity protected either by a Device specific or Model specific authentication tag.
0x02	CUSTOM_AUTH_PENDING	Calibration parameter integrity check is pending either by a Device specific or Model specific tag. Mainline FW: Until the tag verification finishes, the parameter will not be applied
0x03	CUSTOM_DEVICE_SPECIFIC_TAG_AUTHENTICATED	Calibration parameter integrity check is verified successfully by a Device specific tag.
0x04	CUSTOM_MODEL_SPECIFIC_TAG_AUTHENTICATED	Calibration parameter integrity check is verified successfully by a Model specific tag
0x08	INVALID_STATE	Calibration data is either not set or its integrity can't be verified. SR250 uses default value.

10.5 Calibration Parameters

Table 105. Calibration Parameters

Parameter name	Tag	Length	Value range/Description
Calibration Parameters			
CHIP_CALIBRATION	0x00	2 Octets	Description details are in the next immediate row
(...) *** Do_Chip_Calibration_Cmd triggers a write operation on SR250's Flash memory ***			Return the calibration outcome per channel. This parameter is channel dependent. Channel IDs bit[8:0] : Calibration code bit[13:9] : RFU bit[15:14] : Calibration status as below • 0b10: Calibration is valid • Other: Calibration is invalid
RF_CLK_ACCURACY_CALIB *** Stored in SR250's Flash ***	0x01	7 Octets	Defines the capacitors values applied to 38.4MHz XTAL. Channel independent and remains same for each channel. • Octet [0]: Number of registers (must be 0x03) • Octet [2-1]: 38.4 MHz XTAL CAP1 • Octet [4-3]: 38.4 MHz XTAL CAP2 • Octet [6-5]: 38.4 MHz XTAL GM CURRENT CTRL Values : [0x00-0xFF] for 38.4 MHz XTAL CAP Values : [0x00-0x3F] for 38.4 MHz XTAL GM
RX_ANT_DELAY_CALIB *** Stored in SR250's Flash ***	0x02	1 + (N*3) Octets	Delay Calibration for each RX Antenna This parameter is channel dependent Channel IDs • Octet[0]: Number of Entries (Must be greater than 0) • [– 1 Octet: RX Antennas IDs as defined by ANTENNA_RX_IDX_DEFINE – 2 Octets each: RX Delay (unsigned Q14.2).] (Repeat) (Default: 0x0000 for all RX antenna IDs) Example to Set RX Delay Calib for Antenna ID =0x01,0x02,0x03 for Channel 5 and 9

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			- Protected lifecycle: SET_DEVICE_CALIBRATION_SEC_CMD - channel Id=5, Calibration Param=0x02 - Number Of entries = 2 - Ant ID : 0x01, Delay Value1 - Ant ID : 0x02, Delay Value2 - ConfigVersion = 0x01 - Cryptotag = 0xXX...0xXX - Other lifecycle: SET_DEVICE_CALIBRATION_CMD - channel Id=5, Calibration Param=0x02 = Rx Ant Delay calib - Number Of entries = 3 - Ant ID : 0x01, Delay Value1 - Ant ID : 0x02, Delay Value2 RX_ANT_DELAY_CALIB unit is cm and step size is 0.25 cm i.e value 1 =0.25 cm
PDOA_OFFSET_CALIB	0x03	1 + N*(1+2) Octets	PDOA Offset Calibration This parameter is channel dependent Channel IDs - Octet[0]: Number of Entries (Must be greater than 0) (Depends on available RAM) [1 Octet each: RX Antennas PAIR IDs as defined by ANTENNAS_RX_PAIR_DEFINE 2 Octets each: PDoA Offsets. (signed value in Q9.7 format)] (Repeat) (Default: 0x0000 for all RX antenna pair ID) Example to Set PDoA Offset Calib for Antenna Pair ID=0x01, 0x02, 0x03 for Channel 5 and 9 SET_DEVICE_CALIBRATION_CMD(ch Id=5, Calib Param=0x10 = PDoA1, No Of Entries=3, Ant ID=0x01,PDOA Value1, 0x02,PDOA Value2, 0x03,PDOA Value3) Repeat the same command for Channel 9 GET_DEVICE_CALIBRATION_CMD is a cyclic fetch logic as explained below

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			First Invocation GET_DEVICE_CALIBRATION_CMD (ch Id=5, Calib Param=0x10) will report the data for PAIR1 Second Invocation GET_DEVICE_CALIBRATION_CMD (ch Id=5, Calib Param=0x10) will report the data for PAIR2 Repeats until MAX_N of ANTENNAS_RX_PAIR_DEFINE See Table 55 for max value of MAX_N Note: Refer Recommendation while using Calibration Parameters
TX_POWER_PER_ANTENNA	0x04	1 + (N*5) Octets	TX POWER This parameter is channel dependent channel_ID : Channel IDs • Octet[0]: Number of Entries (Must be greater than 0) • [– 1 Octet : TX Antenna IDs as defined by ANTENNAS_TX_IDX_DEFINE – 2 Octets : TX_POWER_DELTA_PEAK - If TX_ADAPTIVE_PAYLOAD_POWER is enabled, this value represents the delta peak ie : PEAK_ID - RMS_ID. This value is a delta and comes from calibration process, It ranges from 0 (max power) to 127 (min power) with steps of 0.25. - If TX_ADAPTIVE_PAYLOAD_POWER is not enabled, this value has no specific use. – 2 Octets : TX_POWER_ID_RMS - If TX_ADAPTIVE_PAYLOAD_POWER is enabled, this value come from RMS calibration as RMS_ID - If TX_ADAPTIVE_PAYLOAD_POWER is not enabled, this value serves as a gain index for TX power control. It ranges from 0 (max power) to 127 (min power) with steps of 0.25.] (Repeat) Default value for all: 0x00 Note: TEST_LOOPBACK_CMD is rejected if TX_POWER_ID_RMS is below than 80
LO_GATING		2 + N*4	channel_ID : Octet[0] channel ID : {5,9,10} Octet[1] number of entries N Octet[2] TX antenna ID as defined by ANTENNA_TX_IDX_DEFINE

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			Octet[3]: RX antennas ID as defined by ANTENNA_RX_IDX_DEFINE Octet[4]: delay calibration data Octet[5]: width calibration data Octet[2+n*4] TX antenna ID as defined by ANTENNA_TX_IDX_DEFINE Octet[3+n*4]: RX antennas ID as defined by ANTENNA_RX_IDX_DEFINE Octet[4+n*4]: delay calibration data Octet[5+n*4]: width calibration data All delays and widths are expressed in nanoseconds, in unsigned Q6.2 format. (Range : 0..63.75ns, resolution of .25ns)
AOA_PHASEFLIP_ANTSPACING	0x05	1+(N*3) Octets	Antenna phase flip and spacing This parameter is channel dependent. Channel IDs - Octet[0]: Number of Entries (Must be greater than 0) [1 Octet: RX Antenna Pair ID 2 Octets: - Bit 15 Phase flip (1 bit) - Bit 14:0 Antenna spacing (15 bits in Q6.9 format)] (Repeat) (Default: 0x0028 for all RX antenna pair IDs)
PLATFORM_ID *** Stored in SR250's Flash ***	0x5C	16 Octets	Channel independent (same value is applied for each channel): Channel IDs 16-bytes string defining the platform name.
CONFIG_VERSION *** Stored in SR250's Flash ***	0x5D	4 Octets	Channel independent (same value is applied for each channel): Channel IDs Version number of the secure parameters (avoiding rollback to wrong settings)
MANUAL_TX_POW_CTRL	0x60	5 Octets	Refer Manual Tx Power Control parameter to for details Channel IDs Octet[0] : TX_PA_GAIN Octet[1] : TX_PPA_GAIN

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			Octet[2] : TX_DIG_GAIN Octet[3] : DAC_GAIN Octet[4] : TX_MIX_GAIN
AOA_ANTENNAS_PDOA_CALIB (Added for 360 FoV)	0x62	1 + N*(1 + S*S*2) Octets	PDOA Calibration tables Octet[0] : No of entries, it shall be set to 1 for Set Calibration Cmd. (Refer to a) of Note for this limitation.) Octet[1] : RX Pair Antennas ID as defined <i>earlier</i> by ANTENNAS_RX_PAIR_DEFINE Octet[2 - M]: PDoA calibration table for specified RX Pair Antennas ID. The older entries would get overwritten, if more than supported data is uploaded to the UWBS. See Table 55 for max supported calibration tables per channel per antennas pair. S is number of steps is Octet[1] of PDOA_CALIB_TABLE_DEFINE Allowed Range: $3 \leq S \leq 17$ (to include 0^0 (default : 11)) N = 1 Always Note: a) In order to support Hosts that can not send more than 255 + 4 bytes, this UCI interface can only be used to set PDoA calibration table for one RX Pair Antennas ID at a time. b) The Get Calibration Cmd response for this ID returns the total number of "RX Pair Antennas IDs" for which PDOA calibration table is set in Octet[0]. Note: Refer Recommendation while using Calibration Parameters
TX_ANT_DELAY_CALIB *** Stored in SR250's Flash ***	0x64	1 + (N*3) Octets	Delay Calibration of TX Part of Antenna Channel IDs - Octet[0]: Number of Entries (Must be greater than 0) [1 Octet : TX Antennas IDs as defined by ANTENNA_TX_IDX_DEFINE 2 Octets : TX Delay offset (signed Q14.2)](Repeat) Example to Set Tx Delay Calib for Antenna ID =0x01,0x02,0x03 for Channel 5 and 9 - Protected lifecycle: - SET_DEVICE_CALIBRATION_SEC_CMD - channel Id=5, Calibration Param=0x0F

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			– Number Of entries = 2 – Ant ID : 0x01, Delay Value1 – Ant ID : 0x02, Delay Value2 – ConfigVersion = 0x01 – Cryptotag = 0xXX...0xXX • Other lifecycle: - SET_DEVICE_CALIBRATION_CMD – channel Id=5, Calibration Param=0x0F = Tx Ant Delay calib – Number Of entries = 3 – Ant ID : 0x01, Delay Value1 – Ant ID : 0x02, Delay Value2
PDOA_MANUFACT_ZERO_OFFSET_CALIB	0x65	1 + (N*1) + (N*2) Octets	Zero offset Manufacturing PDOA Calibration • Octet[0]: Number of Entries (Must be greater than 0) (Depends on available RAM) • [– Array of 1 Octet each: RX Antennas PAIR IDs as defined by ANTENNAS_RX_PAIR_DEFINE – Array of 2 Octets each: Manufacture PDOA1 Offset] (Repeat) Example to Set PDoA1 Calib for Antenna Pair ID=0x01, 0x02, 0x03 for Channel 5 and 9 SET_DEVICE_CALIBRATION_CMD (ch Id=5, Calib Param=0x10 = PDoA1, No Of Entries=3, Ant ID=0x01,Offset Calib Val1, 0x02,Offset Calib Val2, 0x03,Offset Calib Val3) Repeat the same command for Channel 9 This value (PDOA_MANUFACT_ZERO_OFFSET_CALIB) will have higher precedence as compared to this PDOA_OFFSET_CALIB. Note: Refer Recommendation while using Calibration Parameters
AOA_THRESHOLD_PDOA	0x66	1 + (N*1) + (N*2) Octets	AoA Threshold PDOA • Octet[0]: Number of Entries (Must be greater than 0) • [– Array of 1 Octet each: RX Antennas PAIR IDs as defined by ANTENNAS_RX_PAIR_DEFINE – Array of 2 Octets each: Threshold Values

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			] (Repeat) Example to Set Threshold for Antenna Pair ID=0x01, 0x02, 0x03 for Channel 5 and 9 SET_DEVICE_CALIBRATION_CMD (ch Id=5, Calib Param=0x15 = Threshold PDoA, No Of Entries=3, Ant ID=0x01,Threshold PDoA Value1, 0x02,Threshold PDoA Value2, 0x03,Thresho Id PDoA Value3) Repat the same command for Channel 9. Note: Refer Recommendation while using Calibration Parameters
TX_TEMPERATURE_COMP_PER_ANTENNA	0x67	1 + N*(1+8) Octets	TX temperature compensation per antenna This parameter is channel dependent. Channel IDs - Octet[0]: Number of Entries (Must be greater than 0) [– 1 Octet : TX Antenna IDs as defind by ANTENNAS_TX_IDX_DEFINE – 1 Octet : RANGE1_TX_POWER_UPPER_BOUND – 1 Octet : RANGE1_TX_POWER_GAIN_OFFSET – 1 Octet : RANGE2_TX_POWER_UPPER_BOUND – 1 Octet : RANGE2_TX_POWER_GAIN_OFFSET – 1 Octet : RANGE3_TX_POWER_UPPER_BOUND – 1 Octet : RANGE3_TX_POWER_GAIN_OFFSET – 1 Octet : RANGE4_TX_POWER_UPPER_BOUND – 1 Octet : RANGE4_TX_POWER_GAIN_OFFSET] (Repeat) Note: - TX_POWER_UPPER_BOUND: Upper bound for the temperature value. It is a signed number to allow the configuration of negative temperature Default value: 0x80(Disable) - TX_POWER_GAIN_OFFSET: Offset to be applied on the Tx power gain table for a specific temperature range. it's a signed number. Default value : 0x00

Table 105. Calibration Parameters...continued

Parameter name	Tag	Length	Value range/Description
			Associated offset from TX_POWER_GAIN_OFFSET will be applied if measured temperature is below the associated TX_POWER_UPPER_BOUND.
SNR_CALIB_CONSTANT_PER_ANTENNA	0x68	1 + N*(2) Octets	SNR Calibration per RX Antenna This parameter is channel dependent Channel IDs - Octet[0]: Number of Entries (Must be greater than 0) [1 Octet : RX Antenna IDs as defined by ANTENNAS_RX_IDX_DEFINE 1 Octet : SNR Calibration (value in dB)] (Repeat) Default value for all: 0x00
RSSI_CALIB_CONSTANT_PER_ANTENNA	0x69	1 + (N*2) Octets	RSSI Offset per RX Antenna for High/Low Power This parameter is channel dependent channel_ID : Channel IDs - Octet[0]: Number of Entries (Must be greater than 0) [1 Octet : RX Antenna IDs as defined by ANTENNAS_RX_IDX_DEFINE 1 Octet : RSSI Calib (value in dB)] (Repeat) Default value for all: 0x00
Reserved for Vendor Specific	0x82 - 0xDF		Reserved for Vendor Specific
Reserved For Extension of ID's			
Extended Vendor Specific Parameters	0xE0	2 Octets	Parameter ID is 2 octets in length. Refer Extended Calibration Parameters.
RFU	0xE1-0xFF	N Octets	RFU

Recommendation while using Calibration Parameters

The calibration parameters are:

- PDOA_OFFSET_CALIB
- AOA_ANTENNAS_PDOA_CALIB
- PDOA_MANUFACT_ZERO_OFFSET_CALIB
- AOA_THRESHOLD_PDOA
- AOA_ANTENNAS_PDOA_CALIB_EXTENDED_SUPPORT

Recommendation is to use the same antenna pairs for ranging session (antenna pairs configured via ANTENNAS_CONFIGURATION_RX/TX) that were set as part of calibration(SET_CALIBRATION_CMD).

Valid configuration Example-1, 3D AoA

SET_CALIBRATION_CMD(AOA_ANTENNAS_PDOA_CALIB, Antenna Pair1,Antenna Pair2) than host should choose ANTENNAS_CONFIGURATION_RX(Antenna Pair1,Antenna Pair2) in case of 3D AoA

Valid configuration Example-2, 3D AoA

SET_CALIBRATION_CMD(AOA_ANTENNAS_PDOA_CALIB, Antenna Pair1,Antenna Pair2) than host should choose ANTENNAS_CONFIGURATION_RX(Antenna Pair1) in case of 2D AoA

InValid configuration Example-3, 3D AoA

SET_CALIBRATION_CMD(AOA_ANTENNAS_PDOA_CALIB, Antenna Pair1,Antenna Pair2) than host should choose ANTENNAS_CONFIGURATION_RX(Antenna Pair1,Antenna Pair3) in case of 3D AoA, in this case Antenna pair3 is not valid pair as it is not calibrated

InValid configuration Example-4, 3D AoA

SET_CALIBRATION_CMD(AOA_ANTENNAS_PDOA_CALIB, Antenna Pair1,Antenna Pair2) than host should choose ANTENNAS_CONFIGURATION_RX(Antenna Pair3) in case of 2D AoA, in this case Antenna pair3 is not valid pair as it is not calibrated

If there is a mismatch between ANTENNAS_CONFIGURATION_RX and SET_CALIBRATION_CMD, FW will proceed with ranging and the computation of AoA is not correct as the

calibration used vs antenna pair selected for ranging is not a match which will lead to incorrect AoA/PDoA results.

UWBS/FW cannot enforce a check to reject this mismatch b/w ANTENNAS_CONFIGURATION_RX and SET_CALIBRATION_CMD as in case of PDoA calibration process to fetch the raw PDoA to generate the PDOA calibration table(i.e AOA_ANTENNAS_PDOA_CALIB), test will not set any PDOA Offset calibration

Table 106. Extended Calibration Parameters

Parameter Name	Len (Octets)	Tag(Ext ID's)	Sub ID's	Description
AOA_ANTENNAS_PDOA_CALIB_EXTENDED_SUPPORT	1 + N*(1 + S*S*2) Octets	0xE0	0x00	This parameter is used to set the PDoA Calibration Table Octet[0] : No of entries, it shall be set 1 for Set Calibration Cmd. Octet[1] : RX Pair Antennas ID as defined earlier by ANTENNAS_RA_PAIR_DEFINE Octet[2 - (S * S* 2)] : PDoA calibration table for specified RX Pair Antennas ID. S will have value of Octet[1] from PDOA_CALIB_TABLE_DEFINE. Allowed Range: 3<= S <= 17 (to include 0 ⁰ (default : 11)

Table 106. Extended Calibration Parameters

Parameter Name	Len (Octets)	Tag(Ext ID's)	Sub ID's	Description
				Refer the legacy 'AOA_ANTENNAS_PDOA_CALIB' parameter description for more details. Note: The Get Calibration Cmd response for this ID returns the total number of "RX Pair Antennas IDs" for which PDOA calibration table is set in Octet[0]. N = 1 Always Note: Refer Recommendation while using Calibration Parameters

Table 107. Manual Tx Power Control parameter

Parameters	Size (Octet)	Description
TX_PA_GAIN	1	Assigns PA gain value to be used for Tx operation (Default: 0x00) • Bit[2:0]: TX_PA_GAIN_SET • Bit[7:3]: RFU
TX_PPA_GAIN	1	Assigns PA drive gain value to be used for Tx operation (Default: 0x00) • Bit[1:0]: TX_PPA_GAIN_SET • Bit[7:2]: RFU
TX_DIG_GAIN	1	Assigns digital gain value to be used for Tx operation (Default: 0x00) • Bit[7:0]: TX_TRIM_PRMBLGAIN_SET
DAC_GAIN	1	Assigns Tx DAC gain value (Default: 0x00) • Bit[1:0]: DAC_GAIN_SET • Bit[7:2]: RFU
TX_MIX_GAIN	1	Assigns Tx Mix gain value (Default: 0x00) • Bit[1:0]: TX_MIX_GAIN_SET • Bit[7:2]: RFU

11 Appendix A

11.1 SR250 Device Information Parameters

AP shall use CORE_GET_DEVICE_INFO_CMD command to retrieve the device information parameters.

Table 108. Device Information Parameters

Parameter Name	Len (Octets)	Tag (IDs)	Description
DEVICE_NAME	N	0x00	UWBS returns a string to indicate the IC version <i>"SR200T_PROD" => SR200T A0</i> <i>"SR200T_A1_PROD" => SR200T A1V1</i> <i>"SR200T_A1V2_PROD" => SR200T A1V2</i> <i>"SR200S_A2_PROD" => SR200S A2</i> <i>"SR200D_A2_PROD" => SR200D A2</i> Note: Read Only parameter
FIRMWARE_VERSION	3	0x01	Octet [0]: Major number Octet [1]: Minor number Octet [2]: RC version <i>Example: Major number 1, Minor number 2 and Patch version 3 are reported as 0x01 0x02 0x03 at UCI transport</i>
VENDOR_UCI_VERSION	3	0x02	NXP Platform UCI version Octet [0]: Major Version Octet [1]: Minor Version Octet [2]: Patch version, This octet also indicates the changes done for any specific customer. Note: The NXP Platform UCI version is encoded in packed binary-coded decimal (BCD). e.g., UCI spec 01.10.20 (decimal) would be reported as major 0x01, minor 0x10 and patch version 0x20.
UWB_CHIP_ID	16	0x03	Chip ID information
UWBS_MAX_PPM_VALUE	1	0x04	UWBS Max PPM value [100]
TX_POWER	2	0x05	TX Power value set using the SET_DEVICE_CALIBRATION_CMD Signed Q9.7 format in dB (e.g. 1.50dB)
UWBS_CAL_MODE	4	0x06	SR250 Chip LifeCycle: 0x55555555: Protected mode 0xA5A5A5A5: Customer mode
FIRA_EXT_UCI_GENERIC_VERSION	3	0x60	FiRa UCI Generic Spec version implemented by UWBS FW: Octet[0] = Major Version; Octet[1] = Minor/Maintenance Version; - bit[7:4] = Minor version;

Table 108. Device Information Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
			- bit[3:0] = Maintenance version; Octet[2] = Patch version;
FIRA_EXT_UCI_TEST_VERSION	3	0x61	FiRa UCI Test Spec version implemented by UWBS FW: Octet[0] = Major Version; Octet[1] = Minor/Maintenance Version ; - bit[7:4] = Minor version; - bit[3:0] = Maintenance version; Octet[2] = Patch version;
UWBS_FW_GIT_HASH	17	0x62	Octet[15-0] = Git hash (coded in ASCII) Octet[16] = equal to '-'
FW_BOOT_MODE	1	0x63	0x01: User Mode
UCI_CCC_VERSION	2	0xA0	Version of the UCI CCC Specification followed.
CCC_VERSION	8	0xA1	Version of the CCC Specification followed. Octet [0]..Octet[7]: CCC version written in ASCII format
ALIRO_SPEC_VERSION	8	0xA2	Version of the Aliro Specification followed. Octet [0]..Octet[7]: Aliro specification version written in ASCII format

11.2 Device Capability Parameters

AP shall use CORE_GET_CAPS_INFO_CMD command to retrieve the device capability parameters.

Table 109. Device Capability Parameters

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY (Sub IDs)	Description
UWBS_MAX_UCI_PAYLOAD_LENGTH	2	0xE3	NA	Parameter to indicate the maximum UCI payload size can handle by UWBS Value = 2048
UWBS_INBAND_DATA_BUFFER_BLOCK_SIZE	1	0xE3	0x01	Parameter to indicate the block size in bytes which the UWBS manages for the overall in-band data transfer memory pool. <i>Note: UWBS shall report the most recent capability value as value may get vary on every release</i>
UWBS_INBAND_DATA_MAX_BLOCKS	1	0xE3	0x02	Parameter to indicate the number of blocks available in the overall in-band data transfer memory pool. <i>Note: UWBS shall report the most recent capability value as value may get vary on every release</i>

11.3 SR250 Device Configuration Parameters

AP shall use CORE_SET_CONFIG_CMD and CORE_GET_CONFIG_CMD to set and retrieve the device core configurations.

Table 110. Device Configuration Parameters

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
WIFI_COEX_FEATURE	2 + N*4	NA	0xF0	This parameter is used to configure the WiFi CoEx feature. <u>Octet[0]: Enable/Disable WiFi CoEx feature</u> 0x00: Disable (default) • b[3:0]: Enable/Disable functionality CoEx – 0x1 : Enable CoEx Interface without Debug and without Warning Verbose – 0x2 : Enable CoEx Interface with Debug Verbose only – 0x3 : Enable CoEx Interface with Warnings Verbose only – 0x4 : Enable CoEx Interface with both Debug and Warning Verbose • b[7:4]: CoEx Interface (GPIO/OneWire) selection: – 0x2 : OneWire Interface – Rest of the values are Reserved <u>Octet[1]: Number of channels N Shall be >= 1 (0 will be rejected by UWBS)</u> <u>N*4 octets to follow</u> <u>Octet[2]: Channel ID</u> <u>Octet[3]: MIN_GUARD_DURATION</u> Minimum gap in milliseconds between the end of previous active grant duration and next medium access request RFU for OneWire interface <u>Octet[4]: MAX_GRANT_DURATION / MAX_WIFI_BLOCK_DURATION (for OneWire CoEx)</u> Maximum duration for which the UWB can request for medium access <u>Octet[5]: ADVANCED GRANT DURATION / GUARD DURATION (for OneWire CoEx)</u> Minimum preparation time in milliseconds required by the WLAN subsystem to relinquish the CoEx Medium for UWB subsystem usage. It shall be set to at least 6ms for GPIO based CoEx Interface.

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				Ranging interval value in ms should be greater or equal to following expression: (ADVANCED GRANT DURATION + ROUND DURATION + MIN_GUARD_DURATION) Note: 1. No default values for Octet[1] to Octet[1 + N*4] 2. Number of channel should be greater than zero else UWBS will reject config. 3. WIFI_COEX_FEATURE - Duplicate config for same channel should not present else UWBS will reject config. 4. WIFI_COEX_FEATURE - Number of channel should not exceed more than supported channel else UWBS will reject config.
...				Note: • Values for Advance Grant Duration and Minimum Guard Duration are only applicable for GPIO based CoEx interface.
DPD_WAKEUP_SRC	1	0xE4	0x02	Refer : • DPD wakeup sources (Default: 0)
DPD_ENTRY_TIMEOUT	2	0xE4	0x04	DPD entry timeout in ms where UWBS shall defer DPD attempt for configured timeout. This parameter is applicable only when UWBS is woken up due to any activity at • Host Interface. [20-2000] (Default: 500ms) <i>Note: DPD feature shall be disabled via LOW_POWER_MODE generic core device config</i>
GPIO_USAGE_CONFIG	3	0xE4	0x08	• Octet[0]: GPIO to be used. Default: 0x00 (Feature Disabled) • Octet[1]: GPIO Trigger Configuration Default: 0x00 (Feature Disabled) Value: 0x01: Rising (Positive) Edge Trigger. Rest all values are RFU. (BLE needs to toggle the GPIOxx from low to high and then high to low)

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				- Octet[2]: Use Case ID Default: 0x00 (Feature Disabled) Value: 0x01 : Capture time stamp for GPIOxx DPD_WAKEUP_SOURCE interrupt based on GPIO trigger configuration. (For Use Case ID: 1 DPD_WAKEUP_SOURCE needs to be set by Host to GPIOxx otherwise GPIO_USAGE_CONFIG will be rejected by FW). Rest all values are RFU. Note: - GPIO trigger configuration is only applicable when the device is in active state. - Device wakeup event through GPIO1 require more than 1 LFO clock duration (5 microseconds), therefore device wakeup from DPD due to GPIO1 toggle is level triggered in case of DPD. - For H1 only, Device will not enter DPD if GPIO1 is held asserted (high). - Minimum pulse width for wakeup using GPIO1 is 10 microseconds (PCRM design needs 2 LFO clock duration to latch all the wakeup events). - GPIOx ISR latency may not be guaranteed due to atomic regions in FW where interrupts could be disabled. - For ROW Customers other than Fit Bit, GPIO_USAGE_CONFIG shall be default value only. Default: 0 (RFU).
TX_BASE_BAND_CONFIG	1	0xE4	0x26	Allows to Enable/Disable Tx base band configurations - b[0]: Enable/Disable DDFS tone generation after Tx frame 0: Disable (default) 1: Enable - b[7:1]: RFU
DDFS_TONE_CONFIG	18*4	0xE4	0x27	Direct Digital Frequency Synthesizer (DDFS) Tone configuration. 18 bytes value description: Octet[0]: Channel Number Octet[1]: Tx antenna selection. Possible values are 1 or 2. Octet[5:2]:

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				- Content of the register TX_DDFS_TONE_0_INC0_SET. 11 bits [0:2047] Increment of phase accumulator for frequency control (frequency = INC * 975 kHz). Octet[9:6]: Content of the register TX_DDFS_TONE_1_IN1_SET 11 bits [0:2047] Increment of phase accumulator for frequency control (frequency = INC * 975 KHz). Octet[13:10]: Duration of the spur, in 124.8 MHz resolution (~ 8 ns) Octet[14]: Content of register GAINVAL_SET Octet[15]: Content of register DDFS_GAINBYPASS_ENBL Octet[17:16]: - Amount of time (x8ns) between the frame and a DDFS burst. 4 Blocks: 18 Octets repeated for each block Octets[17:0] correspond to Block1 Octets[35:18] correspond to Block2 Octets[53:36] correspond to Block3 Octets[71:54] correspond to Block4
TX_PULSE_SHAPE_CONFIG	4	0xE4	0x28	Octet[0]: Preamble Tx pulse shape ID Octet[1]: Payload Tx pulse shape ID Note: If DC suppression is enabled via TX_BASE_BAND_CONFIG then pulse shape ID 8 will be applied if preamble ID 10 is used. Octet[2]: STS Tx pulse shape ID Octet[3]: RFU Values: Octet[0:2] : - [50(default), 56] Octet[3] : 0

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
HOST_MAX_UCI_PAYLOAD_LENGTH	2	0xE4	0x31	Defines host capability of handling max UCI payload. UWBS shall use this parameter to send the UCI response / notification to host. FW shall use PBF bit if UCI payload goes more than HOST_MAX_UCI_PAYLOAD_LENGTH size. 255 <= HOST_MAX_UCI_PAYLOAD_LENGTH <= UWBS_MAX_UCI_PAYLOAD_LENGTH (capability parameter) (Default: 255) <i>Note: UWBS shall construct the UCI packet based on this parameter.</i>
NXP_EXTENDED_NTF_CONFIG	1	0xE4	0x33	0x00 : FIRA generic Response / Notification (Default) 0x01 : Vendor extended Response / Notification. <i>Note: UWBS shall send response/notification as per "FiRa Consortium - UCI Generic Specification" if set to 0x00</i>
INITIAL_RX_ON_OFFSET_ABS	2	0xE4	0x35	Defines the negative offset when Rx should be enabled to receive the first message of a ranging round on Controlee compared to expected reception time. The time is given in microseconds and has to be combined with INITIAL_RX_ON_OFFSET_REL, i.e. actual offset time is INITIAL_RX_ON_OFFSET_ABS + INITIAL_RX_ON_OFFSET_REL * RANGING_INTERVAL (Default: 100us)
INITIAL_RX_ON_OFFSET_REL	2	0xE4	0x36	Defines the negative offset when Rx should be enabled to receive the first message of a ranging round on Controlee compared to expected reception time. The time is given in ppm relative to RANGING_INTERVAL and has to be combined with INITIAL_RX_ON_OFFSET_ABS, i.e. actual offset time is INITIAL_RX_ON_OFFSET_ABS + INITIAL_RX_ON_OFFSET_REL * RANGING_INTERVAL. (Default: 100ppm) Example: INITIAL_RX_ON_OFFSET_ABS = 100 (us) INITIAL_RX_ON_OFFSET_REL = 100 (ppm) RANGING_INTERVAL = 100 (ms) Actual Offset: 100us + 100ppm*100ms = 100us + 10us = 110us
CLK_CONFIG_CTRL *** Stored in SR250's Flash ***	5	0xE4	0x3A	<u>Octet[0] : INPUT_CLK_CTRL</u> • bit[0]: 32kHz clock source – 0: Use on board crystal (default)

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
*** If never set, CORE_GET_CONFIG will return a status failed ***				– 1: Use external 32k clock • bit[2:1]: External 32k clock waveform – 0b00: Sinus (default) – 0b01: 1.8V square – 0b10: 1.2V square • bit[3]: RF clock source – 0: Use on board crystal (default) – 1: Use external clock • bit[5:4]: RF clock frequency – 0b00: Use 38.4MHz – 0b01: Use 26MHz – 0b10: Use 19.2MHz • bit[6]: External clock waveform – 0: Sinus (default) – 1: Square • bit[7]: RFU <u>Octet[2:1] : SLOW_CLK_WAITING_TIME</u> Maximum waiting time in us until 32.768 kHz clock is present. Default: 1000us. <u>Octet[4:3] : RF_CLK_WAITING_TIME</u> Maximum waiting time in us until RF clock is present. Default: 1000us. Refer Clock Settings section for more information Note: Once we set CLK_CONFIG_CTRL we need to do CORE_DEVICE_RESET_CMD.
DDFS_CONFIG_PER_PULSE_SHAPE	4	0xE4	0x3B	Allow to override the TX_BASEBAND_CONFIG parameter for specific pulse shapes. Octet[0]: DDFS config for FiRa PS Octet[1]: DDFS config for CCC0 PS Octet[2]: DDFS config for CCC1 PS

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				Octet[3]: DDFS config for CCC2 PS (Default: 0x00 00 00 00 for Octet[0:3]) <u>Content of each byte:</u> • bit[0]: override DDFS config – 0: No override – 1: TX_BASEBAND_CONFIG is overridden for this pulse shape (DDFS config will depends on bit[4:1]) • bit[1]: DDFS config for SP0 frame – 0: Disabled – 1: Enabled • bit[2]: DDFS config for SP1 frame – 0: Disabled – 1: Enabled • bit[3]: RFU • bit[4]: DDFS config for SP3 frame – 0: Disabled – 1: Enabled • bit[7:5] RFU
AOA_MODE	1	0xE4	0x3C	Used to define the type of computation to be considered by UWBS for obtaining AoA. <u>Octet[0]: AoA mode</u> 0x01: AOA_COARSE 0x02: AOA_3D (Default: 0x01)
RFU		0xE4	0x3D - 0x3F	Reserved for Future Use
PDOA_CALIB_TABLE_DEFINE	2	0xE4	0x46	PdoA Calibration Table Definition :

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				<u>Octet[0] – Calibration Step Size</u> : The calibration table step size in degrees which indicates the step size between two consecutive points. Allowed Range : 10°, 12°(default) and 15° <u>Octet [1] – Number of Steps</u>: The number of calibration steps needs to be an odd number. Allowed Range : 3<=M<=17(to include 0°)(default : 11) Note: For steps greater than 17, 'AOA_ANTENNAS_PDOA_CALIB_EXTENDED_SUPPORT' calibration parameter should be used. Example : With number of step size as 12° and number of steps as 11, the achieved calibration span is -60, -48, -36, -24, -12, 0, 12, 24, 36, 48, 60. Note: - Total Calibration Span = ((Number of Steps – 1) * Step Size). Example : If steps are 11 and step size is 12° then total span is 120°(-60° to 60° including 0°). - Same configuration will be applied by UWBS for all supported channels - The command should be configured before setting any PdoA/AoA Calibration Parameters such as – AOA_ANTENNAS_PDOA_CALIB – AOA_MULTIPPOINT_PDOA_CALIB – AOA_ANTENNAS_MULTIPPOINT_CALIB – AOA_ANTENNAS_PDOA_CALIB_EXTENDED_SUPPORT
ANTENNA_RX_IDX_DEFINE (GPIO Configuration)	1 + N*6	0xE4	0x60	Define RX antenna Identifier - Octet[0] : Number of Entries. (N). This must be between 1 and "MAX_N". - Array of entries: [– Octet[X + 0] : RX Antennas ID. Index of the Antenna (0 is invalid). Value shall be between 1 to MAX_N – Octet[X + 1] : Receiver used. 0x02: RXB port

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				0x03: RXA2 port 0x04: RXA1 port Value 0 shall be Invalid. – Octet[X + 3, X + 2]: GPIO Filter Mask This mask defines which GPIOs shall be changed during the state transition, if a GPIO bit is set to 0 the GPIO shall not change its state. It's a 2 byte value in Little Endian format. – Octet[X + 5 X + 4]: GPIO State / Value This mask defines the GPIO state, if the corresponding GPIO bit is 0 in the GPIO Filter mask, the state shall be ignored and not changed. It's a 2 byte value in Little Endian format.] See ANTENNAS_SELECTION_GPIO_BIT_MASK for bit masks and values. See Table 55 for max value of MAX_N
ANTENNAS_RX_PAIR_DEFINE	1 + N * 6	0xE4	0x62	Define identifiers for each RX antenna pairs.
...				Define identifiers for each Rx antenna pairs (This configuration is applicable for AoA Rx modes only) <u>Octet[0] : N Entries</u> Must be between 1 and "MAX_N". [• Octet[X+0] : Antennas PAIR ID Must start from 1 • Octet[X+1] : Antenna ID 1 connected to RXC • Octet[X+2] : Antenna ID 2 connected on RXB • Octet[X+3] : Antenna ID 3 connected on RXA • Octets[X+5:X+4] : RFU: Set to "0"] (for each antenna pair ID) Antenna ID value must be aligned with as defined by ANTENNA_RX_IDX_DEFINE. One of the Antenna ID must be equal to 0 (only 2 antenna must be set per entry).

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
...				See Table 55 for max value of MAX_N
ANTENNA_TX_IDX_DEFINE	1 + N * 6	0xE4	0x63	Define TX antenna Identifiers Octet[0] : Number of Entries (N) Must be between 1 and "MAX_N". [- Octet[X+0] : TX Antenna ID - Must start from 1 - Octet[X+1] : TX Antenna port 0x1: TRA1 0x2: TRA2 - Octet[X+3:X+2] : GPIO Filter Mask - If a bit is set to 1, it enables the associated GPIO (2 byte value in Little Endian format) - Octet[X+5:X+4] : GPIO Value - If corresponding bit in filter mask is set, it defines the GPIO state. - If corresponding bit in filter mask is not set, it's ignored (2 byte value in Little Endian format).](for each antenna TX ID) See ANTENNAS_SELECTION_GPIO_BIT_MASK for bit masks and values. See Table 55 for max value of MAX_N
DEFAULT_SESSION_VENDOR_APP_CONFIG	Max 251 bytes	0xE4	0x65	The format will be exactly same as SET_VENDOR_APP_CONFIG as detailed below Refer DEFAULT_SESSION_VENDOR_APP_CONFIG Refer : Recommendation for race condition handling b/w device and session configs Note : All error handling w.r.t vendor sessions configs provided as part of DEFAULT_SESSION_VENDOR_APP_CONFIG shall be performed when CORE_SET_CONFIG(DEFAULT_SESSION_VENDOR_APP_CONFIG) is received by FW,

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				- FW will report CORE_SET_CONFIG_RSP(UCI_STATUS_SUCCESS) if all the vendor session parameters are valid, DEFAULT_SESSION_VENDOR_APP_CONFIG configs are retained in FW and applied on every SESSION_INIT - FW will report CORE_SET_CONFIG_RSP(FAILED) with the same format of as of SET_VENDOR_APP_CONFIG_RSP if any of the vendor Vendor APP TLV validation fails, DEFAULT_SESSION_VENDOR_APP_CONFIG configs are NOT retained in FW - These configs received via "DEFAULT_SESSION_VENDOR_APP_CONFIG" will be applied to all session's irrespective of session type. - UWBS has a constraint of nested TLV payload for DEFAULT_SESSION_VENDOR_APP_CONFIG as not to exceed 250 bytes. - Incase DEFAULT_SESSION_VENDOR_APP_CONFIG is the only TLV in CORE_SET_CONFIG that has failed and leading to syntac error then FW will reject/not apply any of the nested TLV of DEFAULT_SESSION_VENDOR_APP_CONFIG and report CORE_SET_CONFIG_RSP(FAILED) as below. - Status - FAILED =0x02 - Number of params set to 0x01, - Param Tag = 0xE465 - Status of parameter failure = 0x05- Status Invalid parameter - When 0xE465 = DEFAULT_SESSION_VEDNOR_APP_CONFIG tag is included as part of CORE_SET_CONFIG, in this scenario if any other core config tag apart from DEFAULT_SESSION_VEDNOR_APP_CONFIG tag fails, then UWBS will not process or apply any of the vendor app configuration specified as part of DEFAULT_SESSION_VEDNOR_APP_CONFIG i.e UWBS will reject DEFAULT_SESSION_VEDNOR_APP_CONFIG tag and hence this tag(DEFAULT_SESSION_VEDNOR_APP_CONFIG) will be included in response with status PH_UCI_STATUS_REJECTED.
UWBS_SLEEP_ENTRY_PREVENTION_NTF		0xE4	0x3D	Defines how long the UWBS shall wait before sending DPD_ENTRY_PREVENTION_NTF to host , this is also the time interval between two subsequent NTF to host, unit is 100ms (default = 1 seconds) 0x00 - DPD_ENTRY_PREVENTION_NTF is disabled

Table 110. Device Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	PTY sub ID	Description
				0x01 - One time NTF enabled (see previous slide for details) 0x02 to 0xFF – Periodic NTFs enabled Note: When LOW_POWER_MODE is disabled UWBS_SLEEP_ENTRY_PREVENTION_NTF config will be ignored, no DPD_ENTRY_PREVENTION_NTF will be sent.
Reserved for Debug configurations				

Table 111. DEFAULT_SESSION_VENDOR_APP_CONFIG

Payload Index	Payload Fields	Length	Description
Octet[0]	Number of Vendor App Configurations	1	The number (n) of APP Configuration Parameters (TLV) to follow. Each APP Configuration Parameter follows TLV structure as below.
Octet [1 to (2+m)*n]	Vendor Session Configuration Parameter [1...n] when Configuration Type = Vendor Session Config	(2+m)*n Octets	- Tag : 1 Octet Vendor APP configuration Parameter ID - Length : 1 Octet Length of Value(m) - Value : m Octets Value of the Vendor APP Configuration parameter

Example UCI cmd/rsp for DEFAULT_SESSION_VENDOR_APP_CONFIG to set ANTENNA_CONFIGURATION_RX :

Existing SET_VENDOR_APP_CONFIG_CMD

2f:00:00:0B:01:00:00:00:01:03:04:01:02:01:02

Parsed: {MT: CMD(0x1)}{PBF: 0x0}{GID: INTERNAL_USE(0xf)}{EXT_PL_LEN: 0x0}{RFU1: 0x0}{OID: SESSION_SET_VENDOR_APP_CONFIG(0x0)}{RFU2: 0x0}{PL_LEN: 0x13}{SESSION_ID: 0x1}{NUM_OF_PARAMS: 0x1}

{[ANTENNAS_CONFIGURATION_RX](#)(0x3): 01020102}

CORE_SET_CONFIG CMD encapsulating Vendor APP config

20:04:00:XX:

01= Num of Params for CORE_SET_CONFIG(in this example =1)

XX= Tag for CORE_SET_CONFIG(in this example tag value of DEFAULT_SESSION_VENDOR_APP_CONFIG)

04= Length of the CORE_SET_CONFIG TLV, Total length of DEFAULT_SESSION_VENDOR_APP_CONFIG

Value for CORE_SET_CONFIG = Following bytes

01= Num of Params for VENDOR_APP_CONFIG

03: Tag of VENDOR_APP_CONFIG = [ANTENNAS_CONFIGURATION_RX](#)=0x03 as per existing UCI

04: Length of VENDOR_APP_CONFIG, len of [ANTENNAS_CONFIGURATION_RX](#) as per existing UCI

01:02:01:02 Value of Vendor APP config(in this case value for [ANTENNAS_CONFIGURATION_RX](#) as per existing UCI)

Recommendation for race condition handling b/w device and session configs :

Scenario

1. CORE_SET_CONFIG(TAG=DEFAULT_SESSION_VENDOR_APP_CONFIG[vendor-app=ANTENNA_CONFIG_RX], ANTENNA_RX_IDX_DEFINE)

- 1a. ANTENNA_RX_IDX_DEFINE will be validated only for the received input payload
- 1b. ANTENNA_CONFIG_RX will be validated against already set/default ANTENNA_RX_IDX_DEFINE

In such scenario FW cannot ensure validation of latest core config(ANTENNA_RX_IDX_DEFINE) vs session/app config(ANTENNA_CONFIG_RX)

To overcome such issue we recommend below 2 sequence from host/MW,

Recommendation-1 Ensure DEFAULT_SESSION_VENDOR_APP_CONFIG is always the last TLV of core_set_config

CORE_SET_CONFIG(ANTENNA_RX_IDX_DEFINE, TAG=DEFAULT_SESSION_VENDOR_APP_CONFIG[vendor-app=ANTENNA_CONFIG_RX])

- 1a. ANTENNA_RX_IDX_DEFINE will be validated for the received input payload
- 1b. ANTENNA_CONFIG_RX will be validated against latest received ANTENNA_RX_IDX_DEFINE from step 1a.

ANTENNA_CONFIG_RX validation considers updated ANTENNA_RX_IDX_DEFINE, no issue

Recommendation-2- Ensure DEFAULT_SESSION_VENDOR_APP_CONFIG is always sent after all core configs are updated via a separate core set config cmd

CORE_SET_CONFIG(ANTENNA_RX_IDX_DEFINE, TLV2, etc)

CORE_SET_CONFIG(TAG=DEFAULT_SESSION_VENDOR_APP_CONFIG[vendor-app=ANTENNA_CONFIG_RX])

ANTENNA_CONFIG_RX validation considers updated ANTENNA_RX_IDX_DEFINE, no issue

11.4 Vendor Application Configurations

AP shall use SET_VENDOR_APP_CONFIG_CMD and GET_NXP_VENDOR_CONFIG_CMD to set and retrieve the APP configurations

Table 112. Vendor App Configuration Parameters

Parameter Name	Len (Octets)	Tag (IDs)	Description
MAC_PAYLOAD_ENCRYPTION	1	0x00	This parameter shall enable/disable encryption of Payload data 0x00 - Plain Text 0x01 - Encrypted(default)
ANTENNAS_CONFIGURATION_TX (*) Must not be updated during Radar session	1+N	0x02	The antenna used for TX. Octet[0] - Define number of TX Antennas to follow (default value is 1). Octet[1] - Tx Antennas ID as defined by ANTENNA_TX_DEFINE (default value is 1). So we transmit by default with Antennas ID 1 (As a pre-requisite: an antenna with ID 1 using ANTENNA_TX_IDX_DEFINE must be explicitly pre-defined). Octet[2] - Tx Antennas ID as defined by ANTENNA_TX_DEFINE
ANTENNAS_CONFIGURATION_RX (*)	1 + 1 + N	0x03	The session specific antenna configuration for Rx • <u>Octet [0] : Mode of RX operation</u> – 0 : Configuration ToA Mode – ToF Only Mode

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
			– Recommended for TEST_PER, TEST_RX, TEST_SSTWR test modes, Optionally TEST_LOOPBACK test modes (AoA may not be reported) – 1 : Configuration AoA Mode – Dual / Single AoA usecase – ToA Mode with implicit Rx mode as per ANTENNAS_RX_PAIR_DEFINE – Recommended for TEST_LOOPBACK test mode, Optionally TEST_PER, TEST_RX, TEST_SSTWR test modes – 2 : Configuration Mode 2: Radar Mode – 3 : Configuration Mode 3: ToA usecase using different Rx antenna pair for RFM • <u>Octet [1]</u> : Number of Antennas or Antenna pairs to follow Default Value: • For all Sessions - Octet[0] : 0x00 = ToA Mode - Octet[1] : 0x01 = Number fo RX Antenna Note: • <i>For H2 Rx port Mapping</i> – RX1 = RXC – RX2 = RXB – RX3 = RXA
If session is for Single RX: • Octet[1] is equal to 1 • Octet[2] is equal to Antenna ID Dual RX: • Octet[1] is equal to 2 • Octet[2] is equal to Antenna ID applicable to RX1 or RX2 • Octet[3] is equal to Antenna ID applicable to RX2 Triple RX:	Configuration Mode 0		Mode(s): • ToF configuration mode • Recommended for TEST_PER, TEST_RX, TEST_SSTWR test modes, Optionally TEST_LOOPBACK test modes (AoA may not be reported) , number of Octets should be based on the size field • Octet[2] : Antenna ID - <i>Any Rx (mandatory)</i> – Antenna ID as defined by ANTENNA_RX_IDX_DEFINE

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
- Octet[1] is equal to 3 - Octet[2] is equal to Antenna ID applicable to RX1 - Octet[3] is equal to Antenna ID applicable to RX2 - Octet[4] is equal to Antenna ID applicable to RX3			– Default: 0x01 • Octet[3] : Antenna ID - <i>Any Rx (optional)</i> – Antenna ID as defined by ANTENNA_RX_IDX_DEFINE • Octet[4] : Antenna ID - <i>Any Rx (optional)</i> – Antenna ID as defined by ANTENNA_RX_IDX_DEFINE
	Configuration Mode 1		• Configuration Mode for 3D / 2D AoA, • Recommended for TEST_LOOPBACK test mode, Optionally TEST_PER, TEST_RX, TEST_SSTWR test modes • Octet [2] : Antennar Pair 1 (Generally Horizontal Pair) – Antennas Pair ID as defined by ANTENNAS_RX_PAIR_DEFINE – Appropriate value shall be set if AoA azimuth result is requested • Octet [3] : Antennar Pair 2 (Generally Vertical Pair) – Antennas Pair ID as defined by ANTENNAS_RX_PAIR_DEFINE – Appropriate value shall be set if AoA elevation result is requested
	Configuration Mode 2		Radar only configuration mode, number of Octets should be based on the size field • Octet[2] : Antenna ID – RXC – Antenna ID as defined by ANTENNA_RX_IDX_DEFINE – May be 0 if RXC is not used – Note : Must be equal to 0 on SR200S and SR200D for RXC – • Octet[3] : Antenna ID – RXB – Antenna ID as defined by ANTENNA_RX_IDX_DEFINE – May be 0 if RXB is not used – • Octet[4] : Antenna ID – RXA – Antenna ID as defined by ANTENNA_RX_IDX_DEFINE – May be 0 if RXA is not used Both RXC and RXB cannot be 0

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
	Configuration Mode 3		Configuration Mode for ToA which will switch RX Antenna during RFM. This configuration mode applies only to DS-TWR responder. • Octet[2] : RX antenna ID 1 for ranging • Octet[3] : RX antenna ID 2 for ranging • Octet[4] : Rx antenna ID 1 for RFM (used for distance measurement during RFM) • Octet[5] : Rx antenna ID 2 for RFM (used for distance measurement during RFM)
	Configuration Mode 5		ToA mode for CSA Octet[1] : Number of antennas to follow for both active RRs. If Octet[1] : 2 Octet[2] : ANT_RX_ID_1 (RR1) Octet[3] : ANT_RX_ID_2 (RR2) If Octet[1] : 4 Octet[2] : ANT_RX_ID_1 (RR1) Octet[3] : ANT_RX_ID_2 (RR1) Octet[4] : ANT_RX_ID_3 (RR2) Octet[5] : ANT_RX_ID_4 (RR2) This configuration shall be only applicable for the responder side.
	Configuration Mode 6		AoA mode for CSA Octet[1] : Number of antenna pairs to follow for both active RRs. If Octet[1] : 2 Octet[2] : ant_rx_pair1 (RR1) Octet[3] : ant_rx_pair2 (RR2) If Octet[1] : 4 Octet[2] : ant_rx_pair1 (RR1) Octet[3] : ant_rx_pair2 (RR1) Octet[4] : ant_rx_pair3 (RR2) Octet[5] : ant_rx_pair4 (RR2) This configuration shall be only applicable for the responder side.

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
RAN_MULTIPLIER (read access only)	1	0x20	Return the possible RAN multiplier value for a new session
STS_LAST_INDEX_USED (read access only)	1	0x21	Parameter used to get the STS index of the UWB session. When GET_VENDOR_APP_CONFIG_CMD issued for this config during SESSION_STATE_ACTIVE the UWBS shall return the last
CIR_LOG_NTF	1	0x30	0x00: Disable (default) 0x01: Enable Note: CIR_LOG_NTF is not supported/rejected when any sessions are active
PSDU_LOG_NTF	1	0x31	0x00: Disable (default) 0x01: Enable
RSSI_AVG_FILT_CNT	4	0x40	This parameter is used to filter out the outliers in RSSI measurements in PER RX Test. If the RSSI filtering count is set to N and total packet count is set to M then UWBS shall report the average of (M-2N) RSSI values in TEST_PER_RX_NTF excluding(N) maximum and (N) minimum RSSI values. Note: M is the total packet count to be received in PER Rx test (default = 0)
CIR_CAPTURE_MODE	2	0x60	CIR sampling position for incoming UWB packet Octet[0] (default is 0x11) bits[7:4] - CIR1 capture mode bits[3:0] - CIR0 capture mode Octet[1] (default is 0x01) bits[7:4] - RFU bits[3:0] - CIR2 capture mode SYNC mode can be enabled on only 1 of the 3 CIRs capture mode fields; CIR capture modes: 0x0 - SYNC 0x1 - STS 0x2 - Both SYNC and STS

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
			0x3 - 0xF - RFU
RX_ANTENNA_POLARIZATION_OPTION	1	0x61	This parameter is used to choose the antenna polarization option when AOA measurement is enabled (AOA_RESULT_REQ = 1). This parameter is not applicable when AOA_RESULT_REQ = 0. bit[0]: Polarization option to be used for the first AoA computation [0]: Straight (Default) [1]: Reverse bit[1]: Polarization option to be used for the second AoA computation when enabled either via DUAL_AOA_PREAMBLE_STS or NUMBER_OF_STS_SEGMENTS = 2. The value is not applicable when second AoA computation is not enabled [0]: Straight (Default) [1]: Reverse bit[7:2]: RFU
SESSION_SYNC_ATTEMPTS	1	0x62	Number of times scheduler shall attempt to sync in controlee session before reporting error notification. This config is applicable for controlee session only. Range: [3 : 255] (Default: 3)
SESSION_SHED_ATTEMPTS	1	0x63	Number of times scheduler shall attempt to schedule ranging round before reporting error notification Range: [1 : 255] (Default: 3)
SCHED_STATUS_NTF	1	0x64	Enable/disable SCHEDULER_STATUS_NTF 0x00 - Disable (default) 0x01 - Enable for including all the sessions information in notification 0x02 - Enable for include only failure sessions information in notification 0x03-0xFF: RFU
TX_POWER_DELTA_FCC	1	0x65	Session specific Tx power ID offset applied on top of Tx POWER calibration parameter. configured via SET_DEVICE_CALIBRATION_CMD. 0:No offset (default) 1 to 127: Attenuation (0.25 dB per steps) 128 to 255: RFU

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
TX_POWER_TEMP_COMPENSATION	1	0x67	This parameter is used to enable/disable Tx power temperature compensation 0x00: Disable (Default) 0x01: Enable
ADAPTIVE_HOPPING_THRESHOLD	1	0x69	This parameter can be used to configure the required number of successful responses(T) from Responders to conclude a successful ranging round. If numbers of responses is less than this given threshold(T) when Initiator device acting as Controller then initiator device triggers a hop to a different round index within the next block. Range: [0<T<= NUMBER_OF_CONTROLEES] (Default: NUMBER_OF_CONTROLEES) <i>Note: This parameter is applicable when HOPPING_MODE = 0xA0 (NXP Adaptive Hopping mode is Enabled)</i>
CONTENTION_PHASE_UPDATE_LENGTH	1	0x6D	This parameter is used to set contention phase update value in number of slots. Contention Phase is increased / decreased with integer multiple of this parameter. Minimum Contention Phase is 2 X CONTENTION_PHASE_UPDATE_LENGTH (Default: 5)
AUTHENTICITY_TAG	1	0x6E	Config to enable/disable authenticity tag in SESSION_INFO_NTF. This config is applicable when STS_CONFIG is set to Dynamic STS. 0x00 = Disable (Default) 0x01 = Enable
RX_NBIC_CONFIG	2	0x6F	Octet [0] : Used to configure NBIC settings • b[0]: Enable / Disable NBIC. 0 = Disable (default) 1 = Enable • b[1:2]: Content of register MA_FILTER_BW_SET. Filter bandwidth setting (default: 0x3) • b[3:4]: Content of register MA_FILTER_BW_START_SET. Starting filter bandwidth setting for estimation (default: 0x3) • b[5:7]: RFU Octet [1]: Content of register PSD_WEIGHT_SET (Default: 0x14) (Default: 0x40 only applicable when NBIC is enabled)

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
MAC_CFG	1	0x70	Config is used to configure the MAC Header and MAC Footer b[0]: MAC Header present b[1]: MAC Footer present b[7:2]: RFU (Default value: 0x03 for FIRA Session) (Default value: 0x00 for Test Mode Session)
SESSION_INBAND_DATA_TX_BLOCKS	1	0x71	Amount of blocks which should be reserved for the given session for inband data transfer for transmitting data. If set to 0, transmitting inband data is not allowed for this session. Note: The sum of this value for all active session must not exceed UWBS_INBAND_DATA_MAX_BLOCKS and will prevent the first session which is causing to exceed the limit to get started. If the session types are Ranging and in-band data session, Data transfer session, In-band data phase and Ranging-with-data phase, SESSION_INBAND_DATA_TX_BLOCKS/SESSION_INBAND_DATA_RX_BLOCKS is set to 1. If the session types are NOT the Ranging and in-band data session, Data transfer session, In-band data phase and Ranging-with-data phase, SESSION_INBAND_DATA_TX_BLOCKS/SESSION_INBAND_DATA_RX_BLOCKS is set to 0.
SESSION_INBAND_DATA_RX_BLOCKS	1	0x72	Amount of blocks which should be reserved for the given session for inband data transfer for receiving data. If set to 0, receiving inband data is not allowed for this session. Note: The sum of this value for all active session must not exceed UWBS_INBAND_DATA_MAX_BLOCKS and will prevent the first session which is causing to exceed the limit to get started. If the session types are Ranging and in-band data session, Data transfer session, In-band data phase and Ranging-with-data phase, SESSION_INBAND_DATA_TX_BLOCKS/SESSION_INBAND_DATA_RX_BLOCKS is set to 1. If the session types are NOT the Ranging and in-band data session, Data transfer session, In-band data phase and Ranging-with-data phase, SESSION_INBAND_DATA_TX_BLOCKS/SESSION_INBAND_DATA_RX_BLOCKS is set to 0.
RFU		0x78	

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
Wrapped RDS	N	0x79	Session Handle (4 octets) Random (12 octets) Wrapped RDS from SE (Maximum 40 octets)
RFRAME_LOG_NTF	1	0x7B	0x00 : Disable (Default) 0x01 : Enable
TX_ADAPTIVE_PAYLOAD_POWER	1	0x7F	This configuration is used to enable/disable adaptive payload power for TX. 0x00 = Disable 0x01 = Enable (default) Values 0x02 to 0xFF = RFU Note: Default value is equal to 1 except for FIRA Test session (Default value is equal to 0 for FIRA Test Session).
ENABLE_FOV	1	0x84	This parameter decides whether to enable or disable 2D AoA FoV Processing. 0x00 : 2D AoA FoV Processing Disabled (Default) 0x01 : 2D AoA FoV Processing Enabled
AZIMUTH_FIELD_OF_VIEW	2	0x85	This parameter indicates if the peer device is in the configured FoV of the device or not. Octet[0] : Horizontal RX Antenna Pair ID as defined in ' ANTENNAS_RX_PAIR_DEFINE UCI parameter'. Value 0 shall be rejected.(Default : 0x01) Octet[1] : FoV Coverage in degrees.(Default : 60) Note: The maximum FoV range supported is within [-90, 90] degrees.
SESSION_INFO_NTF_FILTER_NUM(*)	1	0x86	Represents a single byte number: 1~255 (default: 1) 0 = Invalid Range 1 = No filtering applied. Host is notified immediately with SESSION_INFO_NTF upon PROXIMITY criteria being met 2~255 = The number of consecutive Ranging Rounds which must meet the PROXIMITY criteria before a Notification is sent to the Host This can be applied to either an Initiator device or Responder device Note: Applicable for SESSION_INFO_NTF_CONFIG = 0x02, 0x05.
ENABLE_FOM	1	0x91	Enable / Disable FoM Processing.

Table 112. Vendor App Configuration Parameters...continued

Parameter Name	Len (Octets)	Tag (IDs)	Description
			Bit [0] – 3D AoA FoM: Enable/Disable 3D AoA FoM Default is 0. Bit [7:1] - RFU Note: When FoM is disabled, both Azimuth FoM and Elevation FoM fields shall be Zero(0).

11.4.1 Proprietary extension for Generic Application Configurations

Below section defines the extended values for FiRa generic Application configuration parameters

Table 113. Extension for Generic Application Configuration

Parameter Name (*): Can be applied during session active state as well	Length (Octets)	Tag (IDs)	Description
Device Type	1	0x00	0x00 = Controlee 0x01 = Controller 0xA0 = CCC Controller 0xA1 = CCC Controlee 0x02-0x09F, 0xA2-0xFF = RFU
SFD_ID	1	0x15	Identifier for SFD sequence Possible values are { 0, 2 and 5 (Proprietary) } for BPRF Mode
PREAMBLE_DURATION	1	0x17	0xA0: RFU 0xA1: 128 symbols 0xA2: 256 symbols 0xA3: 512 symbols
HOPPING_MODE	1	0x2C	This parameter is used to enable/disable the hopping. 0x00 = Hopping Disable 0x01 = Hopping Enable (default = 0x00) CCC Ranging Session 0x00: No Hopping

Table 113. Extension for Generic Application Configuration...continued

Parameter Name (*): Can be applied during session active state as well	Length (Octets)	Tag (IDs)	Description
			0xA0: Continuous Hopping mode with AES-based hopping sequence 0xA1: Continuous hopping mode with default hopping sequence 0xA2: Adaptive hopping mode with AES based hopping sequence. 0xA3: Adaptive Hopping mode with default hopping sequence. 0x02-0x9F = RFU 0xA4-0xFF = Vendor specific modes <i>Note: This config is applicable only for Controller and ignored in case of Controlee</i>
UWB_CONFIG_ID	2	0xA4	UWB Configuration ID [0x0000 – 0xFFFF] Note: Upon receiving SESSION_START_CMD, If UWB_CONFIG_ID is not configured then the BPRF set# is selected with the allowed SFD_ID value and the corresponding UWB_CONFIG_ID is updated by the implementation (i.e., the UWBS shall set UWB_CONFIG_ID to 0 if SFD_ID is set 0, or to 1 if SFD_ID is set to 2).
CSA_MAC_MODE	1	0xA9	This configuration is used to configure- - The number of active ranging rounds in a RANGING_DURATION. - Offset between two active ranging rounds in a RANGING_DURATION. [b7-b6]: Number of active ranging round(s) 0 = One active ranging round (default). 1 = Two active ranging rounds (CSA use case). 2 and 3 = RFU. [b5-b0]: Offset between two active ranging rounds. 1 to (Nround – 1) Note: Bits [b5, b0] SHALL be set if [b7, b6] set to decimal value 1. Otherwise, bits [b5, b0] will be ignored.

Table 113. Extension for Generic Application Configuration...continued

Parameter Name (*): Can be applied during session active state as well	Length (Octets)	Tag (IDs)	Description
			Nround is calculated based on RANGING_DURATION, SLOTS_PER_RR and SLOT_DURATION.

Note: Refer FiRa Generic UCI specification for more information

11.4.1.1 UL-TDOA specific Proprietary extension

Table 114. UL-TDOA specific Extension for Generic Application Configurations

Parameter Name (*): Can be applied during session active state as well	Length (Octets)	Tag (IDs)	Description
UL_TDOA_DEVICE_ID	1/3/5/9	0x38	Note: In case of Legacy / Vendor specific UL-TDOA Tag session, this configuration parameter is NOT applicable.
UL_TDOA_TX_TIMESTAMP	1	0x39	Note: In case of Legacy / Vendor specific UL-TDOA Tag session, this configuration parameter is NOT applicable.

11.5 Configuring GPIO PIN(s) state

This section tells how to configure and read GPIO PIN state

11.5.1 Setting GPIO PIN state

Host shall use SET_GPIO_PIN_STATE_CMD to drive the state of the GPIO Manually based on the GPIO_MASK.

A) Set the Pin state to High or Low based on the GPIO Mask being set by the HOST.

B) If the Pin state is Hi-Z state: Don't set the pin state to zero or one. Retain the defaults.

Table 115. SET_GPIO_PIN_STATE_CMD

Payload Field	Length (Octets)	Description
GPIO_PIN_DIRECTION	1	Value: 0x00: Pin Direction Input 0x01: Pin Direction Output

Table 115. SET_GPIO_PIN_STATE_CMD...continued

Payload Field	Length (Octets)	Description
		Note: Host will not set the Pin state to High/Low using SET_GPIO_PIN_STATE_CMD if the pin direction is Input.
GPIO_MASK	2	GPIO Filter Mask Refer ANTENNA_SELECTION_GPIO_BITMASK This mask defines which GPIOs shall be changed during the state transition, if a GPIO bit is set to 0 the GPIO shall not change its state. It's a 2-byte value in Little Endian format
GPIO_VALUE	1	Value: 0x00: Set GPIO Pin state to Low (as per GPIO_MASK set by Host Refer: ANTENNA_SELECTION_GPIO_BITMASK 0x01: Set GPIO Pin state to High. 0x02: Hi-Z state (FW will not change the GPIO Pin state. Default state will be retained. No Manual control over the GPIO)

- SET_GPIO_PIN_STATE_CMD command is only allowed in Customer lifecycle,
- Command is allowed only in DEVICE_STATE_READY.

Table 116. SET_GPIO_PIN_STATE_RSP

Payload Filed	Length (Octets)	Value/Description
Status	1	Status values. Refer Status Codes

11.5.2 Reading GPIO PIN state

Host shall use GET_GPIO_PIN_STATE_CMD to get the state of the GPIO Pin based on the GPIO Mask set by the Host

Table 117. GET_GPIO_PIN_STATE_CMD

Payload Filed	Length (Octets)	Value/Description
GPIO_MASK	2	GPIO Filter Mask Refer ANTENNA_SELECTION_GPIO_BITMASK This mask defines which GPIOs state shall be notified to the Host. It's a 2-byte value in Little Endian format

- GET_GPIO_PIN_STATE_CMD command is only allowed in Customer lifecycle,
- Command is allowed only in DEVICE_STATE_READY.

Table 118. GET_GPIO_PIN_STATE_RSP

Payload Field	Length (Octets)	Value/Description
Status	1	Status value = 0 (PH_UCI_STATUS_OK). Below fields are applicable only if UCI status is OK.
GPIO_STATUS	$N * (2 + 1 + 1)$	N refers to number of GPIO Mask being set by the Host via GET_GPIO_PIN_STATE_CMD Octet[0:1] = GPIO_MASK1 Refer: ANTENNA_SELECTION_GPIO_BITM ASK Octet [2] = GPIO_MASK1 Pin direction (0: Refers to Pin Direction Input, 1: Refers to Pin Direction Output) Octet [3] = GPIO_MASK1 Pin State (0: Pin State Low, 1: Pin State High) Octet $[(N*4) - 4] : (N*4) - 3$ = GPIO_MASKN Octet $[(N*4) - 2]$ = GPIO_MASKN Pin direction Octet $[(N*4) - 1]$ = GPIO_MASKN Pin State

11.6 TIMESYNC_TIMESTAMP_NTF

If GPIO_USAGE_CONFIG use case ID is 0x01 (Capture RTC Time stamp if GPIOx is used as a DPD Wake up source) then Time sync notification will be shared to the Host.

Table 119. TIMESYNC data notification

Payload Field	Length(Octets)	Description
Status	1	Status value = 0 (PH_UCI_STATUS_OK).
Time Stamp	8	Absolute time of UWBS when the DPD_WAKEUP_SOURCE Interrupt was received in microseconds
Accuracy	1	Value: 0x00 – Accuracy Low, RTC timestamp captured when UWBS wakeup from DPD (Approximate DPD wakeup time is: 2 milli seconds) Value 0x01 – Accuracy High, RTC timestamp captured when UWBS is in active state (Approximate time for GPIOxx ISR processing: 0.003 milli seconds) Note: In case UWBS not in active state, there is a chance of missing the wakeup(IRQ's are disabled when UWBS is in the process of either entering DPD or exiting out from DPD) , In this case host/BLE has to re-trigger the SYNC GPIO again.

11.7 Debug Configurations

11.7.1 Generic proprietary debug notification

Debug NTF sent if an incorrect/corrupted cmd is received by the UWBS. The Data part of this NTF holds the header information of the received incorrect/corrupted CMD sent by the Host.

Table 120. PROP_GEN_DEBUG_NTF

Payload Field(s)	Length	Value/ Description
PROP_GEN_DEBUG_NTF		
Status	1 Octet	Status values. Refer Status Codes
Data	4 Octets	Header of the received incorrect/corrupted cmd.

11.8 Configuring GPIO PIN(s) state

This section tells how to configure and read GPIO PIN state

11.8.1 Setting GPIO PIN state

Host shall use SET_GPIO_PIN_STATE_CMD to drive the state of the GPIO Manually based on the GPIO_MASK.

- A) Set the Pin state to High or Low based on the GPIO Mask being set by the HOST.
- B) If the Pin state is Hi-Z state: Don't set the pin state to zero or one. Retain the defaults.

Table 121. SET_GPIO_PIN_STATE_CMD

Payload Field	Length (Octets)	Description
GPIO_PIN_DIRECTION	1	Value: 0x00: Pin Direction Input 0x01: Pin Direction Output Note: Host will not set the Pin state to High/Low using SET_GPIO_PIN_STATE_CMD if the pin direction is Input.
GPIO_MASK	2	GPIO Filter Mask Refer ANTENNA_SELECTION_GPIO_BITMASK

Table 121. SET_GPIO_PIN_STATE_CMD...continued

Payload Field	Length (Octets)	Description
		This mask defines which GPIOs shall be changed during the state transition, if a GPIO bit is set to 0 the GPIO shall not change its state. It's a 2-byte value in Little Endian format
GPIO_VALUE	1	Value: 0x00: Set GPIO Pin state to Low (as per GPIO_MASK set by Host Refer: ANTENNA_SELECTION_GPIO_BITMASK 0x01: Set GPIO Pin state to High. 0x02: Hi-Z state (FW will not change the GPIO Pin state. Default state will be retained. No Manual control over the GPIO)

- SET_GPIO_PIN_STATE_CMD command is only allowed in Customer lifecycle,
- Command is allowed only in DEVICE_STATE_READY.

Table 122. SET_GPIO_PIN_STATE_RSP

Payload Filed	Length (Octets)	Value/Description
Status	1	Status values. Refer Status Codes

11.8.2 Reading GPIO PIN state

Host shall use GET_GPIO_PIN_STATE_CMD to get the state of the GPIO Pin based on the GPIO Mask set by the Host

Table 123. GET_GPIO_PIN_STATE_CMD

Payload Filed	Length (Octets)	Value/Description
GPIO_MASK	2	GPIO Filter Mask Refer ANTENNA_SELECTION_GPIO_BITMASK This mask defines which GPIOs state shall be notified to the Host. It's a 2-byte value in Little Endian format

- GET_GPIO_PIN_STATE_CMD command is only allowed in Customer lifecycle,
- Command is allowed only in DEVICE_STATE_READY.

Table 124. GET_GPIO_PIN_STATE_RSP

Payload Field	Length (Octets)	Value/Description
Status	1	Status value = 0 (PH_UCI_STATUS_OK). Below fields are applicable only if UCI status is OK.
GPIO_STATUS	$N * (2 + 1 + 1)$	N refers to number of GPIO Mask being set by the Host via GET_GPIO_PIN_STATE_CMD Octet[0:1] = GPIO_MASK1 Refer: ANTENNA_SELECTION_GPIO_BITMASK Octet [2] = GPIO_MASK1 Pin direction (0: Refers to Pin Direction Input, 1: Refers to Pin Direction Output) Octet [3] = GPIO_MASK1 Pin State (0: Pin State Low, 1: Pin State High) Octet $[(N*4) - 4] : [(N*4) - 3]$ = GPIO_MASKN Octet $[(N*4) - 2]$ = GPIO_MASKN Pin direction Octet $[(N*4) - 1]$ = GPIO_MASKN Pin State

11.9 Debug Configurations

11.9.1 Generic proprietary debug notification

Debug NTF sent if an incorrect/corrupted cmd is received by the UWBS. The Data part of this NTF holds the header information of the received incorrect/corrupted CMD sent by the Host.

Table 125. PROP_GEN_DEBUG_NTF

Payload Field(s)	Length	Value/ Description
PROP_GEN_DEBUG_NTF		
Status	1 Octet	Status values. Refer Status Codes
Data	4 Octets	Header of the received incorrect/corrupted cmd.

11.10 Proprietary Test Configurations

Below configuration shall be set using TEST_CONFIG_SET_CMD command which is defined in "**FiRa Consortium - UCI Test Specification**". All Extended APP configuration are starts with "**0xE5**".

Table 126. TEST_CONFIG_SET_CMD

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Proprietary sub ID	Description
RSSI_CALIBRATION_OPTION	1	0xE5	0x01	This parameter is used to adjust the receive gain index for AGC. This parameter to be set before performing PER Rx Test. 0x00 - Calibration level 1 0x01 - Calibration level 2 (default = 0x01)
AGC_GAIN_VAL_RX	2	0xE5	0x02	This parameter is used for AGC gain setting of Rx during RSSI calibration. This config is applicable only when RSSI_CALIBRATION_OPTION = 0. <u>Octet[0]: Gain index override enable/disable</u> • bit[0]: Override control flag – 0: Override disable – 1: Override enable (default) • bit[1-7]: RFU <u>Octet[1]: Gain index value</u> • bit[0-5] : AGC gain index value (default = 0x2) • bit[6-7]: RFU
TEST_SESSION_STS_KEY_OPTION	1	0xE5	0x03	This parameter is used to choose between the IEEE or proprietary key and data input for DRBG (Deterministic Random bit generator) used for STS generation in test sessions. <u>Octet[0]:</u> • bit[0]: values as below – 0: Proprietary option for key, data values – 1: IEEE option for key, data set values as mentioned in Annex H of P802.15.4z-D8 specification (default) • bit[1-7]: RFU
TX_POWER_CALIBRATION_MODE	1	0xE5	0x10	0x00: Disable (default) 0x01: Enable
CARRIER_FREQUENCY_OFFSET	4	0xE5	0x11	Adds the given offset in Hz to the configured Carrier Frequency (which depends on selected RF Channel) during test mode. The value is encoded as signed 32bit integer with a range from -250000000 (-250MHz) up to 250000000 (250MHz). Default is 0 (no offset added).

Table 126. TEST_CONFIG_SET_CMD...continued

Parameter Name	Len (Octets)	Tag (Ext. IDs)	Proprietary sub ID	Description
				Note: • This parameter is applicable only in Continuous wave mode (i.e., TEST_PERIODIC_TX_CMD with NUM_PACKETS is set to 0) and it will be ignored for other test modes.
RFU		0xE5	0xA0-0xFF	Reserved for Radar Test Configurations

11.11 GID and OID Definitions

Table 127. Proprietary GID and GID Definition

OID	Message Name	Description
0b100000	CCC_RANGE_DATA_NTF	While the UWB session is ongoing, the UWBS shall send ranging result with the CCC_RANGE_DATA_NTF notification to the Host when the ranging round is complete.
GID: 1001 (Proprietary group)		
0b001010	RADAR_RX_NTF	RADAR notification
GID: 1010 (IOT Proprietary group)		
0b000010	TRNG_DATA_CMD TRNG_DATA_RSP	TRNG_DATA command is used for True Random Number generation.
0b000011	WRITE_MODULE_MAKER_INFO_CMD WRITE_MODULE_MAKER_INFO_RSP	WRITE_MODULE_MAKER_INFO command is used to write module maker info. Note: Supported in Customer Lifecycle.
0b000100	READ_MODULE_MAKER_INFO_CMD READ_MODULE_MAKER_INFO_RSP	READ_MODULE_MAKER_INFO command is used to read module maker info. Note: Supported in all Lifecycle.
0b000101	PROP_SET_PROFILE_CMD PROP_SET_PROFILE_RSP	Used to set the specific UCI parameters received from MW and 'Profile'.
0b001010	TEST_CHAINED_TX_CMD TEST_CHAINED_TX_RSP	Command requesting for UCI chained frames.
0b111010	DEBUG_CFX_LAST_EVENTS_NTF	DSP last events notification
0b111011	DEBUG_ARM_LAST_EVENTS_NTF	ARM MCU last events notification
GID: 1110 (Customer Proprietary 1 group)		
0b000000	CORE_DEVICE_INIT_CMD CORE_DEVICE_INIT_RSP	Command is used to initialize the UWBS for specific platform
0b000001	DBG_DATA_LOGGER_NTF	Notification used by the UWBS to share DSP log
0b000010	DBG_GET_ERROR_LOG_CMD DBG_GET_ERROR_LOG_RSP	Command used to dump the crash log
0b000011	SE_GET_BINDING_COUNT_CMD SE_GET_BINDING_COUNT_RSP	Command to fetch the available binding count
0b000101	SE_COMM_ERROR_NTF	Report SE communication failure
0b000110	BINDING_STATUS_NTF	Notify the binding status between SR250 and SE
0b000111	SCHEDULER_STATUS_NTF	To notify Schedule status for sessions
0b001000	UWB_SESSION_KDF_NTF	To notify KDF output values
0b001011	QUERY_TEMPERATURE_CMD	Query current temperature in UWBS

Table 127. Proprietary GID and GID Definition...continued

OID	Message Name	Description
	QUERY_TEMPERATURE_RSP	
0b010011	UWB_WLAN_COEX_NTF	Send warning NTF when WLAN max grant duration is exceeded
0b011000	PROP_GEN_DEBUG_NTF	Debug NTF sent if an incorrect/corrupted cmd is received by the UWBS. Refer Table 120
0b011101	TIMESYNC_TIMESTAMP_NTF	If GPIO_USAGE_CONFIG use case ID is 0x01 then Time sync notification will be shared to the Host
0b011010	SET_GPIO_PIN_STATE_CMD	Drive the state of the GPIO Manually based on the GPIO_MASK. Applicable only in Customer lifecycle
0b011011	GET_GPIO_PIN_STATE_CMD	Get the state of the GPIO Pin based on the GPIO Mask set by the Host
0b100100	DATA_DOWNLOAD_START_CMD DATA_DOWNLOAD_START_RSP	Command sent to start the Data Download refer Section 5.1
0b100101	DATA_DOWNLOAD_CMD DATA_DOWNLOAD_RSP DATA_DOWNLOAD_NTF	Command sent to download the data refer Section 5.2
0b100110	DATA_DOWNLOAD_STOP_CMD DATA_DOWNLOAD_STOP_RSP	Command sent to stop the data download refer Section 5.3
0b100111	DATA_DOWNLOAD_QUERY_CMD DATA_DOWNLOAD_QUERY_RSP	Command sent to Query all the downloaded Data. refer Section 5.6
0b101000	DATA_DOWNLOAD_DELETE_CMD DATA_DOWNLOAD_DELETE_RSP	Command sent to delete downloaded Data. refer Section 5.5
0b101001	DATA_DOWNLOAD_COMPLETE_NTF	Notification sent to indicate the Data Download is complete refer Section 5.4
GID: 1111 (Customer Proprietary 2 group)		
0b000000	SET_VENDOR_APP_CONFIG_CMD SET_VENDOR_APP_CONFIG_RSP	Set NXP proprietary session app configurations
0b000010	GET_ALL_UWB_SESSIONS_CMD GET_ALL_UWB_SESSIONS_RSP	Retrieve all UWB sessions
0b000011	GET_VENDOR_APP_CONFIG_CMD GET_VENDOR_APP_CONFIG_RSP	Get NXP proprietary session app configurations
0b0110101	UWBS_SLEEP_ENTRY_PREVENTION_NTF	Contains IDs of tasks that prevent UWBS from entering sleep.
0b100000	DO_CHIP_CALIBRATION_CMD DO_CHIP_CALIBRATION_RSP DO_CHIP_CALIBRATION_NTF	Calibrate the Chip of UWBS

Table 127. Proprietary GID and GID Definition...continued

OID	Message Name	Description
0b100001	SET_DEVICE_CALIBRATION_CMD SET_DEVICE_CALIBRATION_RSP	Set the calibrated value for requested parameter
0b100010	GET_DEVICE_CALIBRATION_CMD GET_DEVICE_CALIBRATION_RSP	Get the calibrated value for requested parameter
0b100011	SET_SECURE_CALIBRATION_CMD SET_SECURE_CALIBRATION_RSP	Set a value to a secure calibration parameter when SR250 is in protected lifecycle
0b110011	PSDU_LOG_NTF	Notification used by SR250 to notify PSDU payload
0b110100	CIR_LOG_NTF	Notification used by SR250 to notify the CIR logs
0b111001	UWB_WLAN_IND_STATUS_NTF	To notify the UWB_WLAN_IND GPIO toggle status
0b111010	WLAN_UWB_IND_STATUS_NTF	To notify the WLAN_UWB_IND GPIO status during ranging round
0b111011	UWB_WLAN_COEX_MAX_GRANT_DURATION_EXCEEDED_WRN_NTF	Send warning NTF when WLAN max grant duration is exceeded

11.12 Status Codes

Table 128. Status Codes

Status Codes	Description	
Status Codes		
0x00-0x4F	[UCI-FORUM] Status Codes	
Proprietary status code		
0x53	STATUS_CALIBRATION_IN_PROGRESS	If calibration process is ongoing
0x54	STATUS_DEVICE_TEMP_REACHED_THERMAL_RUNAWAY	Over temperature detected
0x55	STATUS_FEATURE_NOT_SUPPORTED	Status to indicate feature not supported.
0x56	STATUS_NUM_PACKET_EXCEEDS_1000_FOR_TEST_PER_RX	NUM_PACKETS > 1000 packet for TEST_PER_RX if Vendor extended notifications enabled
0x57	STATUS_CALIBRATION_NOT_CONFIGURED	Status to indicate precondition is not set during set calibration

Table 128. Status Codes...continued

Status Codes	Description	
0x58	STATUS_UNDERFLOW_TIMEOUT	Status indicating SR250 received an incomplete UCI frame from the HOST.
0x59	STATUS_DEVICE_LOW_VBAT	Notification of low VBat to host
0x81	STATUS_RANGING_PHY_RX_SECDEC_FAILED	Rx SECDEC decoder failed
0x82	STATUS_RANGING_PHY_RX_RSDEC_FAILED	Rx RS decoder failed
0x83	STATUS_RANGING_PHY_RX_DEC_FAILED	Rx decode chain failed
0x84	STATUS_RANGING_PHY_RX_ERR_FAILED	RX data buffer overflow happened
0x85	STATUS_RANGING_PHY_RX_PHR_DECODE_FAILED	PHR decoder failed
0x86	STATUS_RANGING_PHY_RX_SYNC_SFD_TIMEOUT	SYNC or SFD timeout
0x87	STATUS_RANGING_PHY_RX_PHR_DATA_RATE_ERROR	In PHR, Data rate error
0x88	STATUS_RANGING_PHY_RX_PHR_RANGING_ERROR	In PHR, Ranging error
0x89	STATUS_RANGING_PHY_RX_PHR_PREAMBLE_DUR_ERROR	In PHR, Preamble duration error
0x8A	STATUS_MAX_ACTIVE_GRANT_DURATION_EXCEEDED	MAX Active Grant Duration Exceeded Warning Notification
0x8B-0x9F	RFU	
0xA2-0xAF	RFU	
0xB0	STATUS_CALIBRATION_FORBIDDEN	Parameter in SET_DEVICE_CALIBRATION_CMD cannot be accessed in PROTECTED mode.
0xB1	STATUS_CALIBRATION_NOT_AUTHORIZED	CMAC error detected in SET_DEVICE_CALIBRATION_SEC_CMD.
0xB2	STATUS_CALIBRATION_INVALID_CONFIG_VERSION	The ConfigVersion used in SET_DEVICE_CALIBRATION_SEC_CMD is older than the one stored in Customer Area.
0xB3	STATUS_CALIBRATION_NO_PLATFORM_ID	Platform ID not set yet, hence SET_DEVICE_CALIBRATION_SEC_CMD can't be used.

Table 128. Status Codes...continued

Status Codes	Description	
0xE1	STATUS_REJECTED_BUFFER_IN_USE	Indicates that the UCI buffer is already in use
0xB4-0xFF	RFU	

Table 129. SE Status Codes

Status Codes	Description
Proprietary SE status code	
0x70	STATUS_NO_SE
0x71	RFU
0x72	STATUS_SE_RECOVERY_FAILURE
0x73	STATUS_SE_RECOVERY_SUCCESS
0x74	STATUS_SE_APDU_CMD_FAIL
0x75	STATUS_SE_AUTH_FAIL

11.13 Pulse Shape configuration

The mapping table at firmware:

Table 130. Pulse Shape Config Default values

Pulse Shape ID \ Pulse Shape Config	Octet[0]	Octet[1]	Octet[2]
50	50	50	50
56	56	56	56
60	60	60	60

12 References

Table 131. References

Reference	Description / Link
UWBS HBCI commands	[HBCI commands]
[UCI-FORUM]	FiRa Consortium - UCI Generic Specification

13 Legal information

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